Waste Management and Cleanliness

Problem: Overflowing garbage bins and inefficient waste collection routes.

Technological University of the Philippines - Manila

GROUP 2 - BET-CpET 4A-M
Embedded Systems - LECTURE

Proposed System Dashboard, Touchscreen
and Mobile App Visual Presentation

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I. Smart Bin Touchscreen Mockup

24 screens across all operational states
Each shown in Light and Dark themes

II. Smart Bin Dashboard

15 views including all tabs, roles, and special modes
Each shown in Light and Dark themes

III. Smart Bin Mobile App

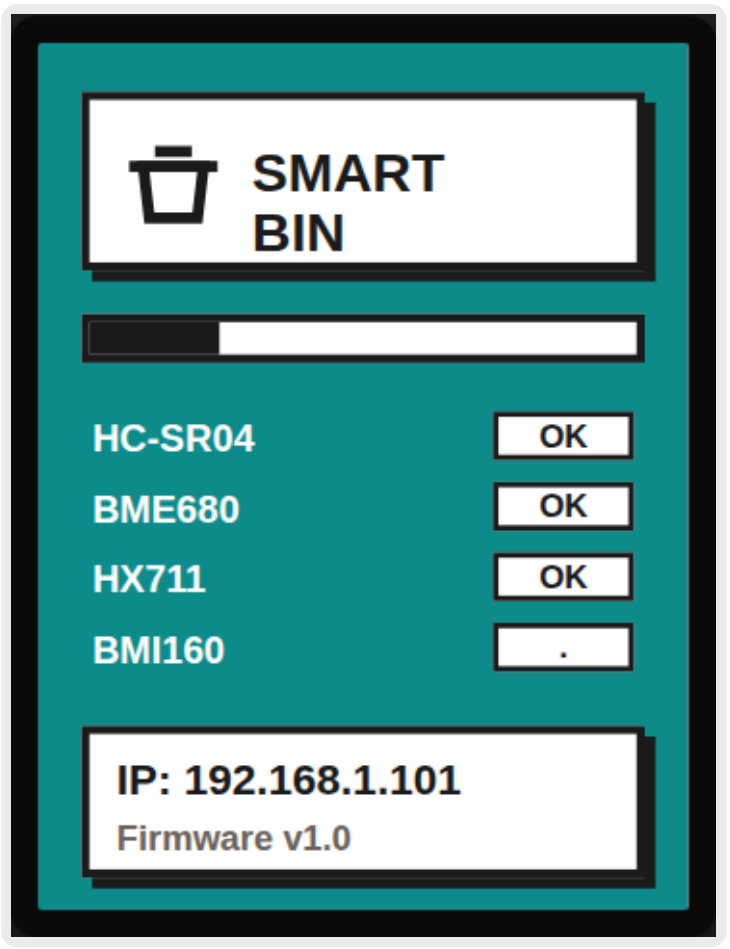
48 screens across 9 sections
Each shown in Light and Dark themes
Sections: Start, Overview, Many Bins, Charts, Settings, Simulation, Alerts, Push, Roles & Theme

Smart Bin Touchscreen Mockup

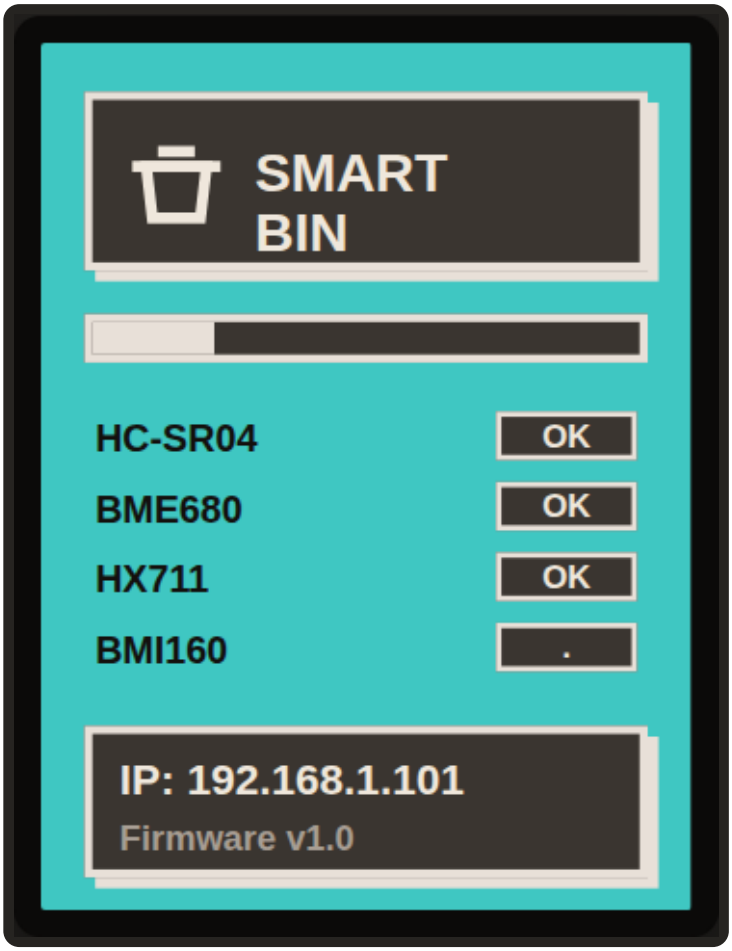
240 x 320 px | 24 Screens | Light & Dark Themes

A8 - Boot Screen

System startup with sensor checks, IP address, and firmware version



Light Theme



Dark Theme

A8 - Calibrating

Operator calibration in progress with spinning ring and step status



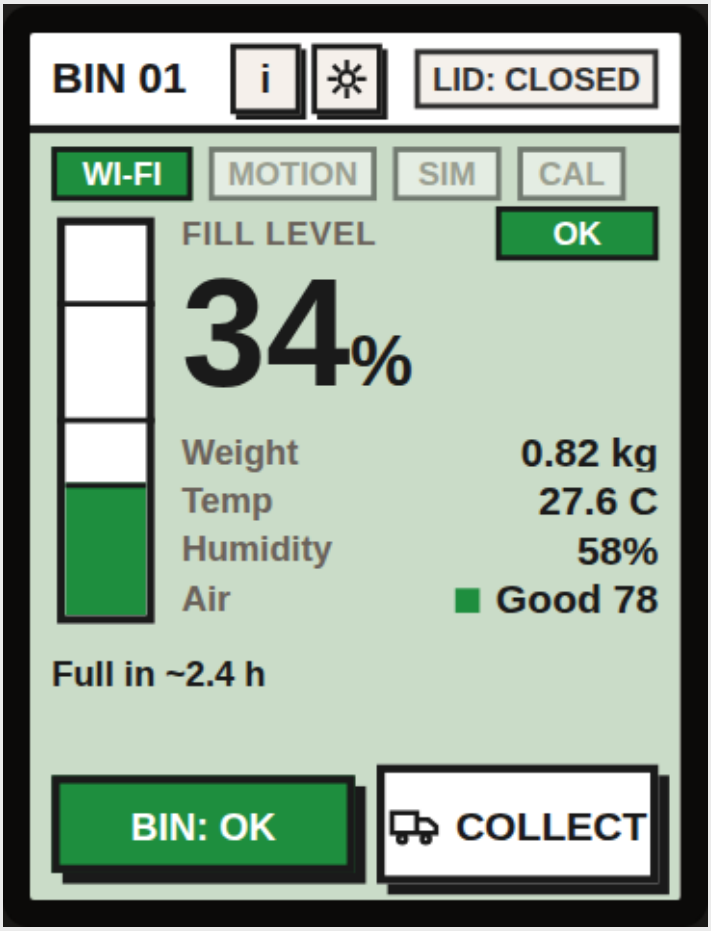
Light Theme



Dark Theme

A1 - Main Status (Normal)

Normal operation at 34% fill with sensor readings and status chips



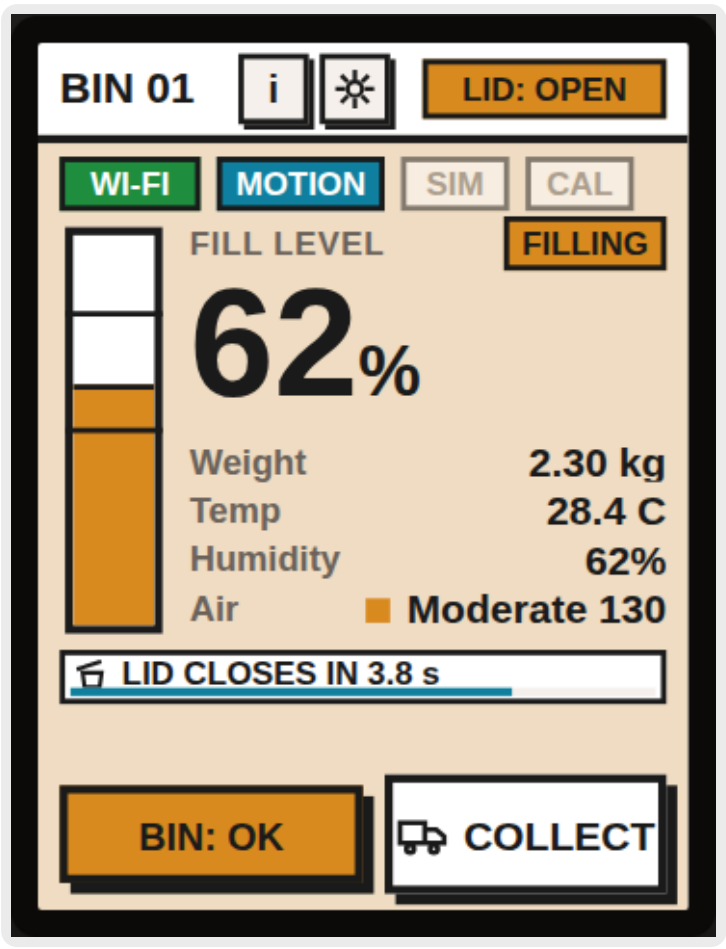
Light Theme



Dark Theme

A1 - Filling (Lid Open)

Motion-triggered lid open at 62% with countdown timer



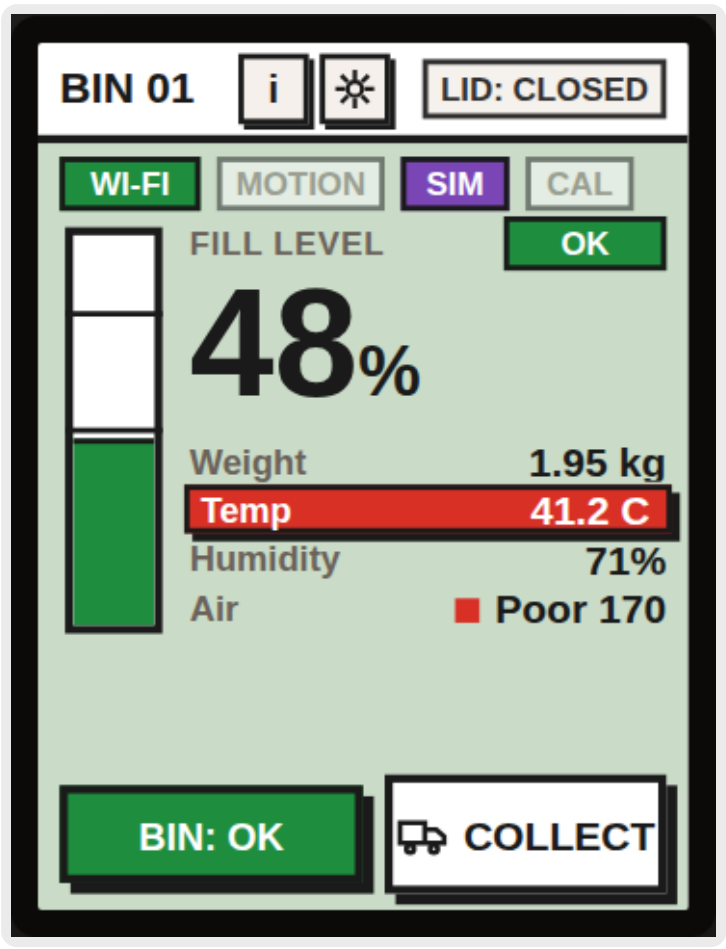
Light Theme



Dark Theme

A1 - Sensor Alarm

Temperature alarm triggered with simulation mode active



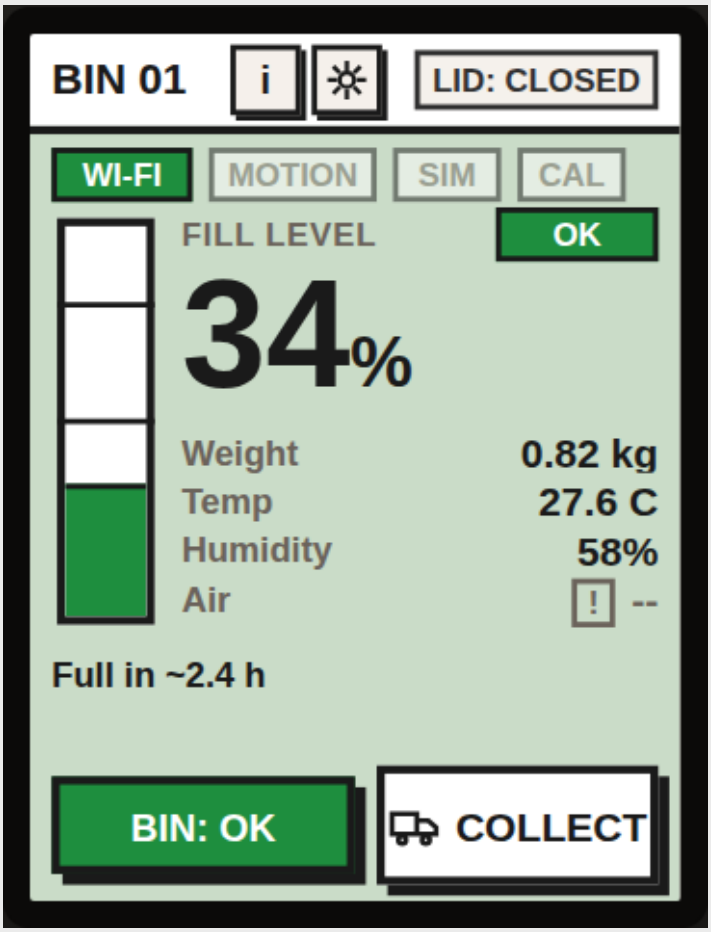
Light Theme



Dark Theme

A1 - Air Sensor Warming Up

BME680 gas sensor stabilizing after power-on



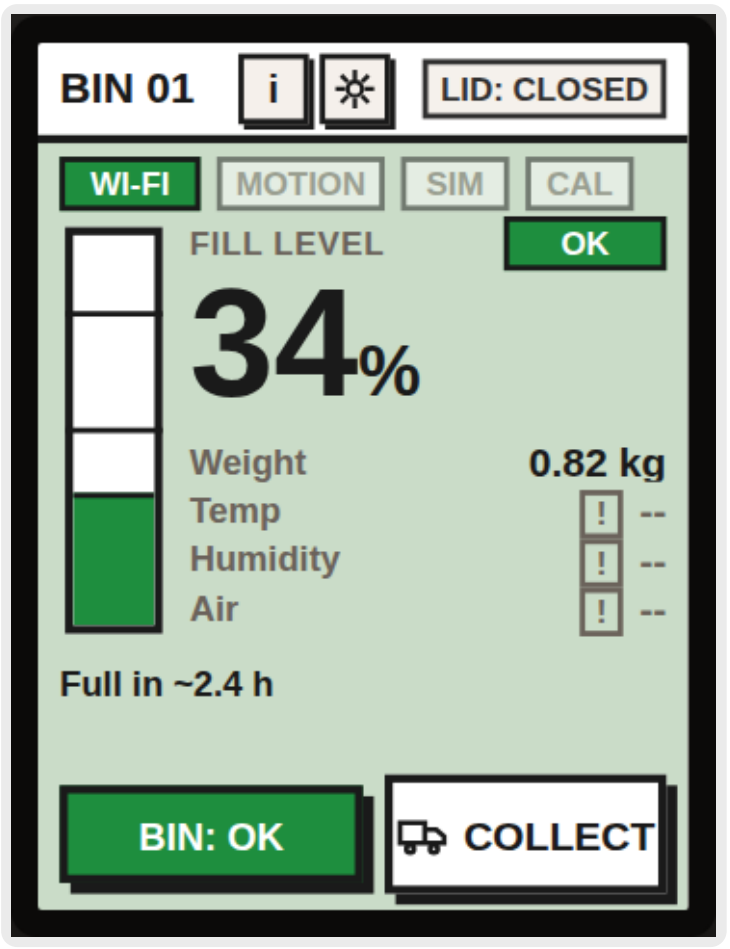
Light Theme



Dark Theme

A1 - BME680 Sensor Failure

Environmental sensor not responding, fill and weight still active



Light Theme



Dark Theme

A2 - Urgent (Bin Full)

Full red alert at 92% fill with audible alarm



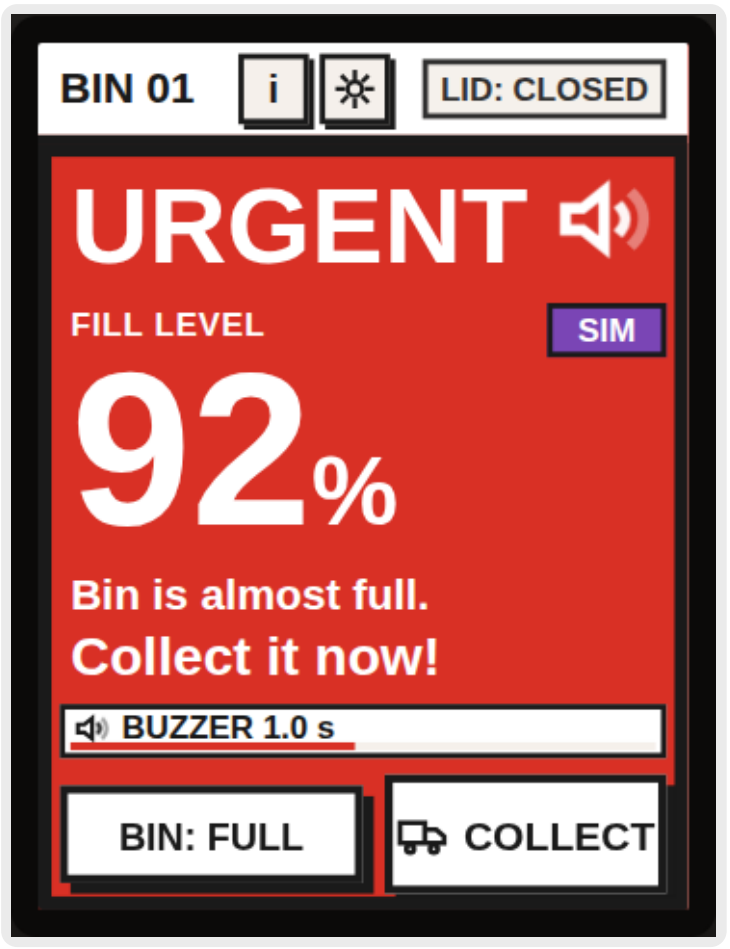
Light Theme



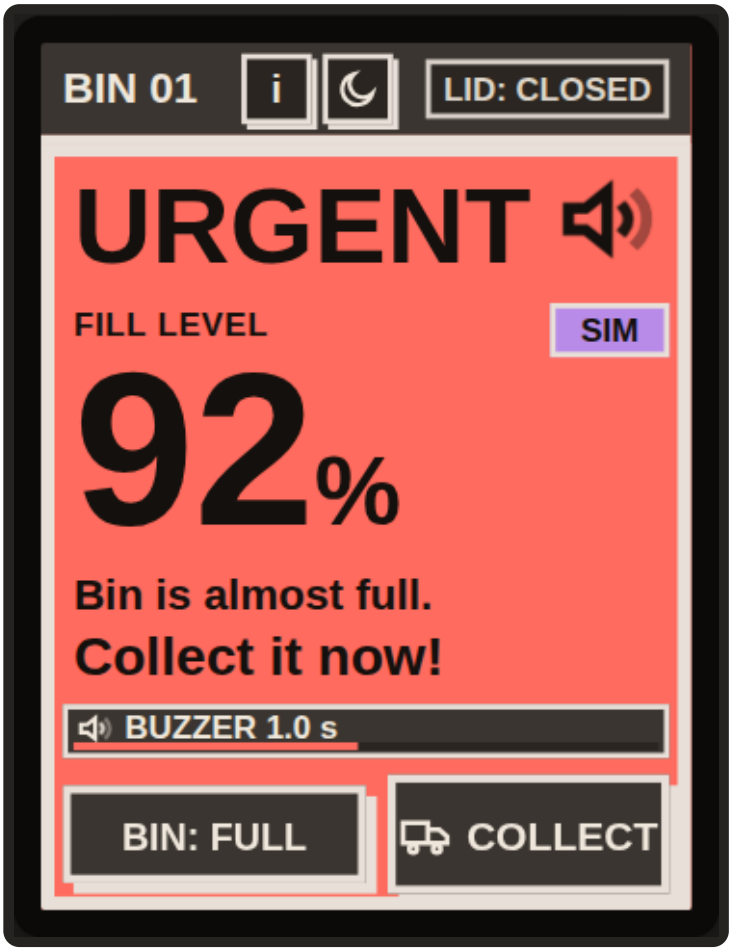
Dark Theme

A2 - Urgent with Buzzer Timer

Buzzer countdown with SIM badge indicator



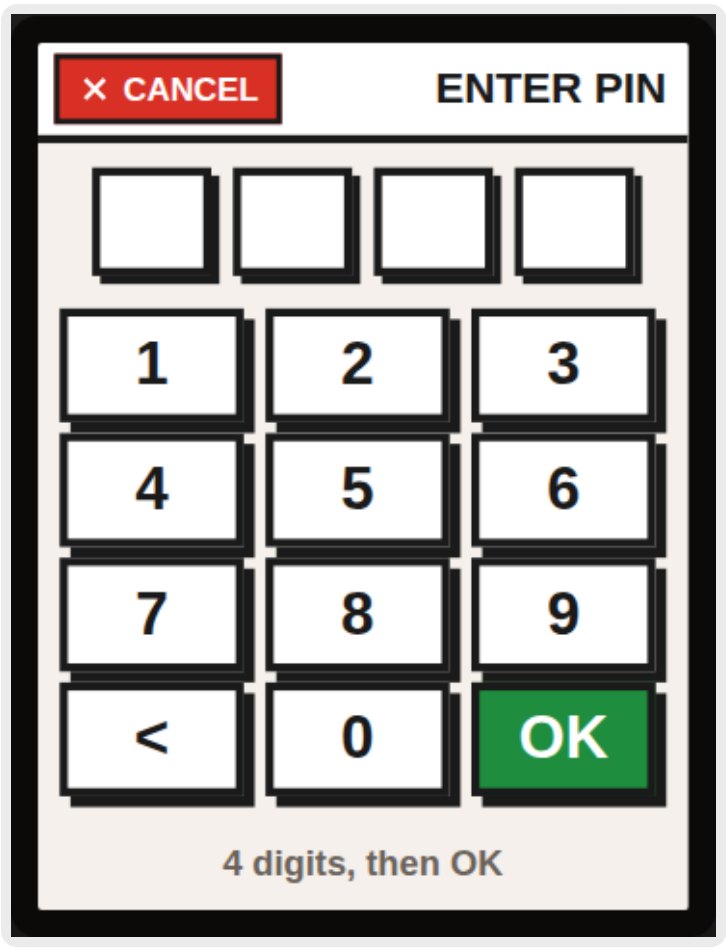
Light Theme



Dark Theme

A3 - Enter PIN

Keypad entry for collection authorization



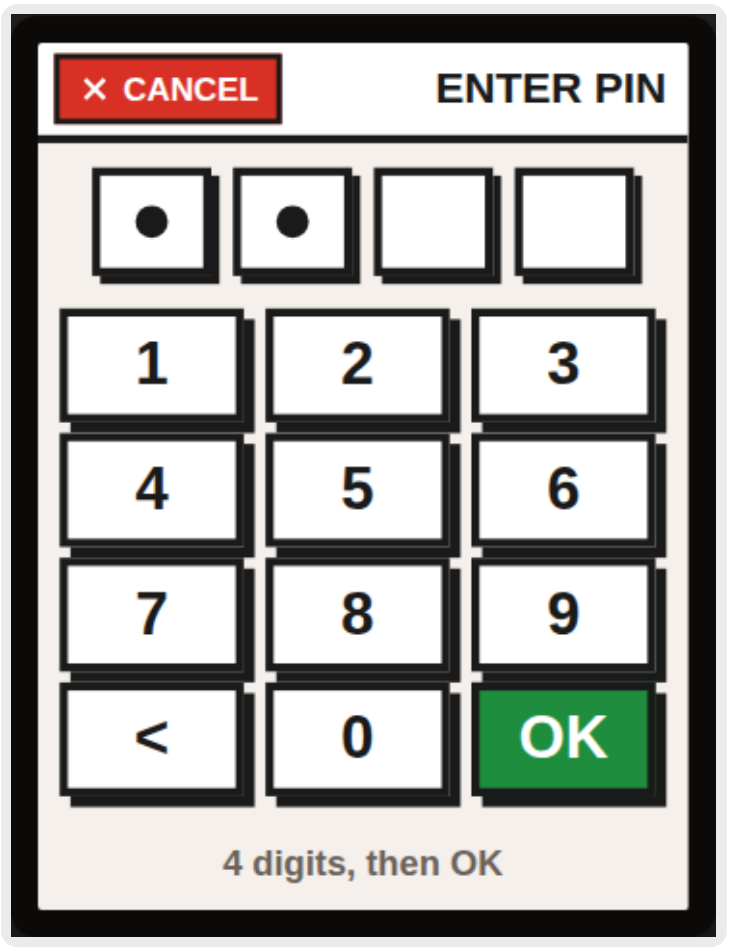
Light Theme



Dark Theme

A3 - PIN Typing

PIN entry in progress with masked dots



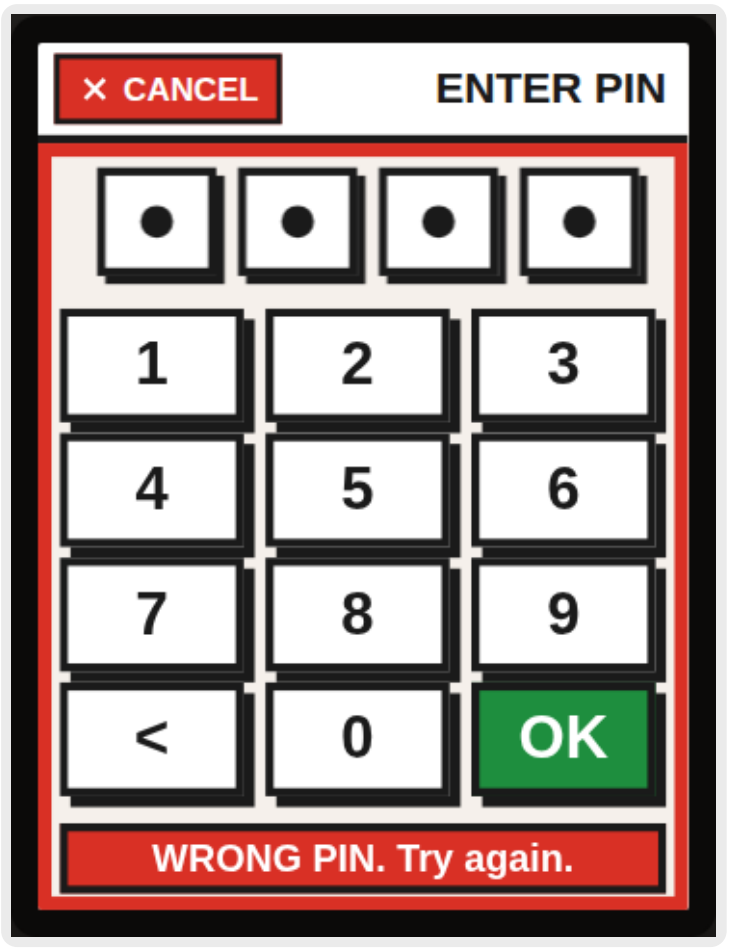
Light Theme



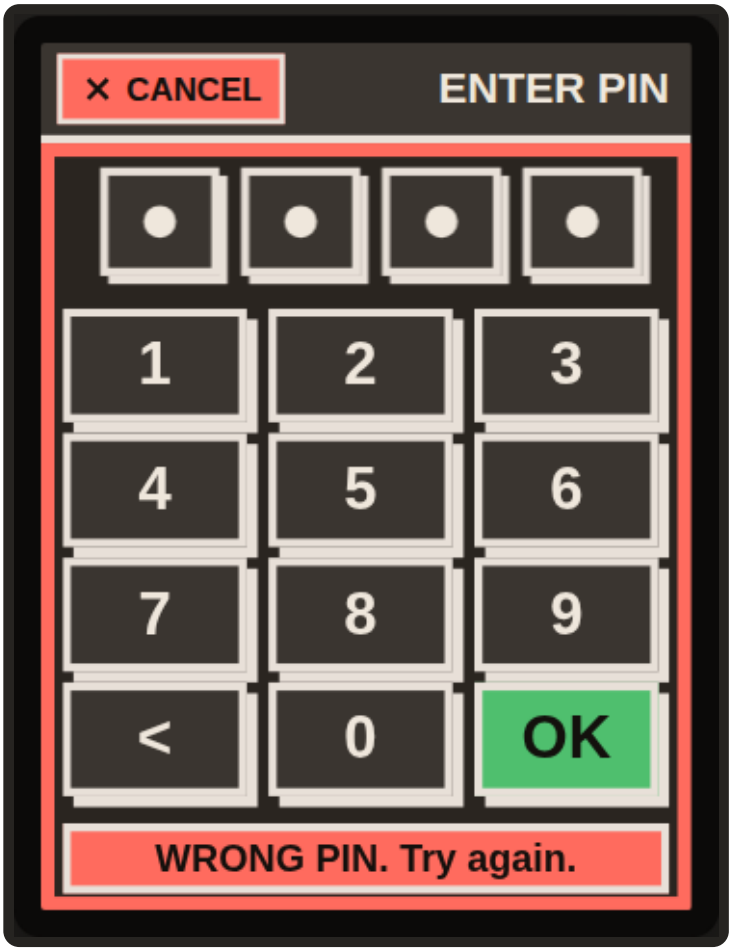
Dark Theme

A3 - Wrong PIN

Incorrect PIN with shake animation and red border



Light Theme



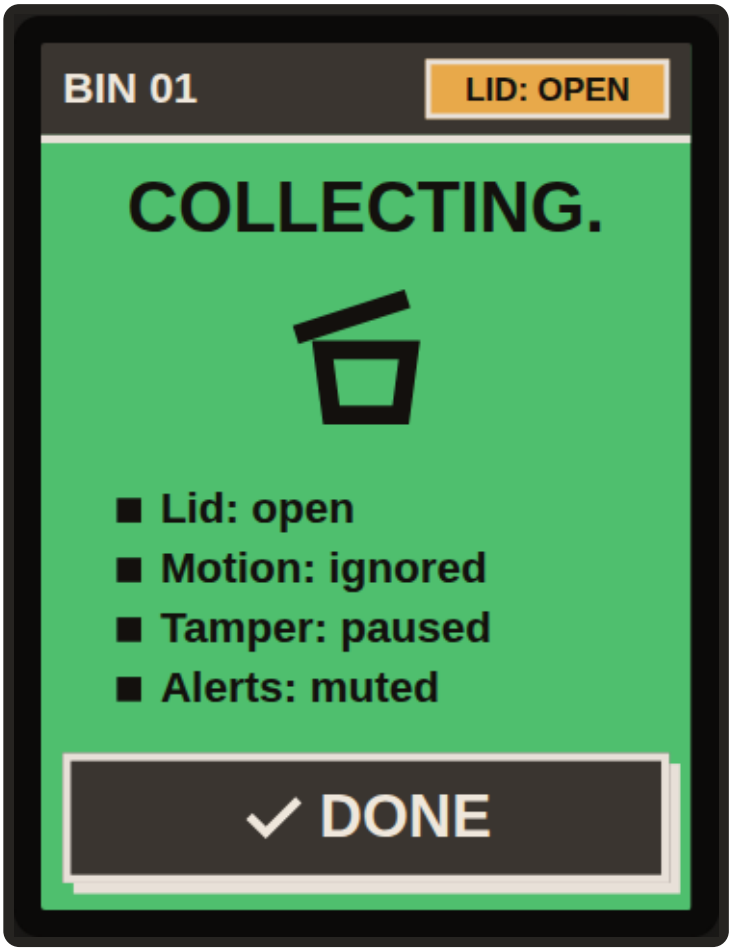
Dark Theme

A4 - Collecting

Active collection in progress, alerts muted



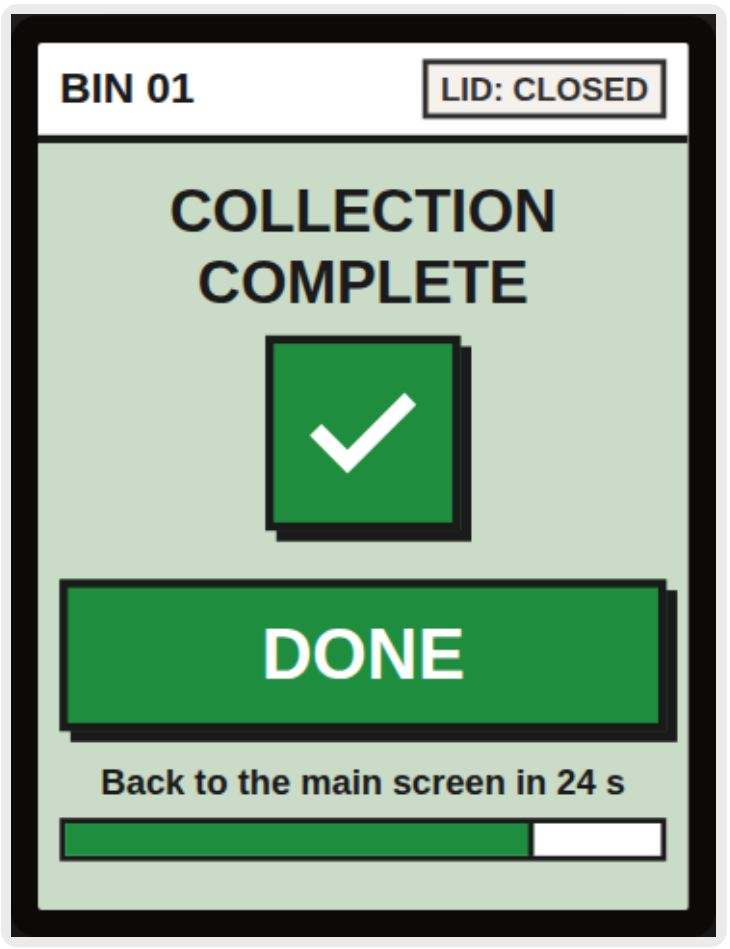
Light Theme



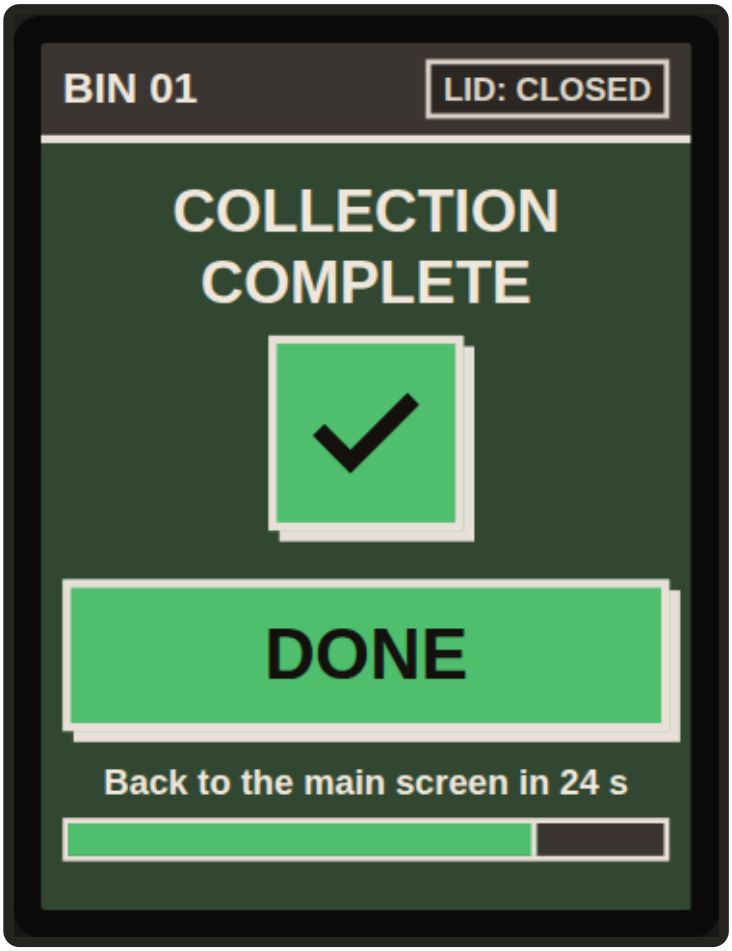
Dark Theme

A5 - Collection Complete

Completion screen with 30-second auto-return countdown



Light Theme



Dark Theme

A6 - Tamper Countdown

Tamper detected with 10-second stabilization countdown



Light Theme



Dark Theme

A6 - Tamper Latched

Sustained tamper alert, requires dashboard reset



Light Theme



Dark Theme

A7 - Locked

Bin locked, only PIN-authorized collection allowed



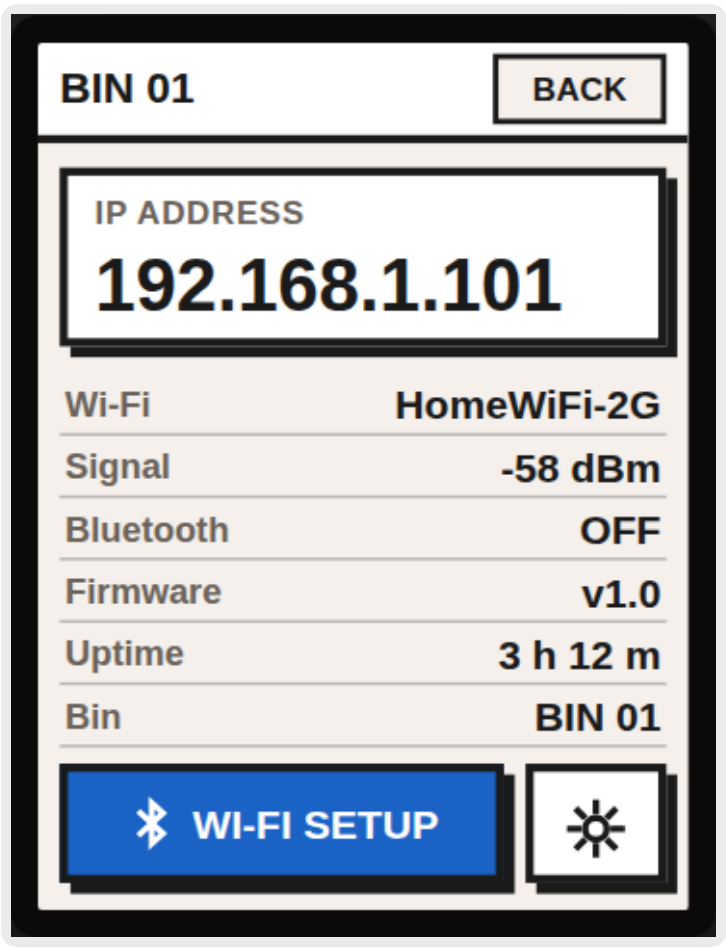
Light Theme



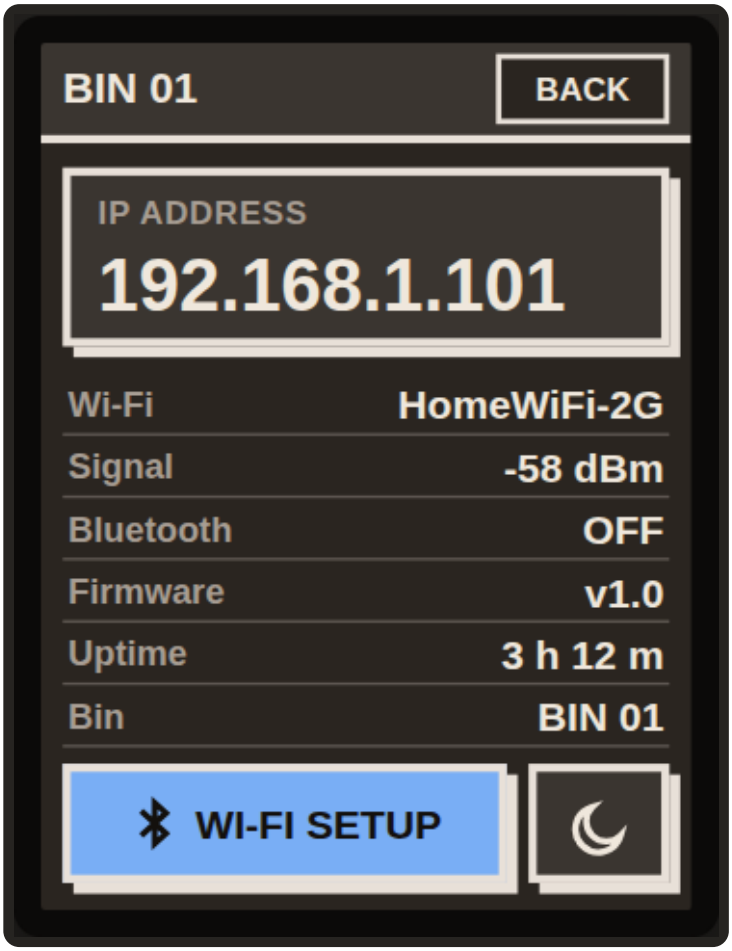
Dark Theme

A9 - Info Screen

System information: IP, Wi-Fi, Bluetooth, firmware, uptime



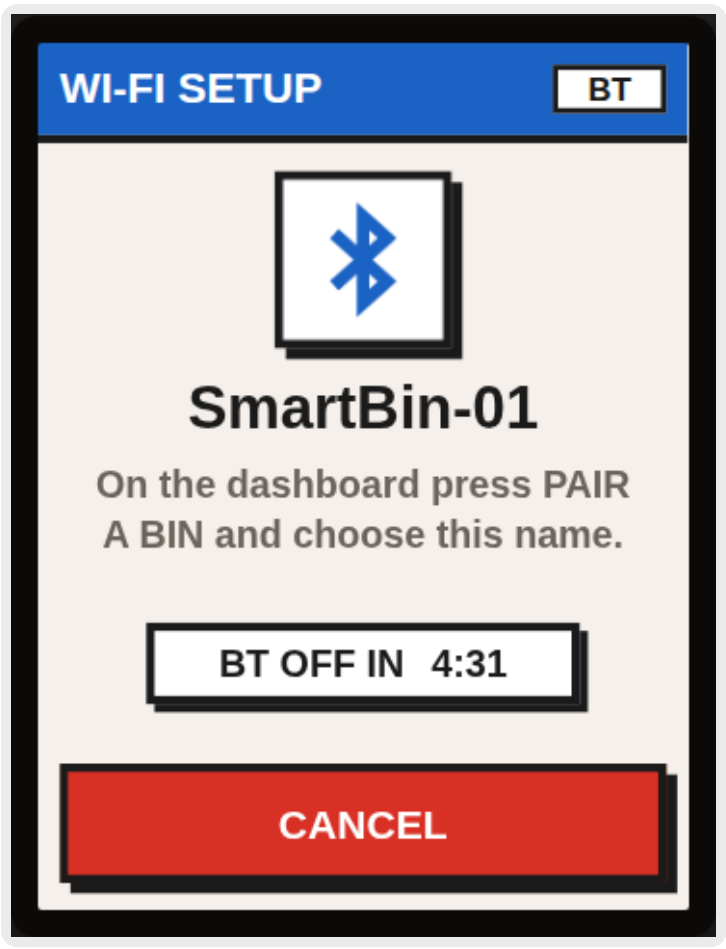
Light Theme



Dark Theme

A10 - Wi-Fi Setup: Waiting

Bluetooth active, waiting for dashboard connection



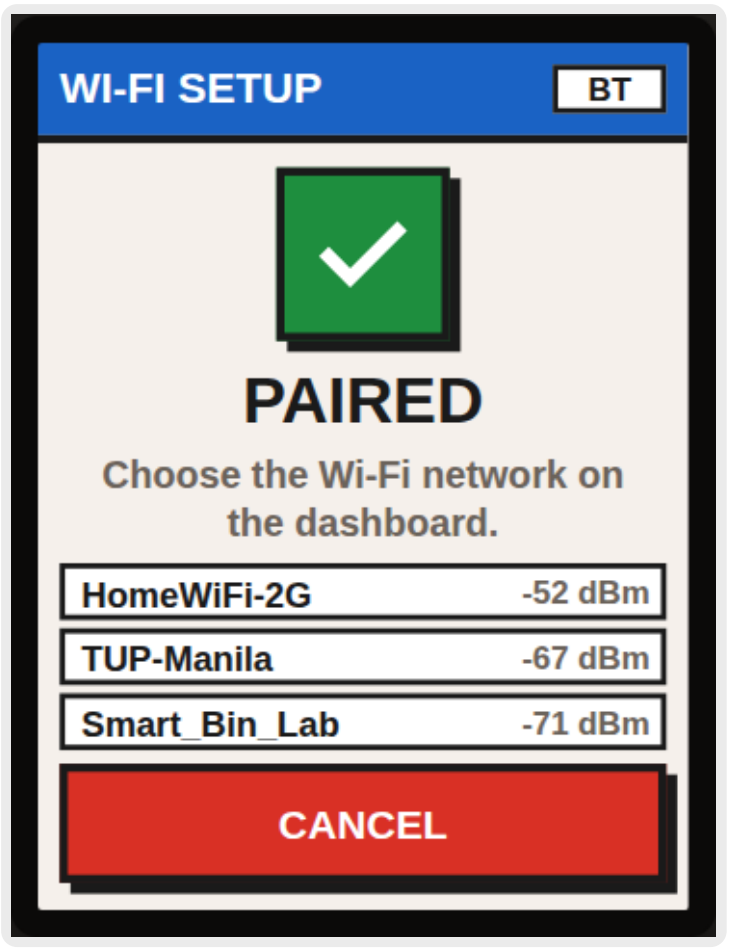
Light Theme



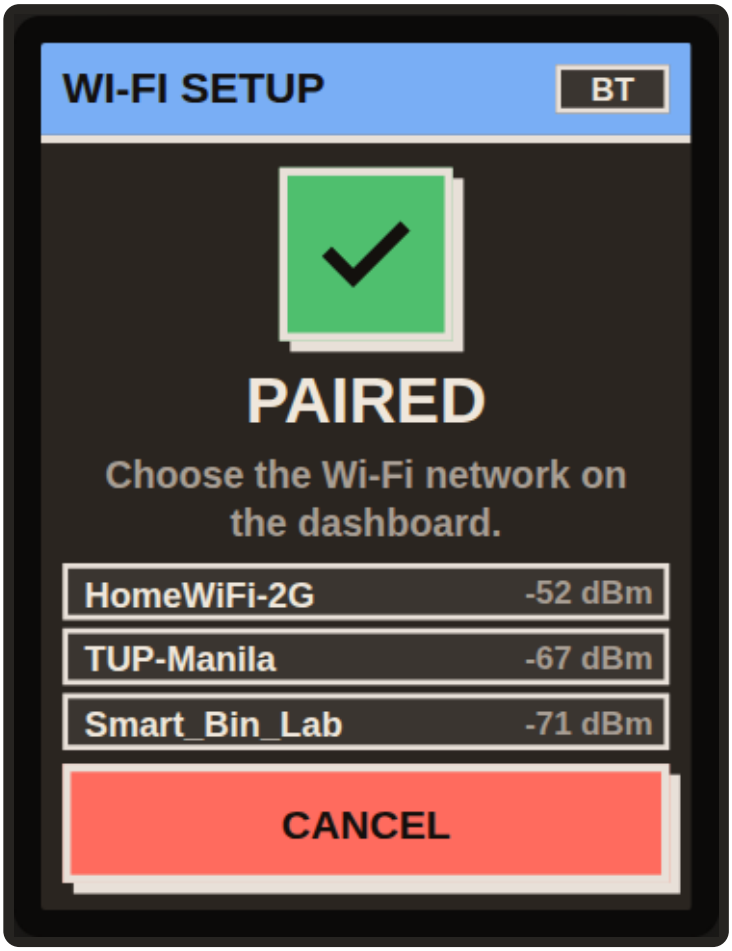
Dark Theme

A10 - Wi-Fi Setup: Paired

Dashboard connected, scanning 2.4 GHz networks



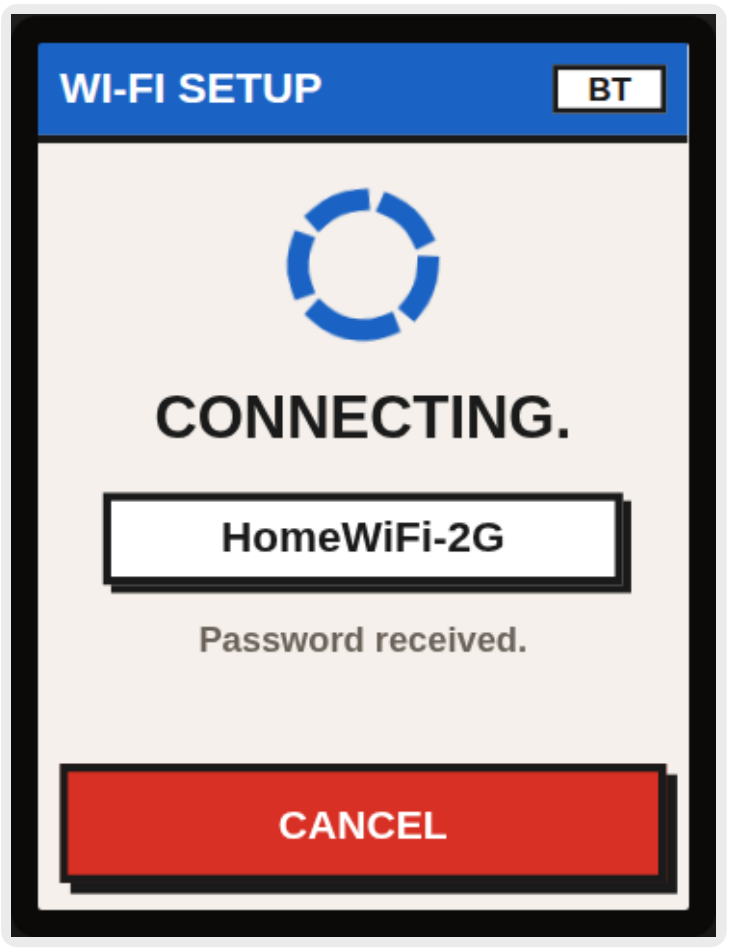
Light Theme



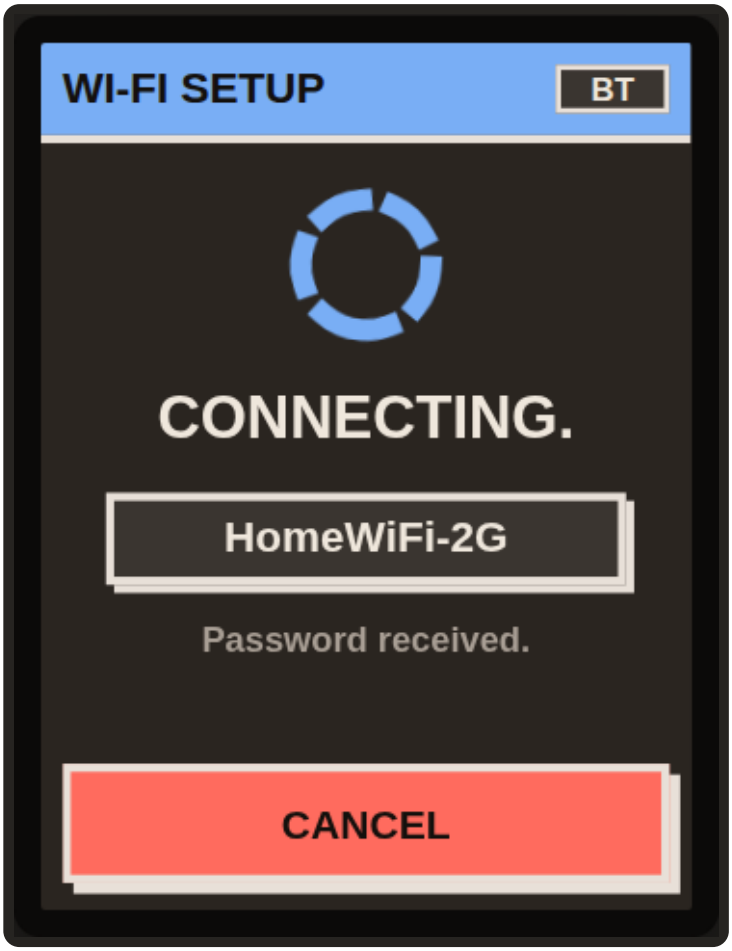
Dark Theme

A10 - Wi-Fi Setup: Connecting

Attempting Wi-Fi connection with received credentials



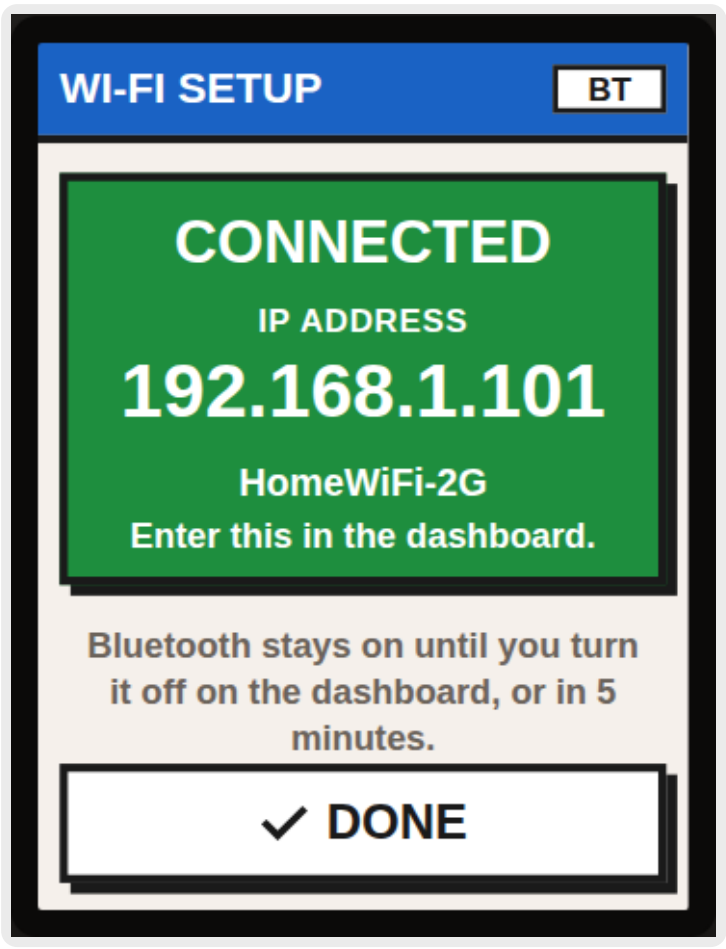
Light Theme



Dark Theme

A10 - Wi-Fi Setup: Connected

Successfully connected, showing IP address



Light Theme



Dark Theme

A10 - Wi-Fi Setup: Failed

Connection failed with error and retry option



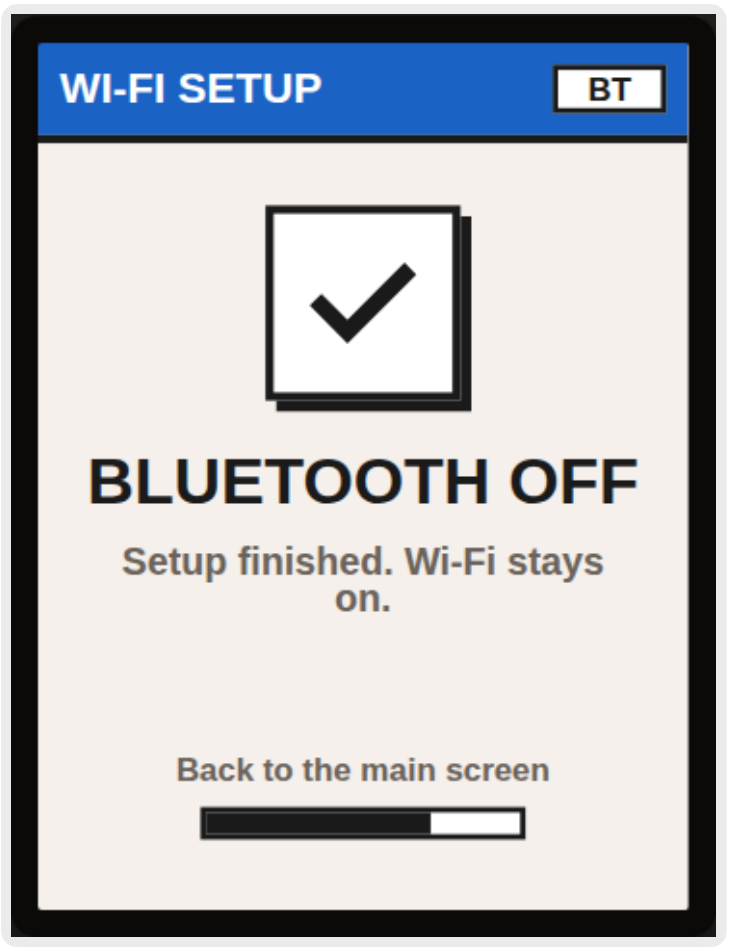
Light Theme



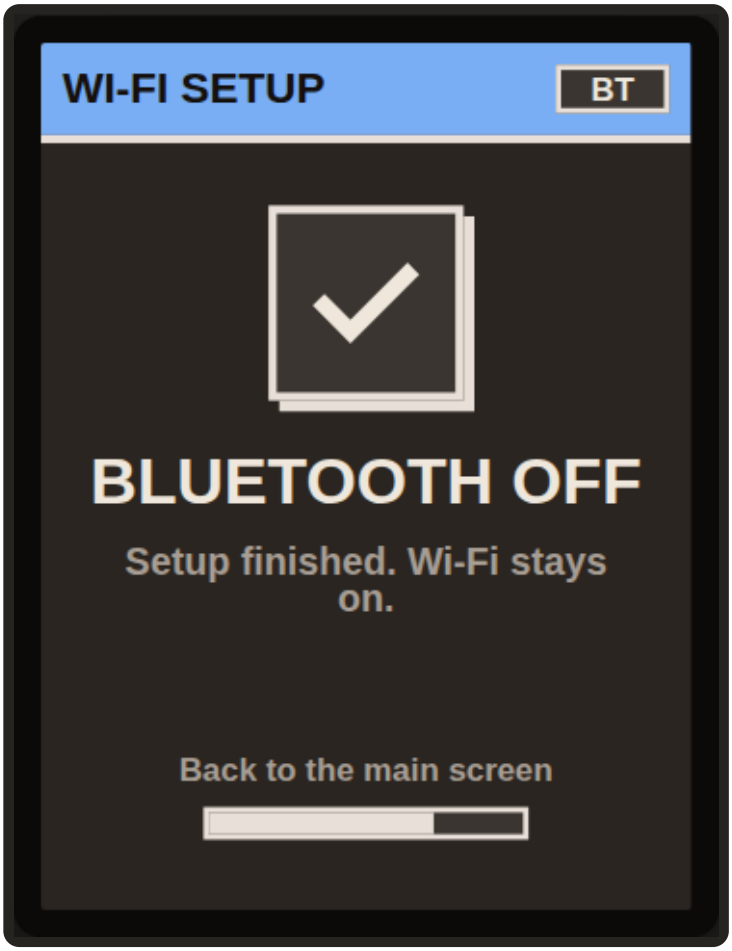
Dark Theme

A10 - Bluetooth Off

Bluetooth disabled confirmation



Light Theme



Dark Theme

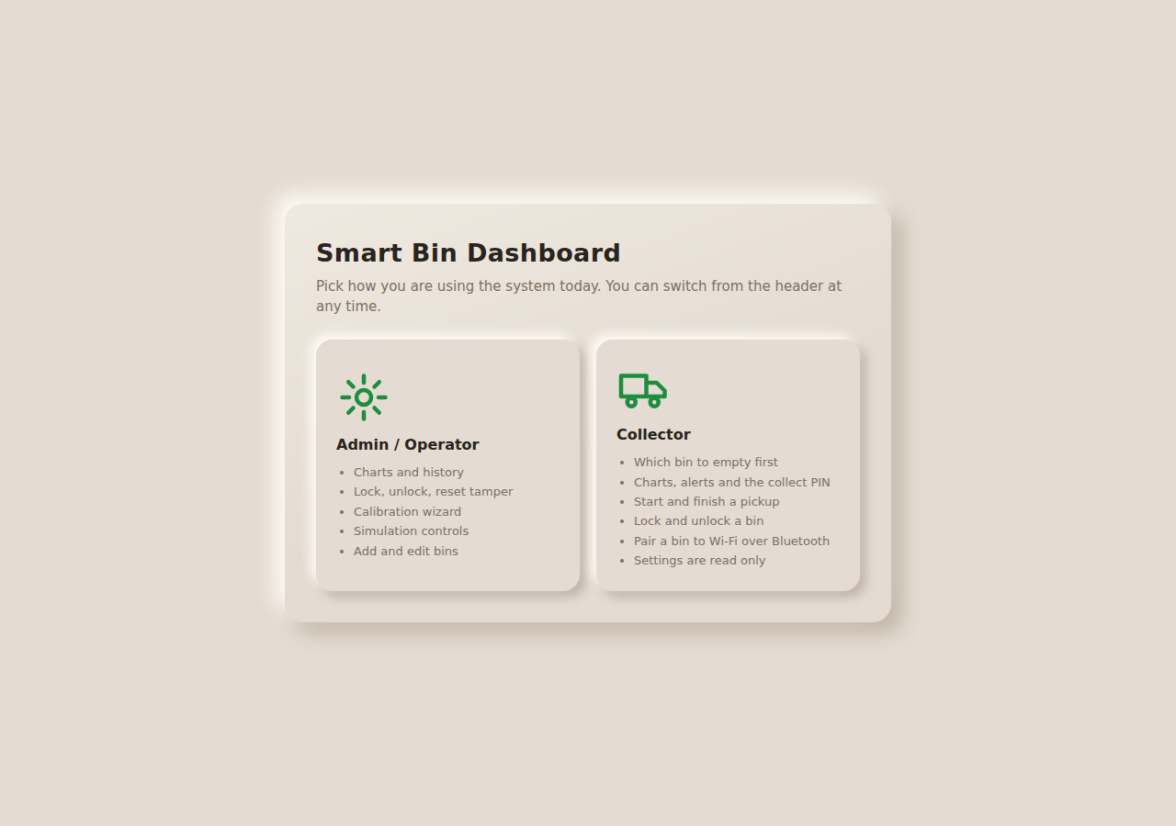
Smart Bin Dashboard

Web Application | Neumorphic Design | 15 Views | Light & Dark Themes

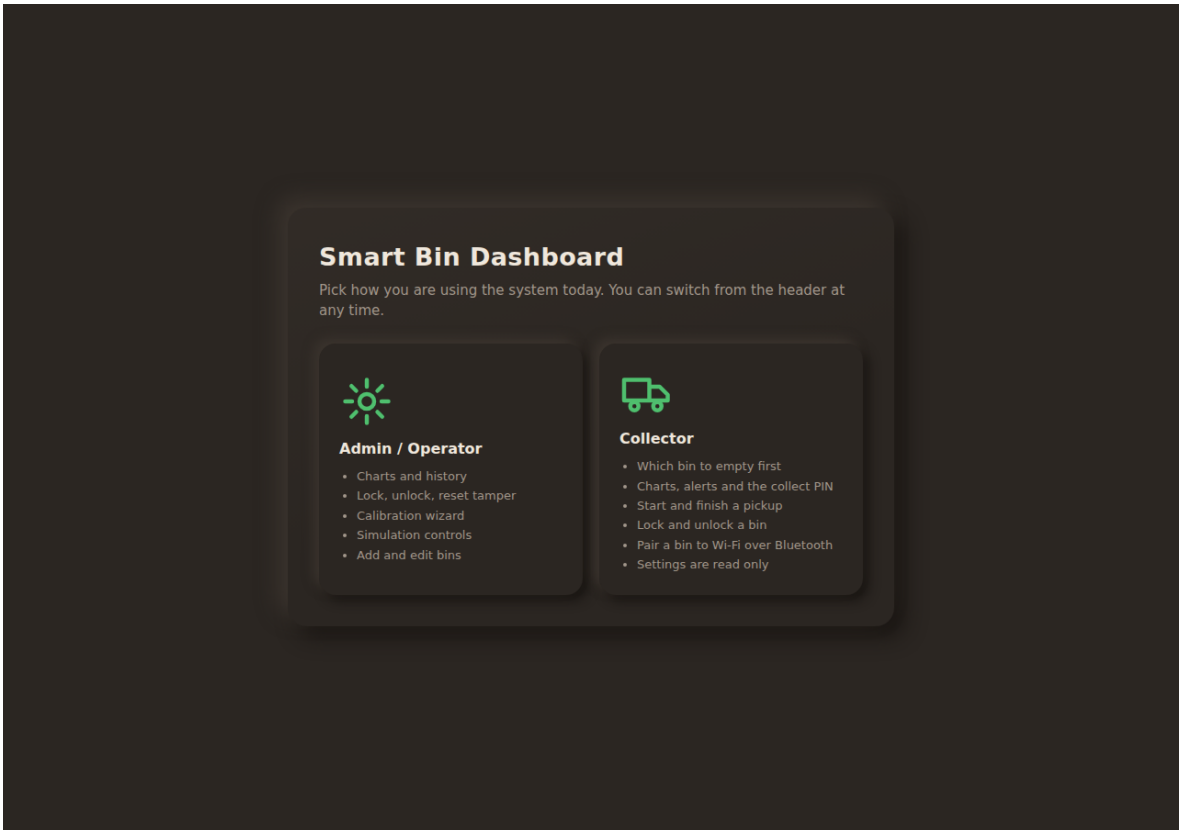
Role Selection

Choose between Admin/Operator and Collector roles

Light Theme



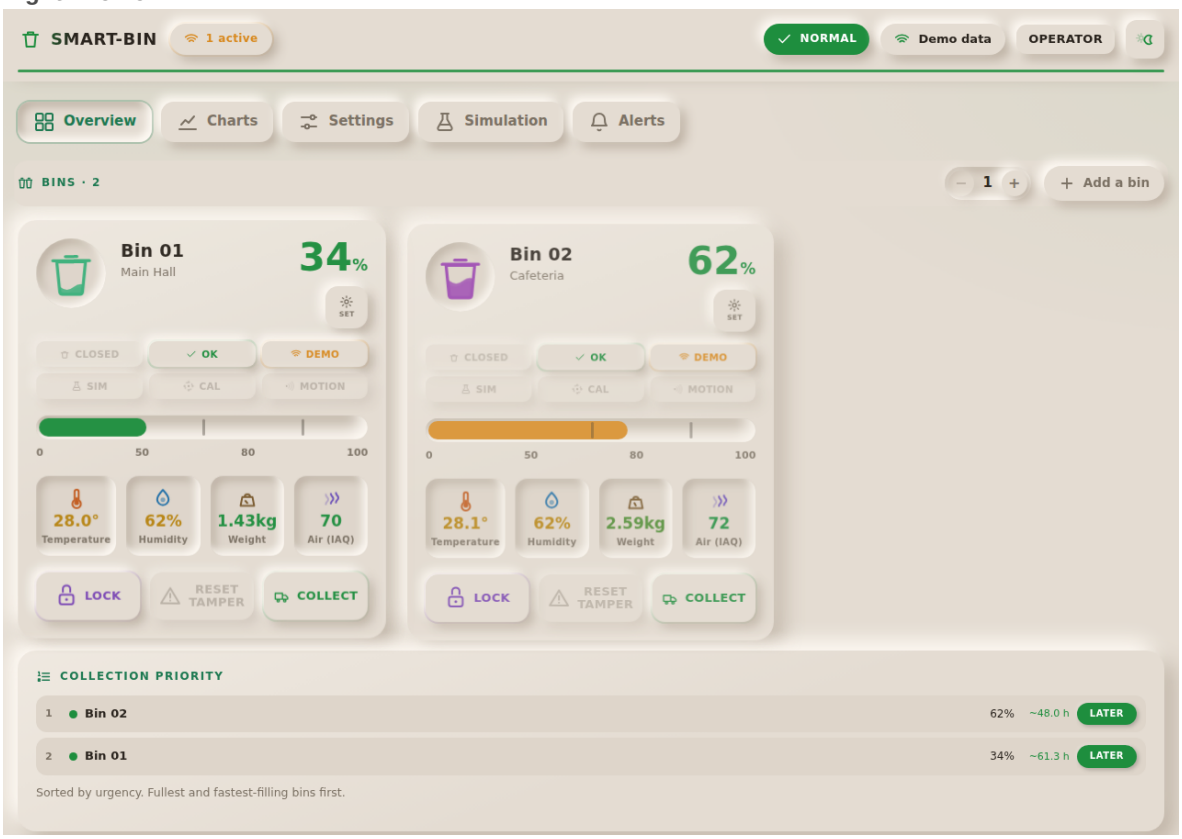
Dark Theme



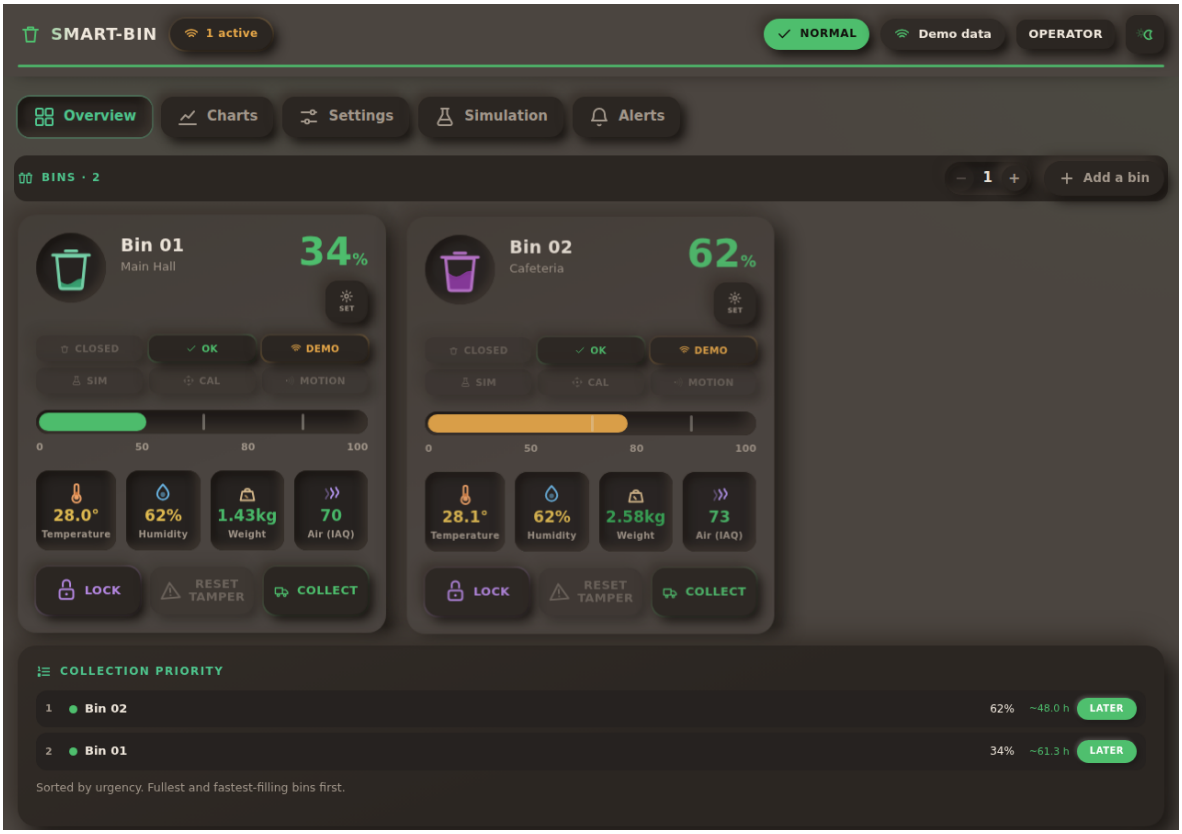
Overview (List View)

Bin cards with fill levels, sensors, and action buttons

Light Theme



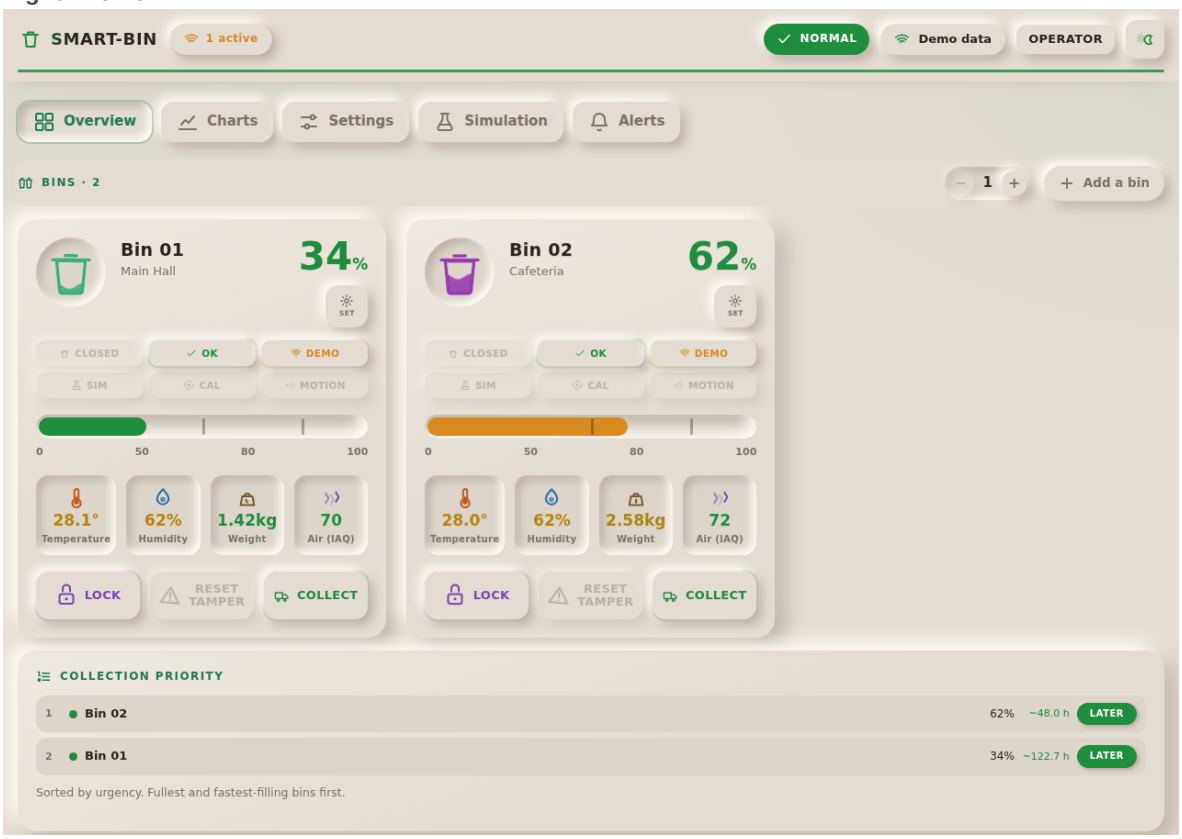
Dark Theme



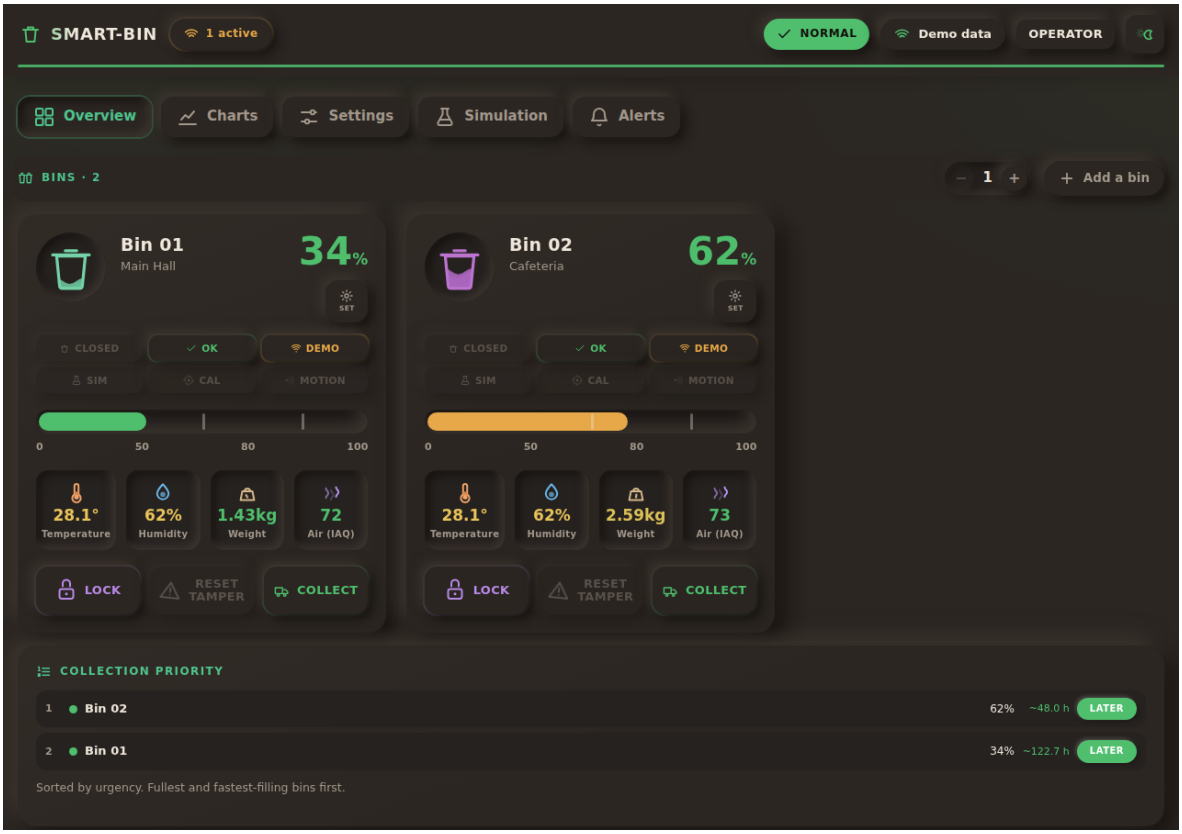
Overview (Grid View)

Compact grid layout for monitoring multiple bins

Light Theme



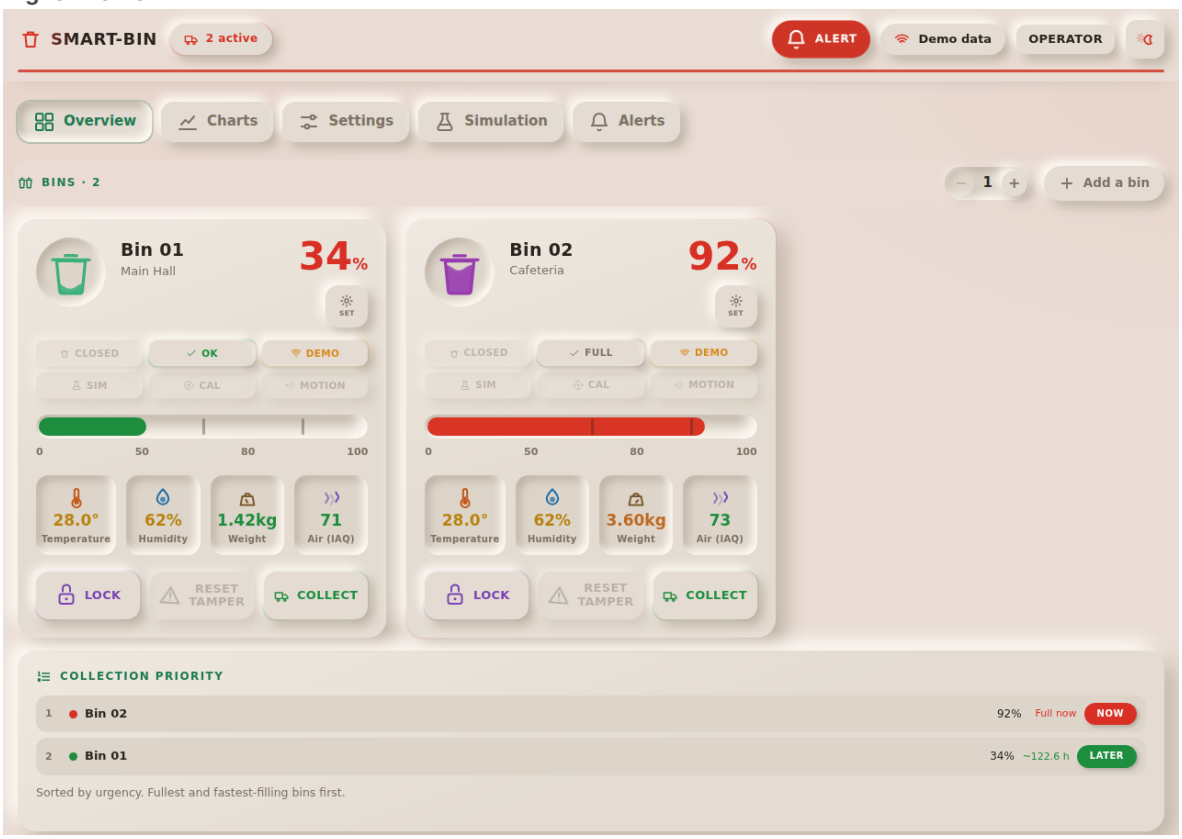
Dark Theme



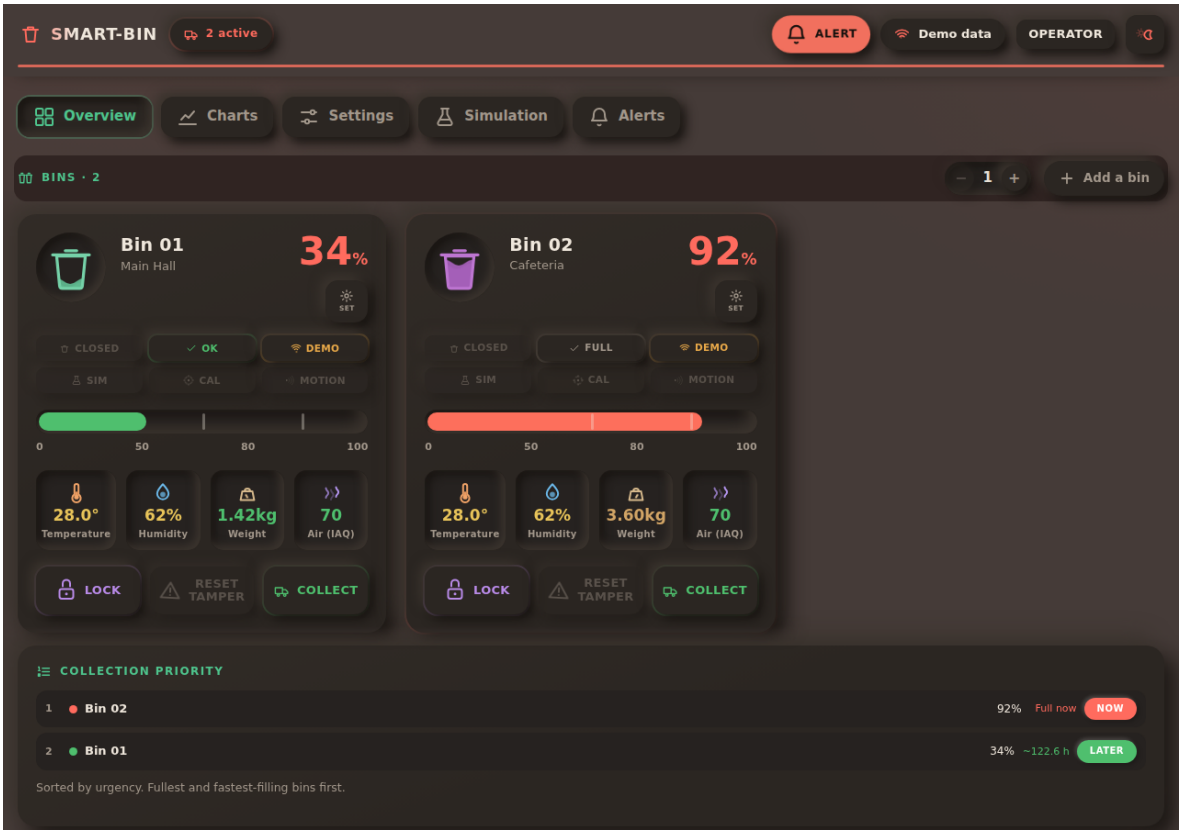
Overview (Alert Mode)

Bin full alert with red accent and glowing status

Light Theme



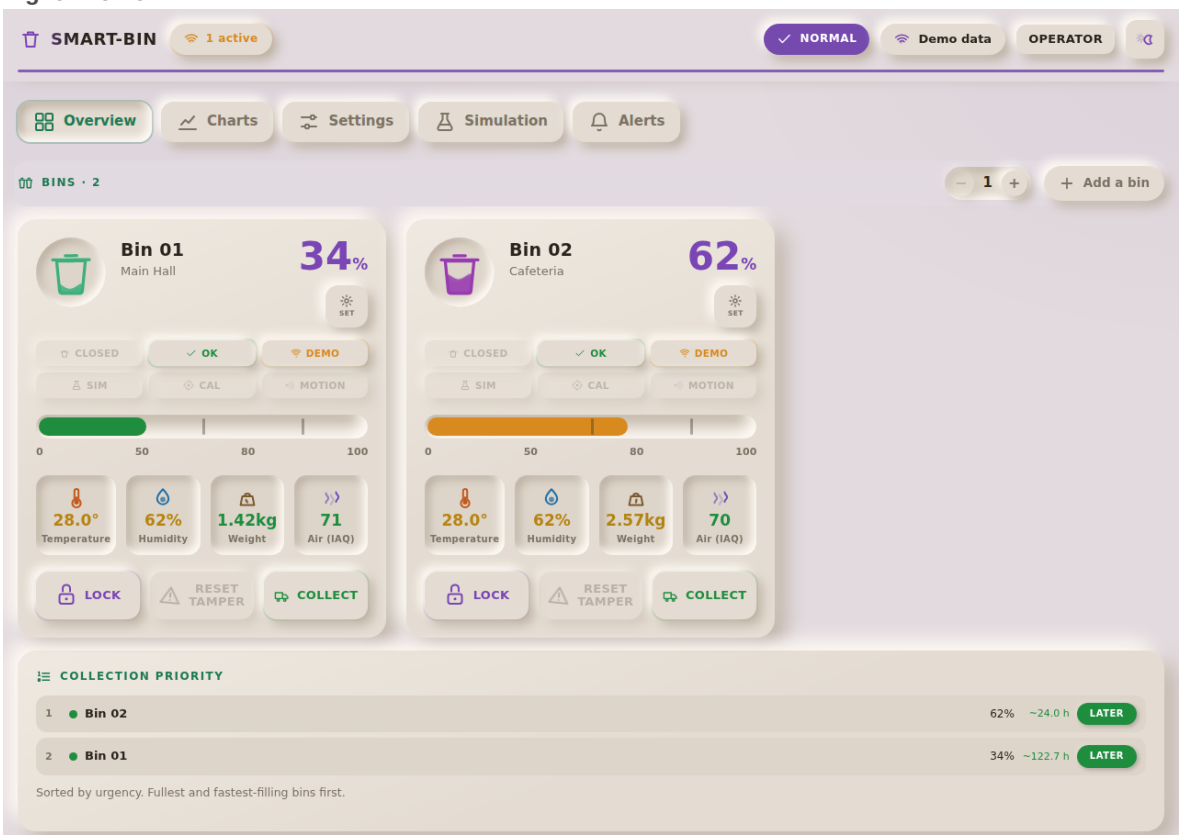
Dark Theme



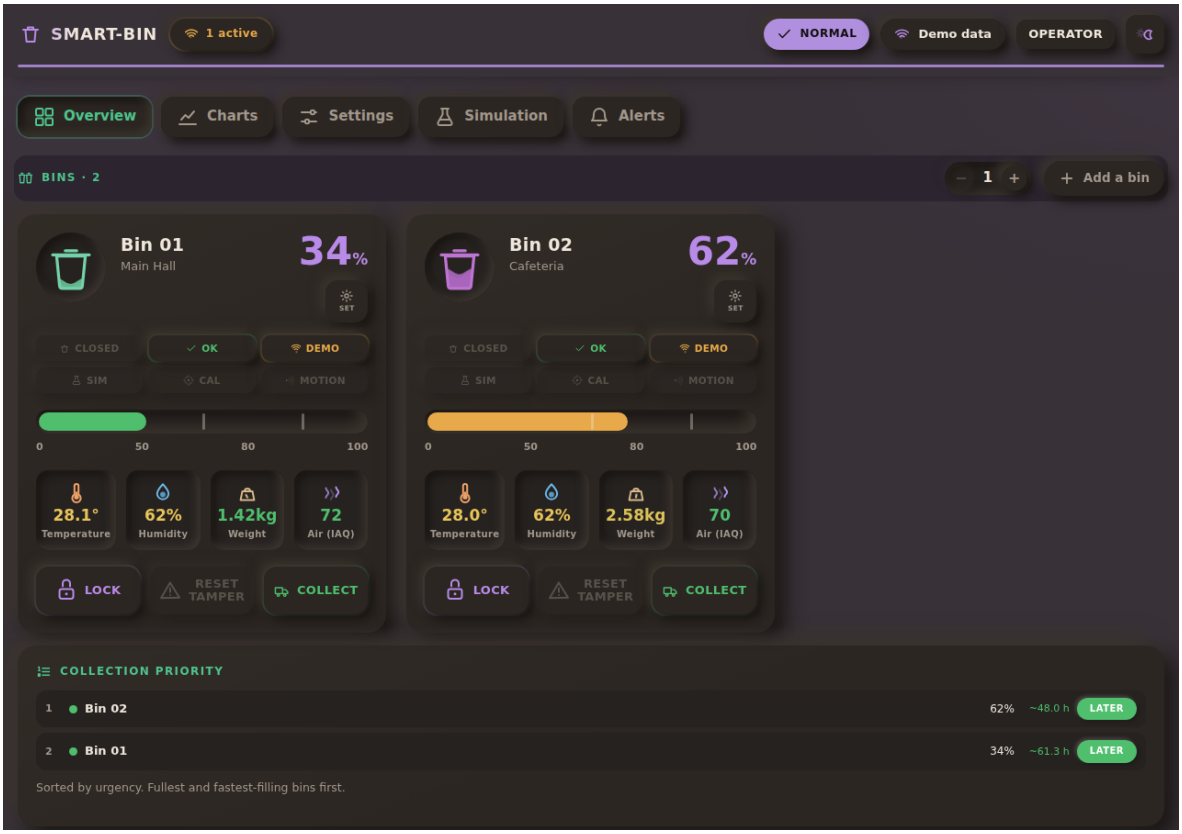
Overview (Simulation Mode)

Dashboard appearance during active simulation

Light Theme



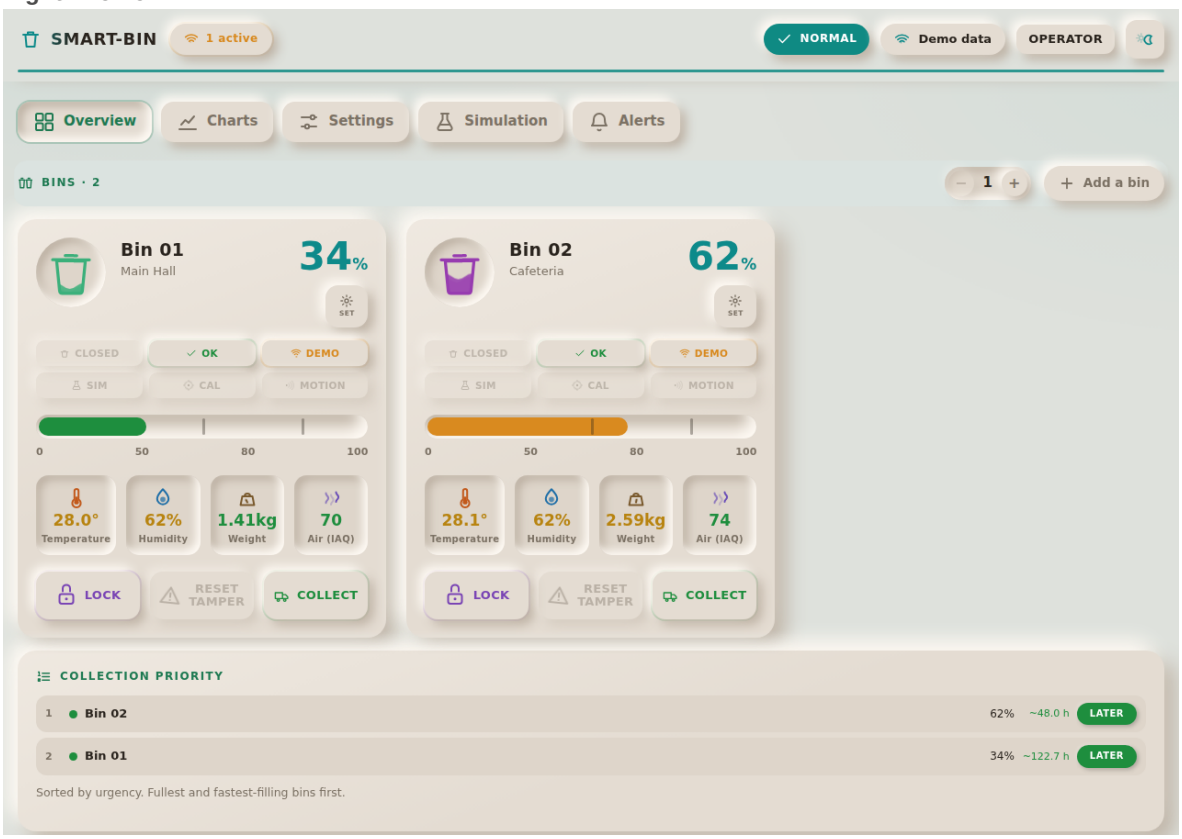
Dark Theme



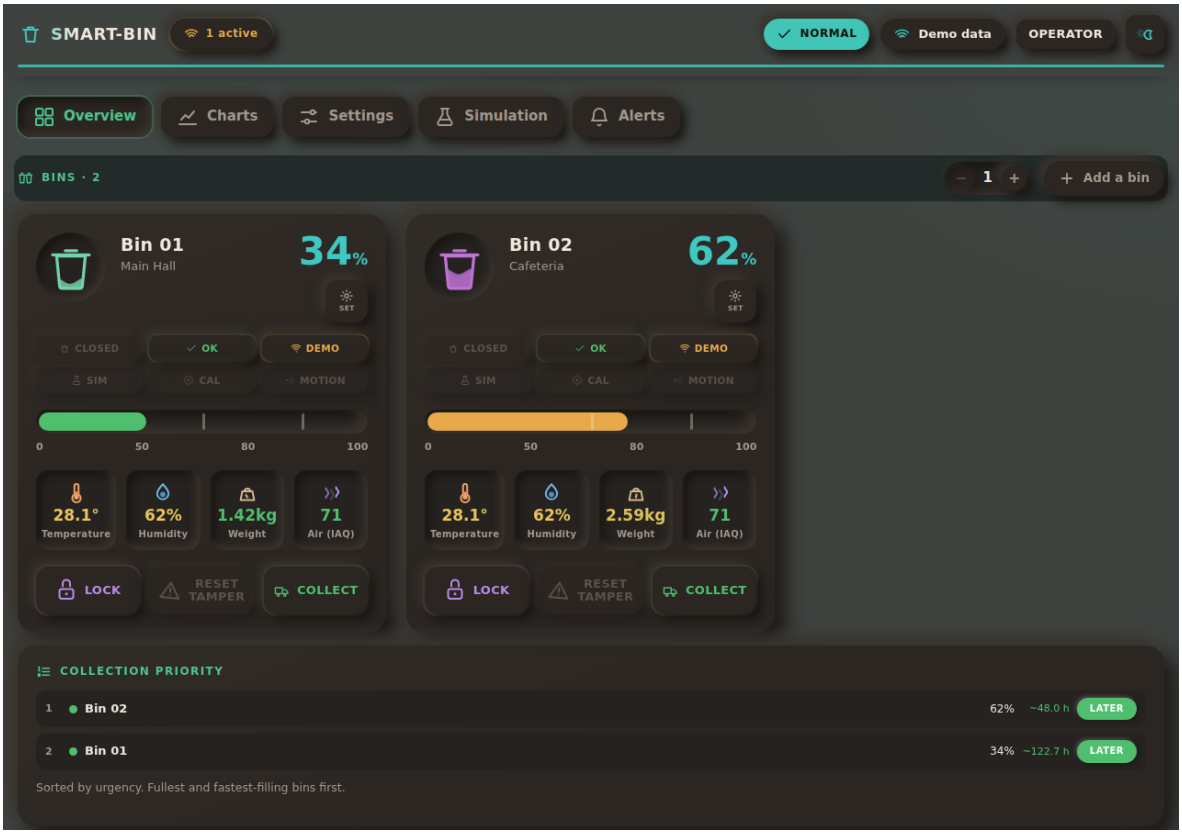
Overview (Calibration Mode)

Dashboard appearance during active calibration

Light Theme



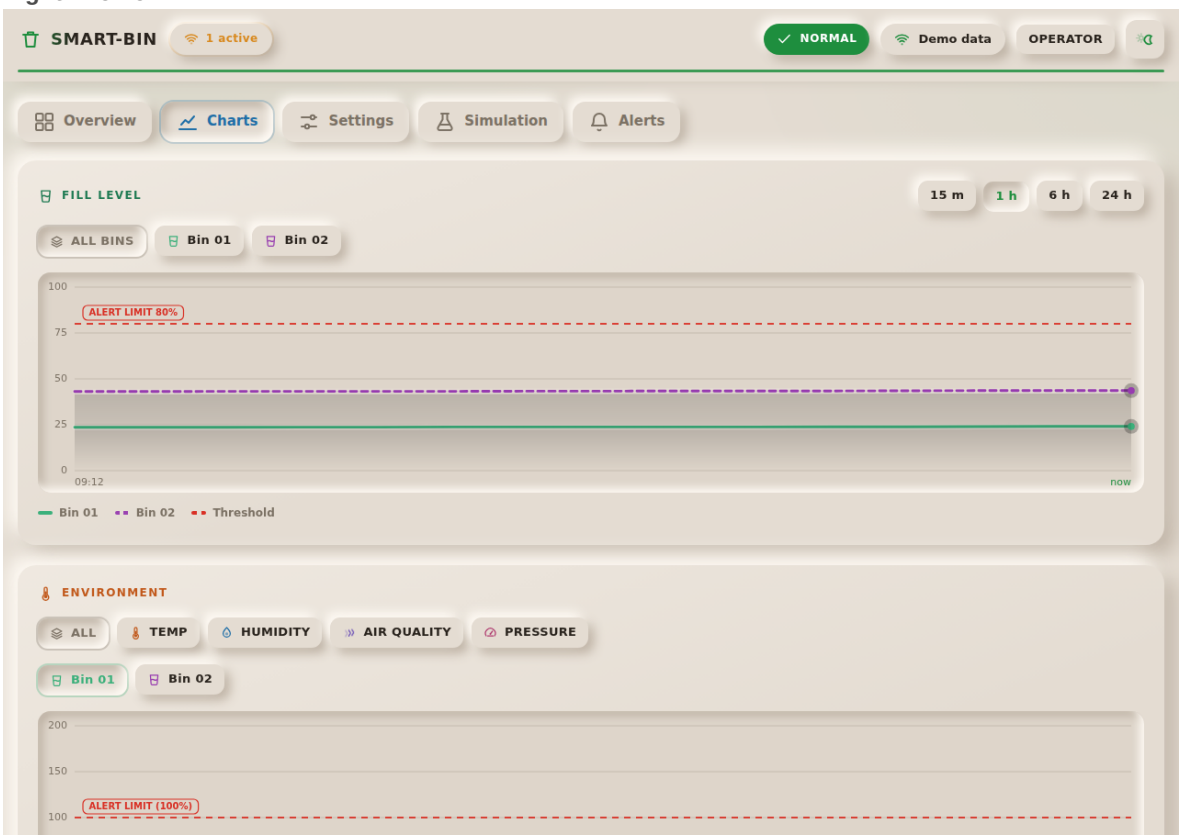
Dark Theme



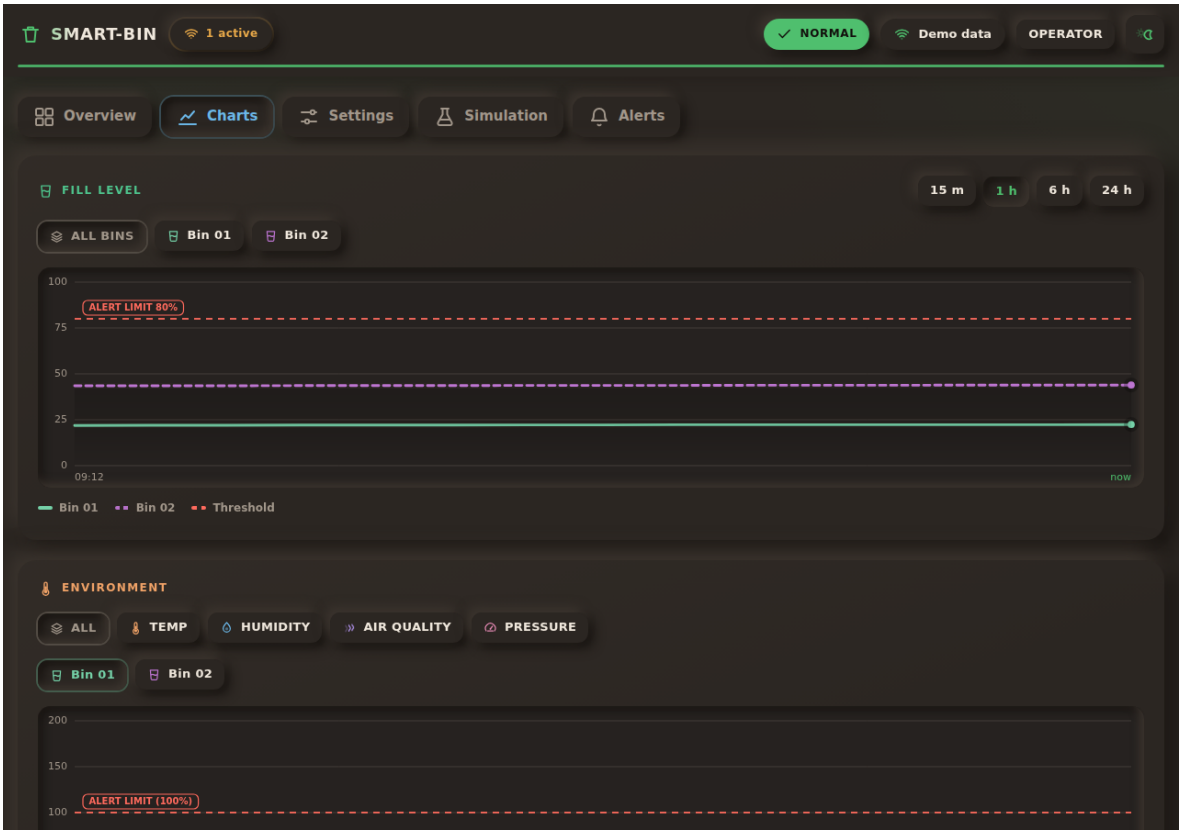
Charts - Fill Level

Fill level trends with range and bin selectors

Light Theme



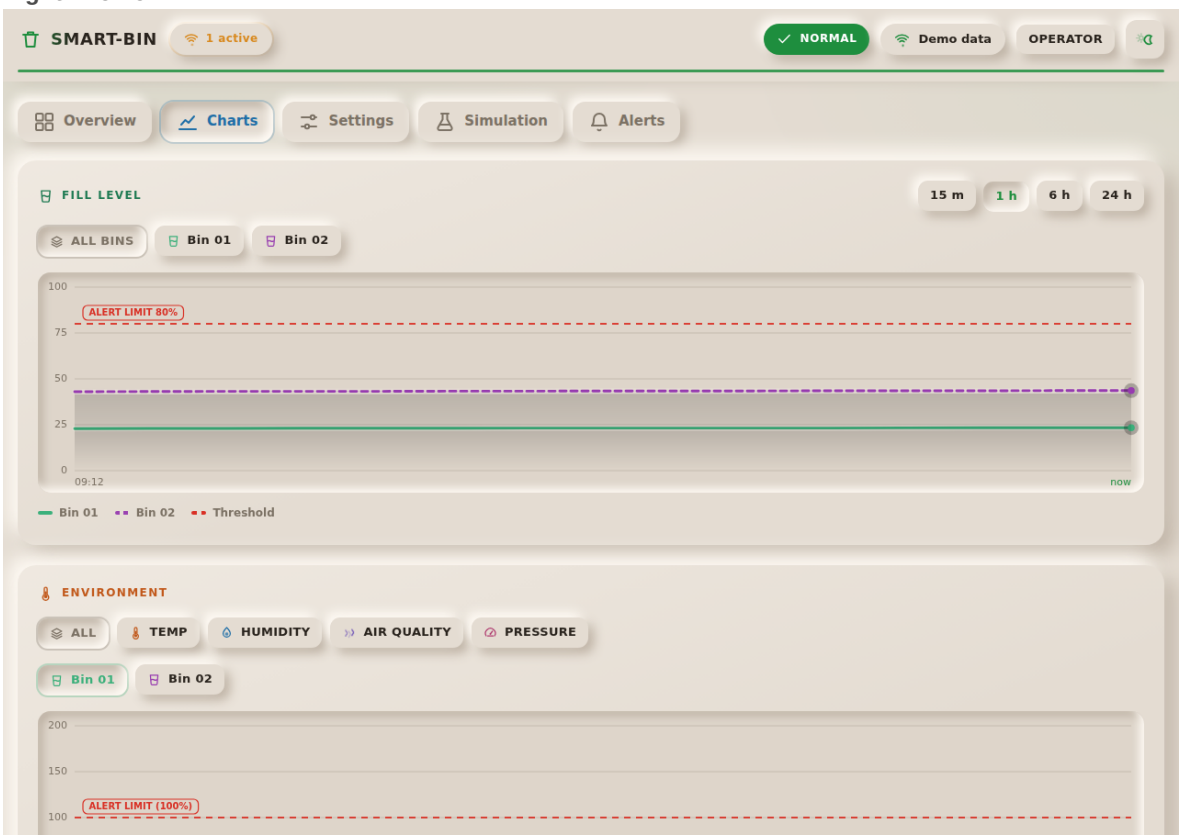
Dark Theme



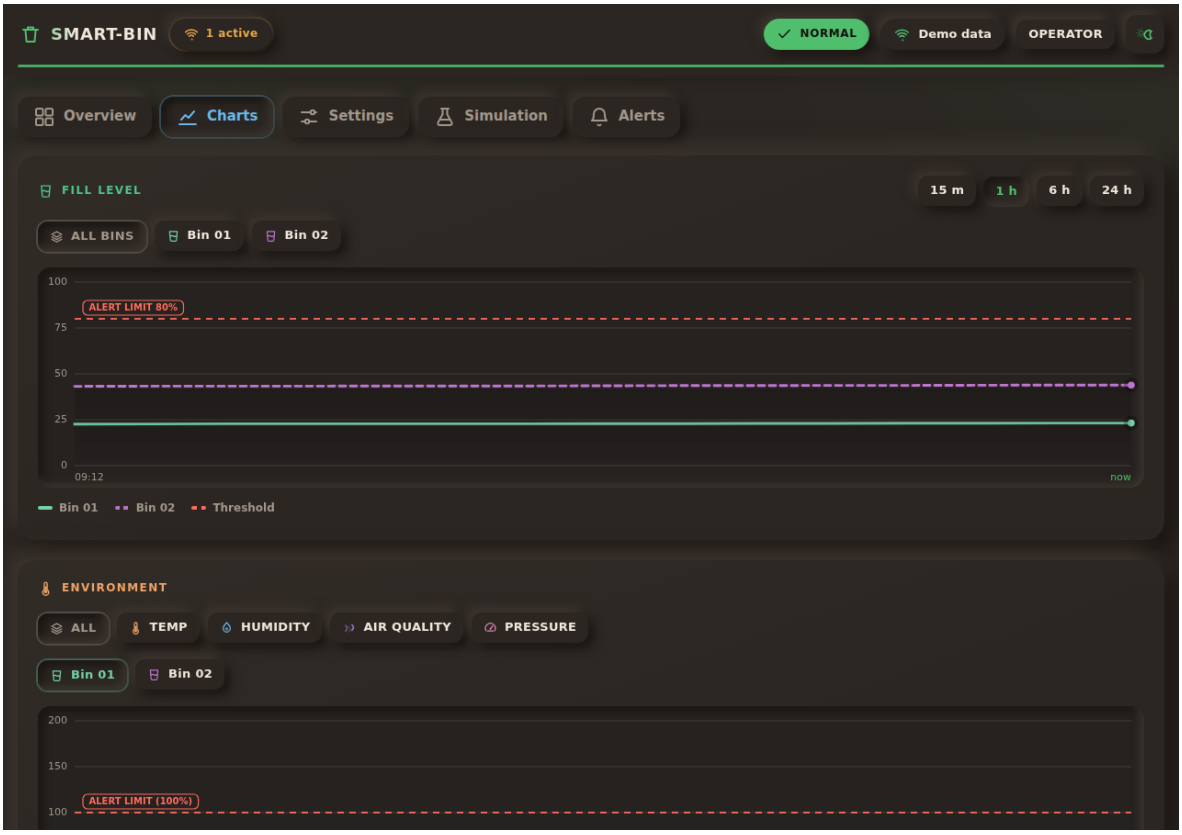
Charts - Environment

Temperature, humidity, gas, and pressure data

Light Theme



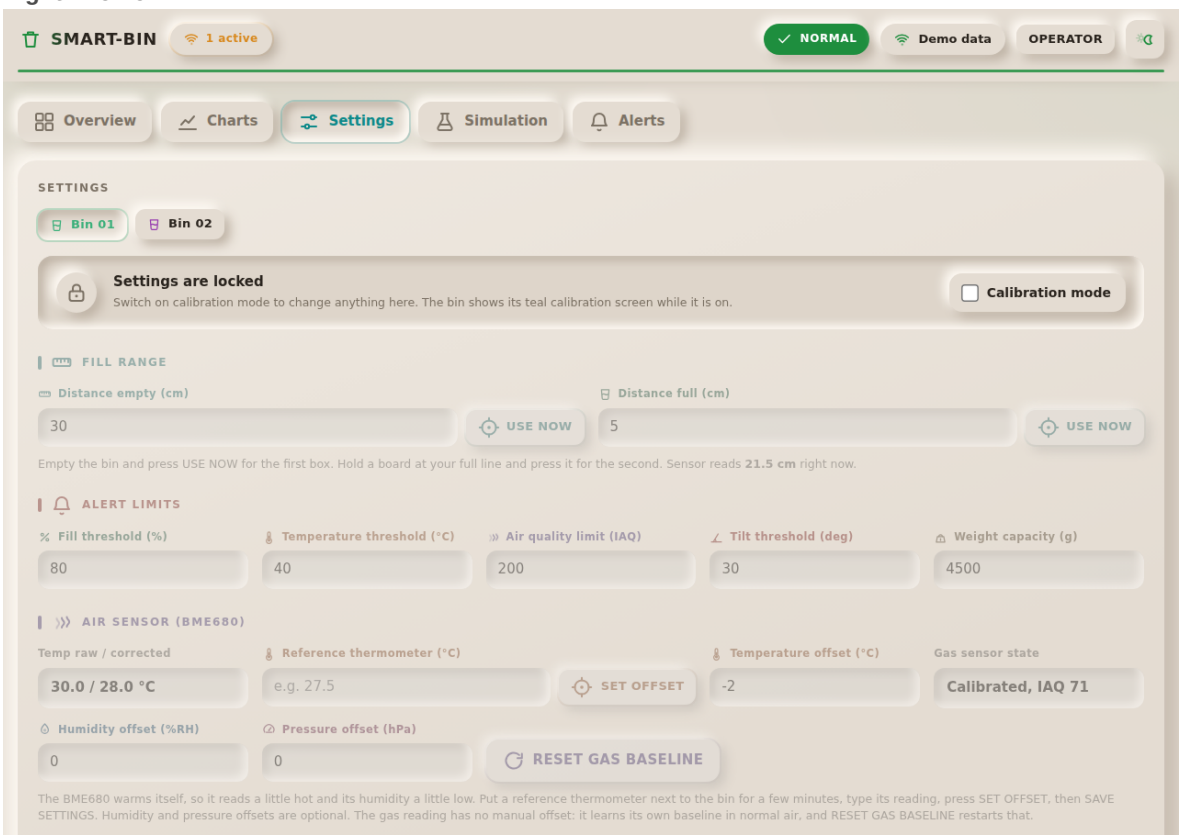
Dark Theme



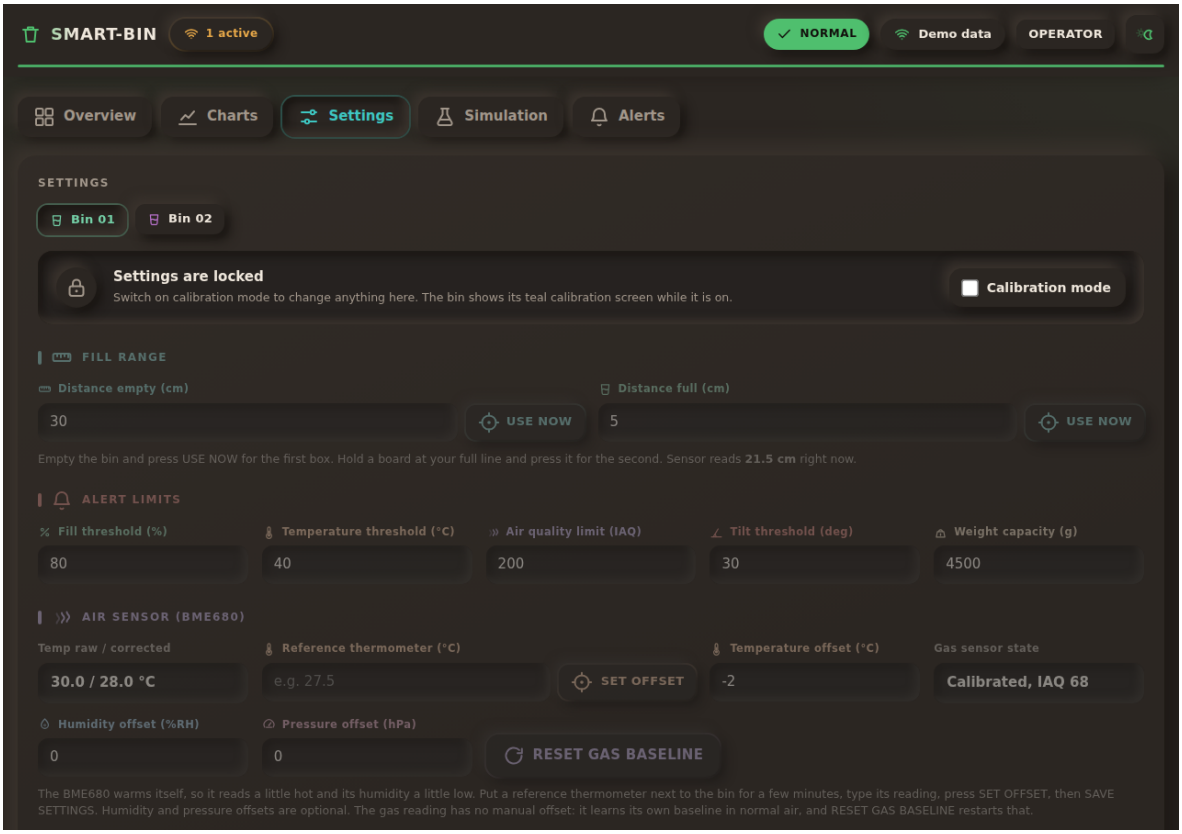
Settings

Bin configuration, thresholds, and system preferences

Light Theme



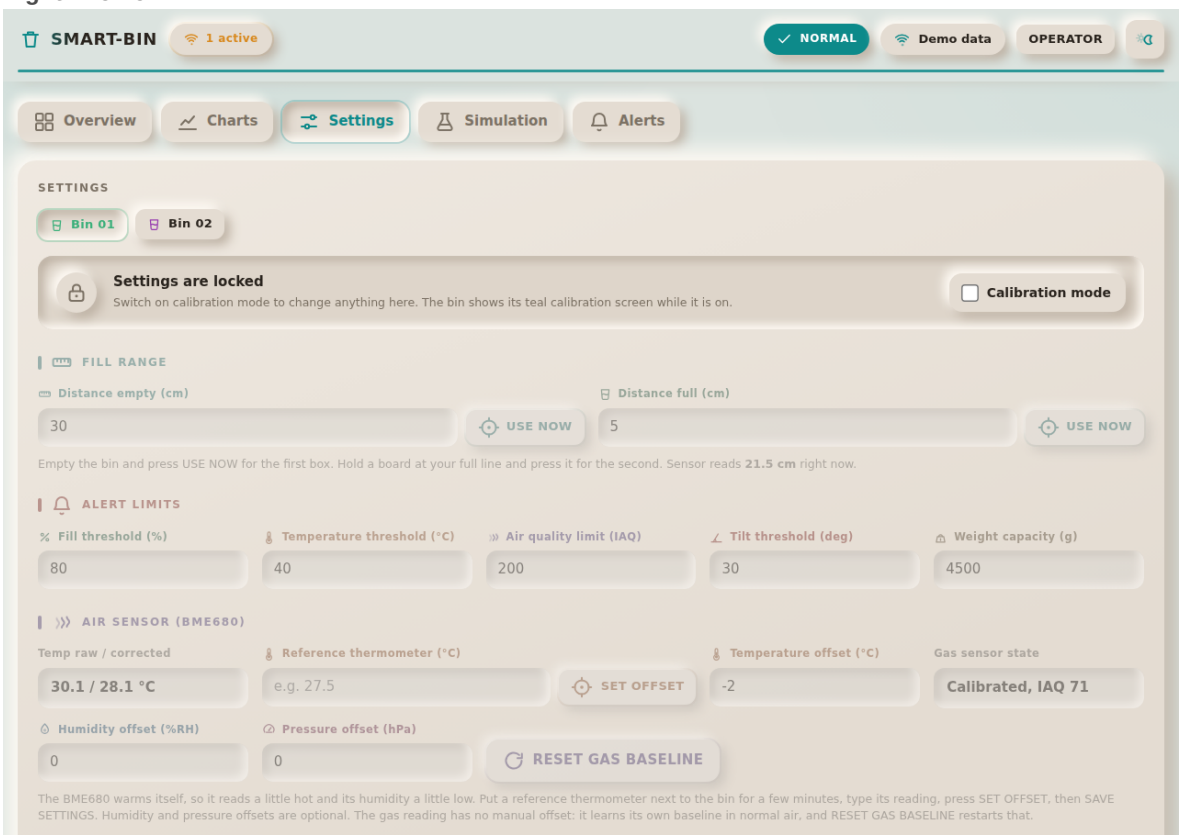
Dark Theme



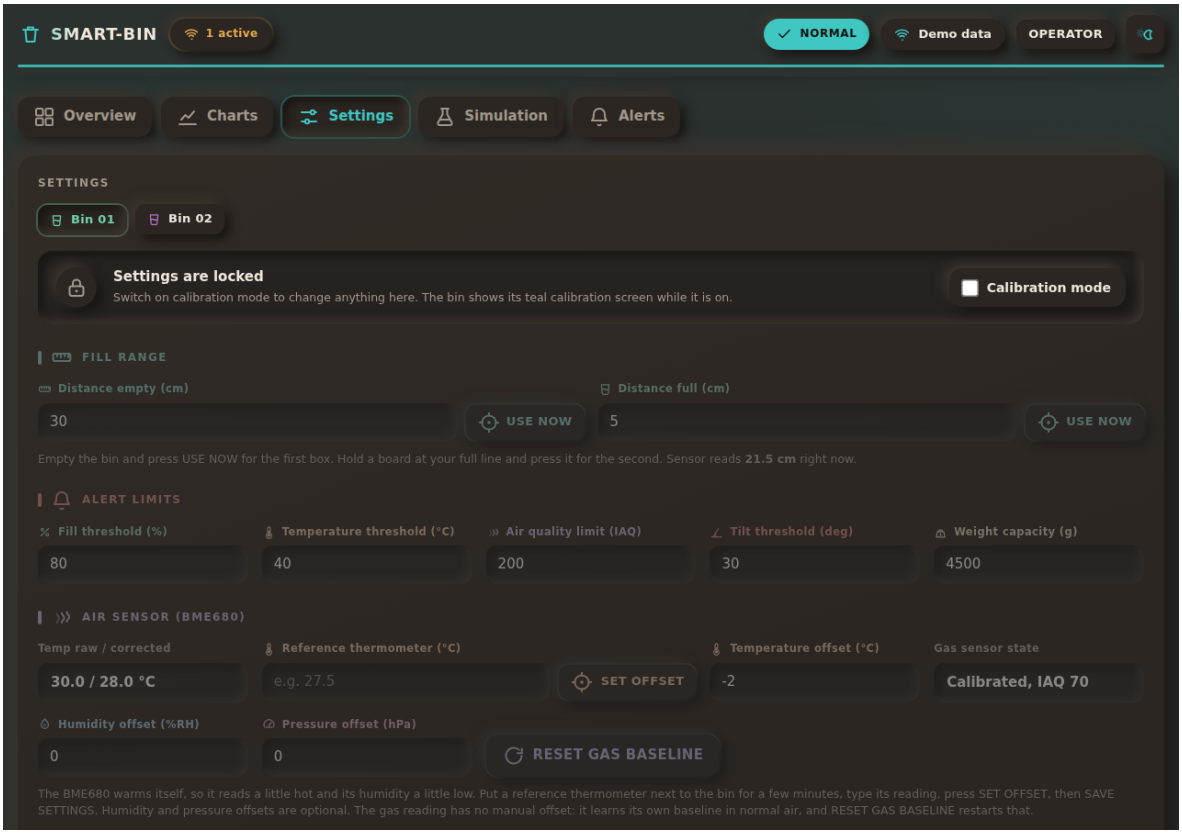
Settings (Calibration Active)

Calibration mode with editable parameters

Light Theme



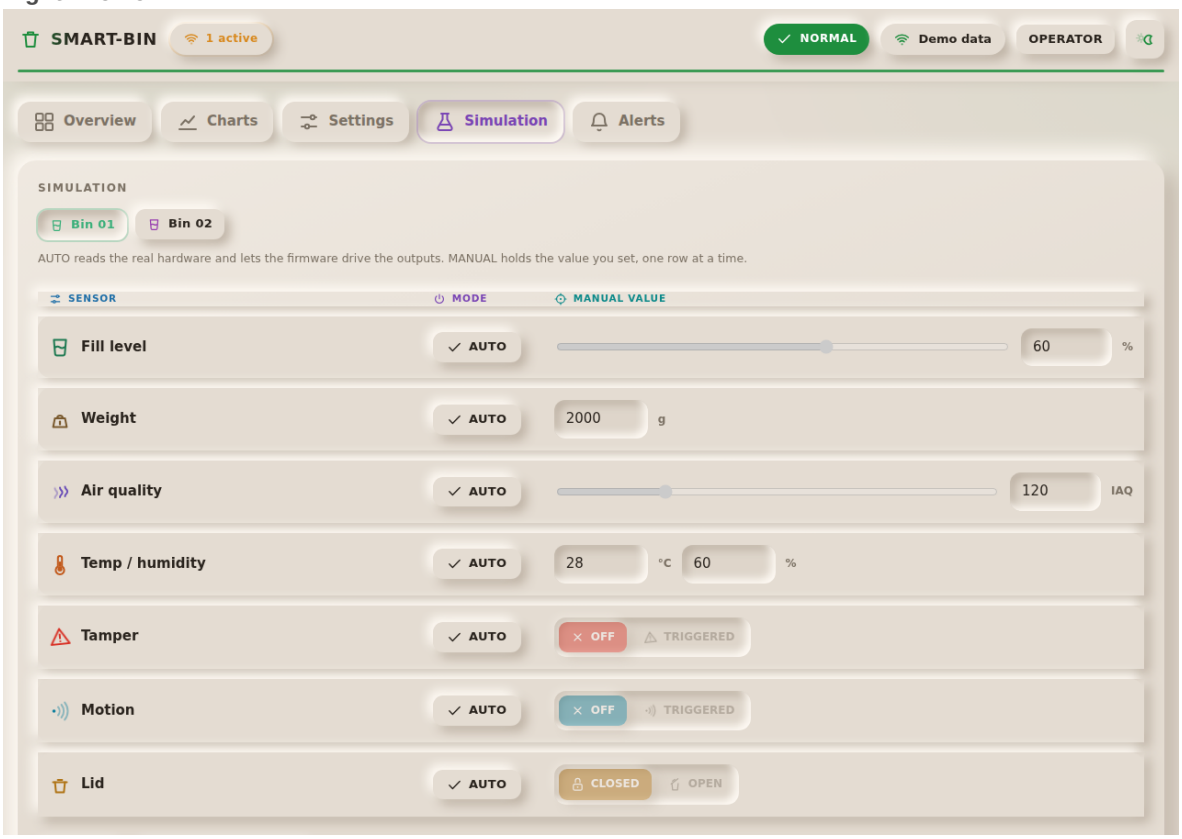
Dark Theme



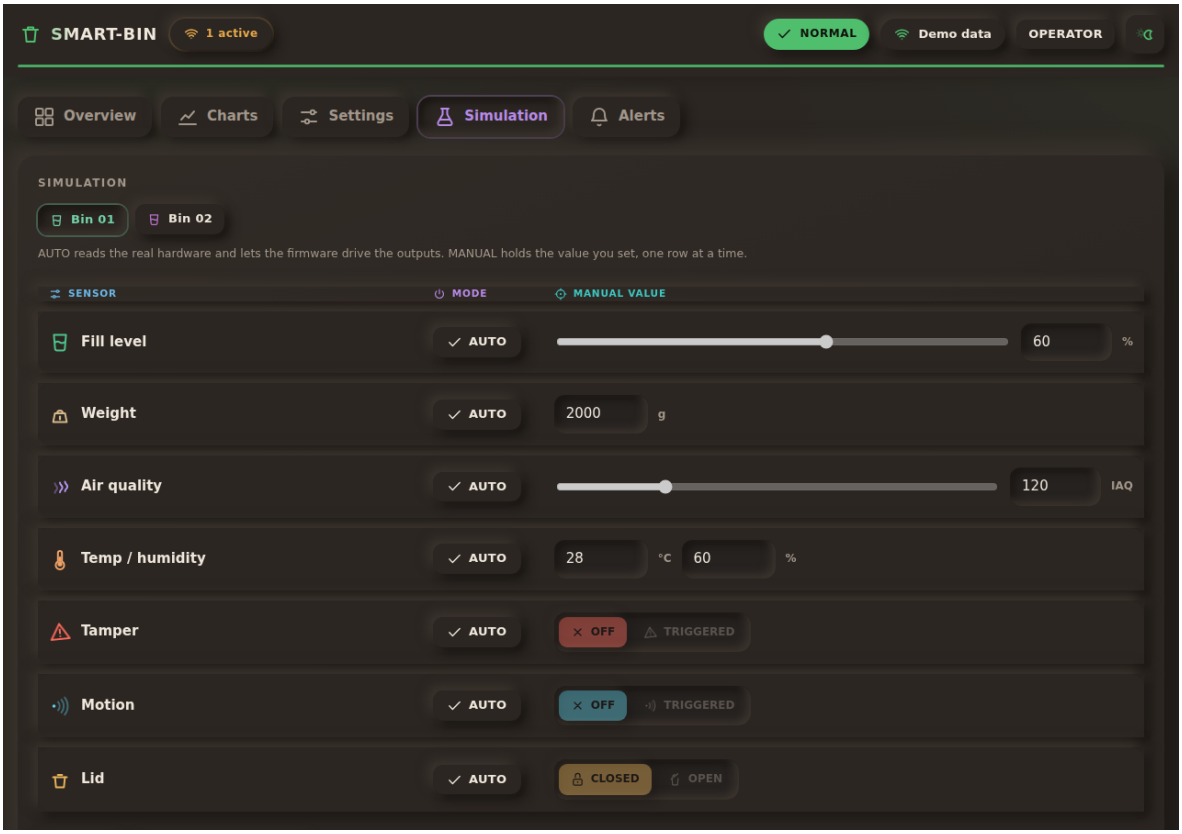
Simulation

Manual sensor value control for testing and demos

Light Theme



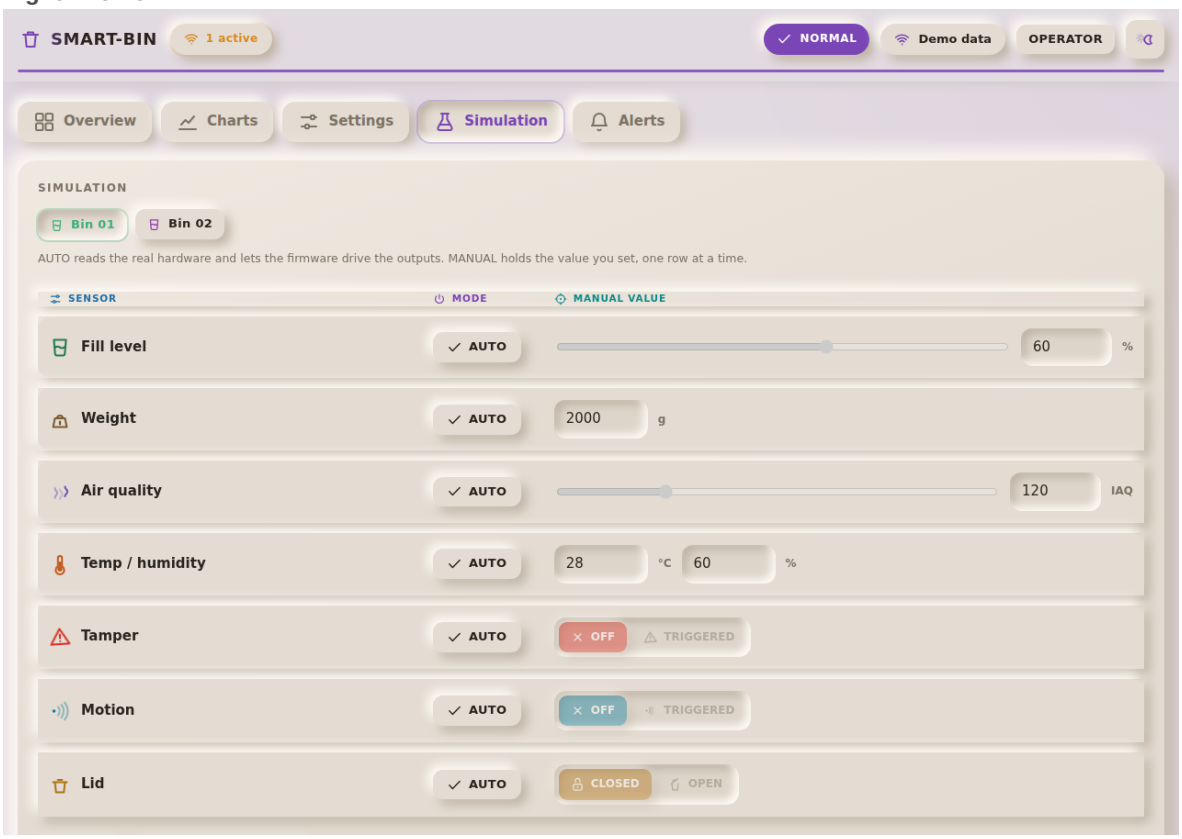
Dark Theme



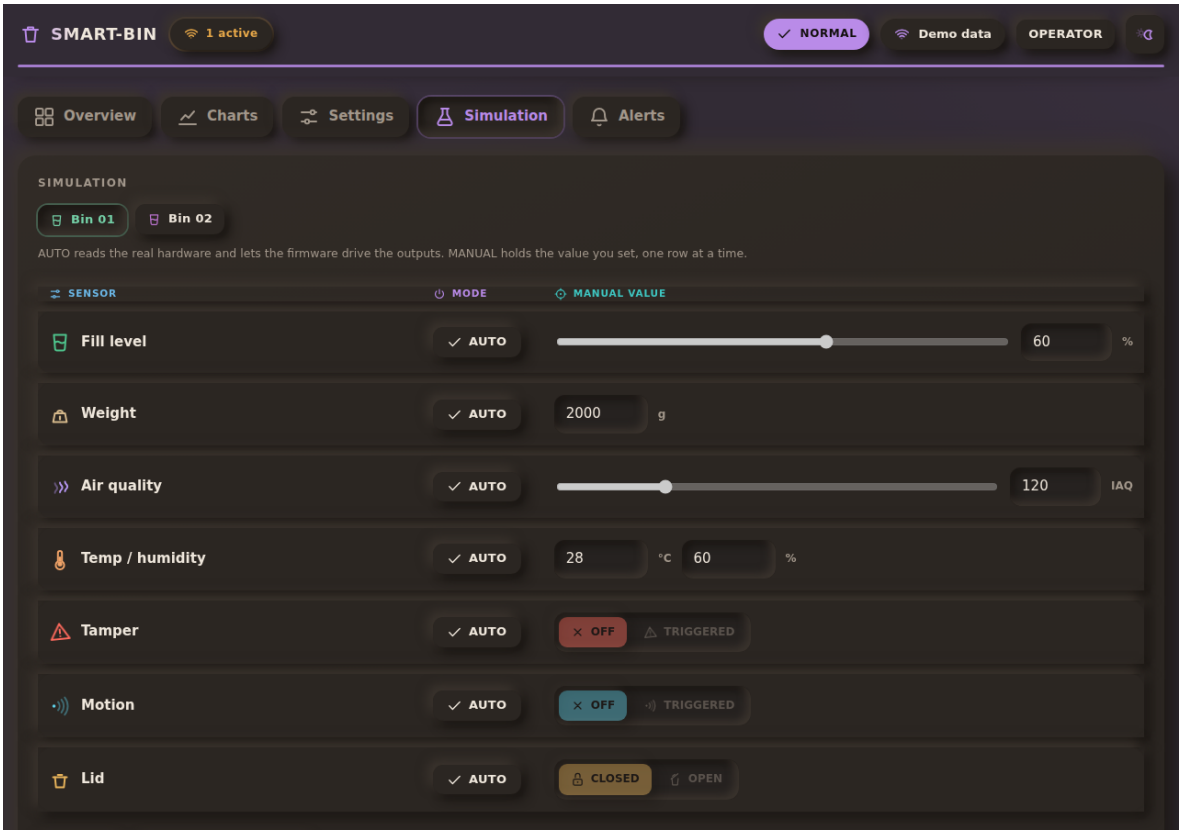
Simulation (Active)

Simulation running with violet accent

Light Theme



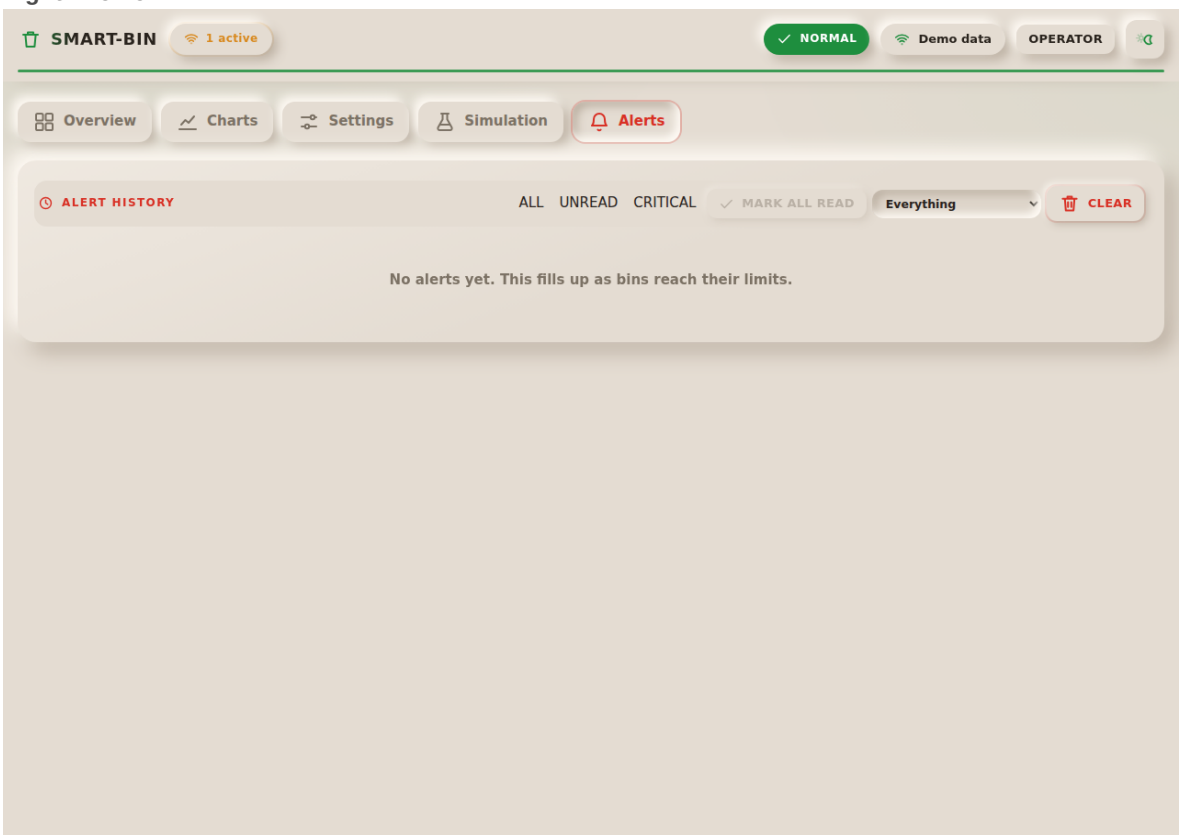
Dark Theme



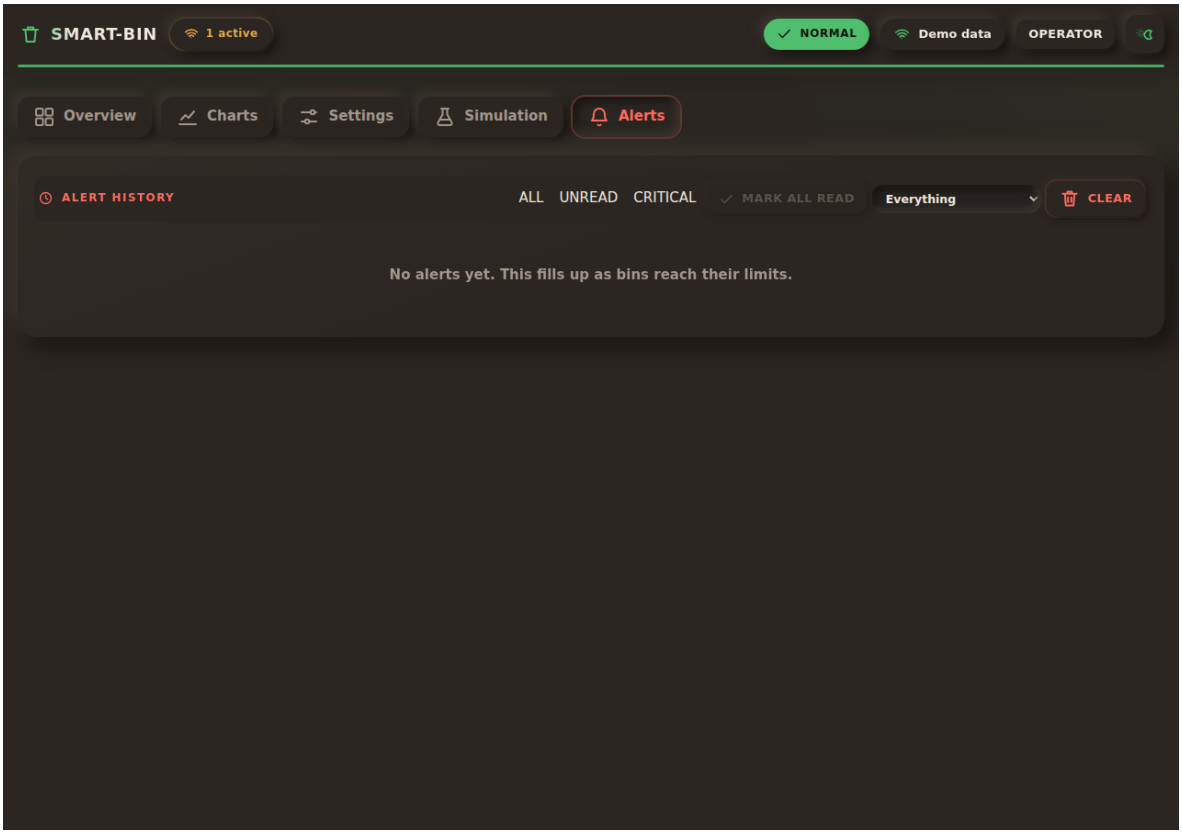
Alerts

Alert history and notification log

Light Theme



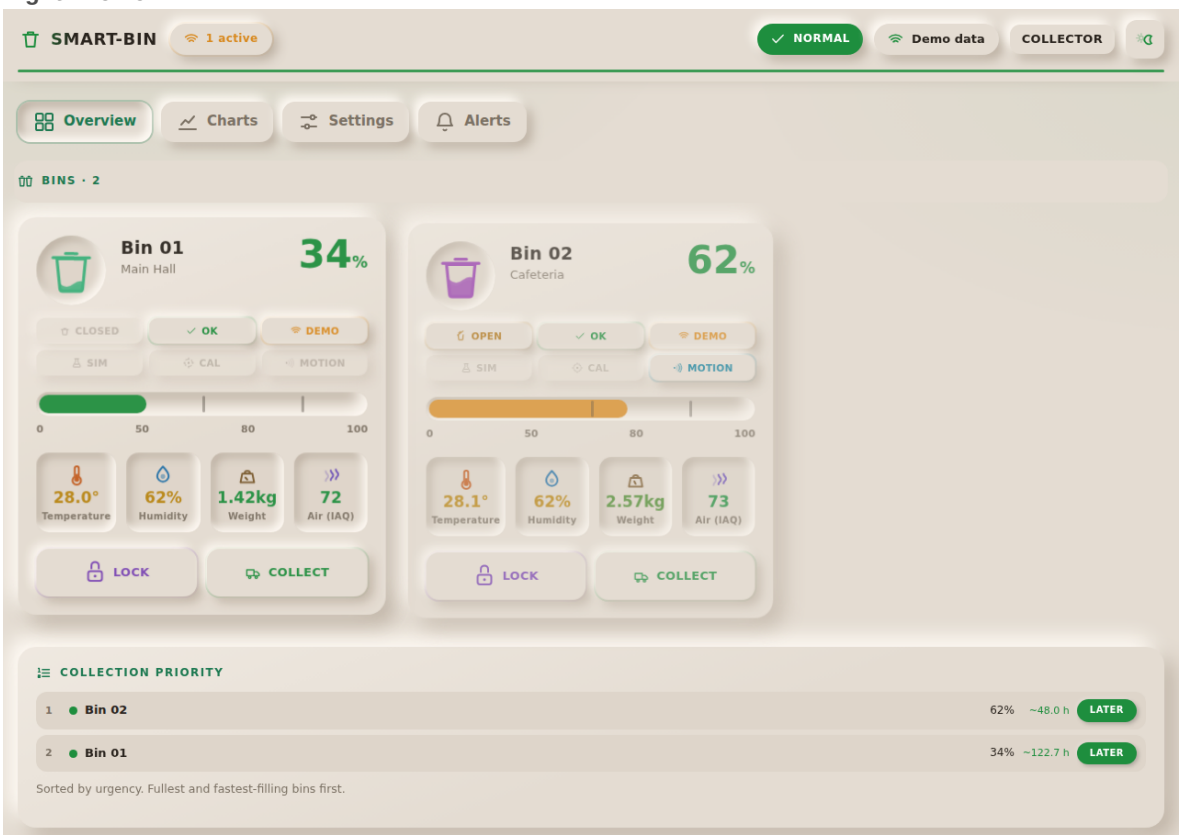
Dark Theme



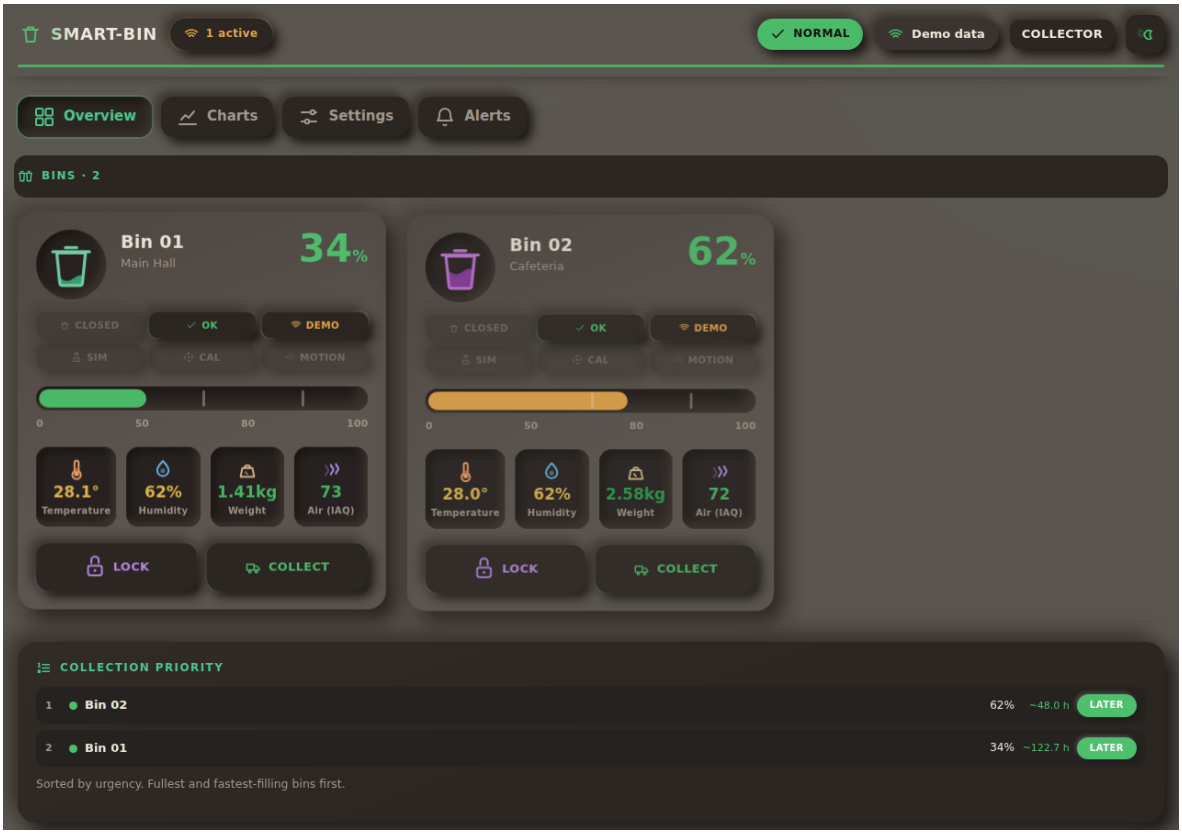
Collector Role - Overview

Simplified view for waste collection personnel

Light Theme



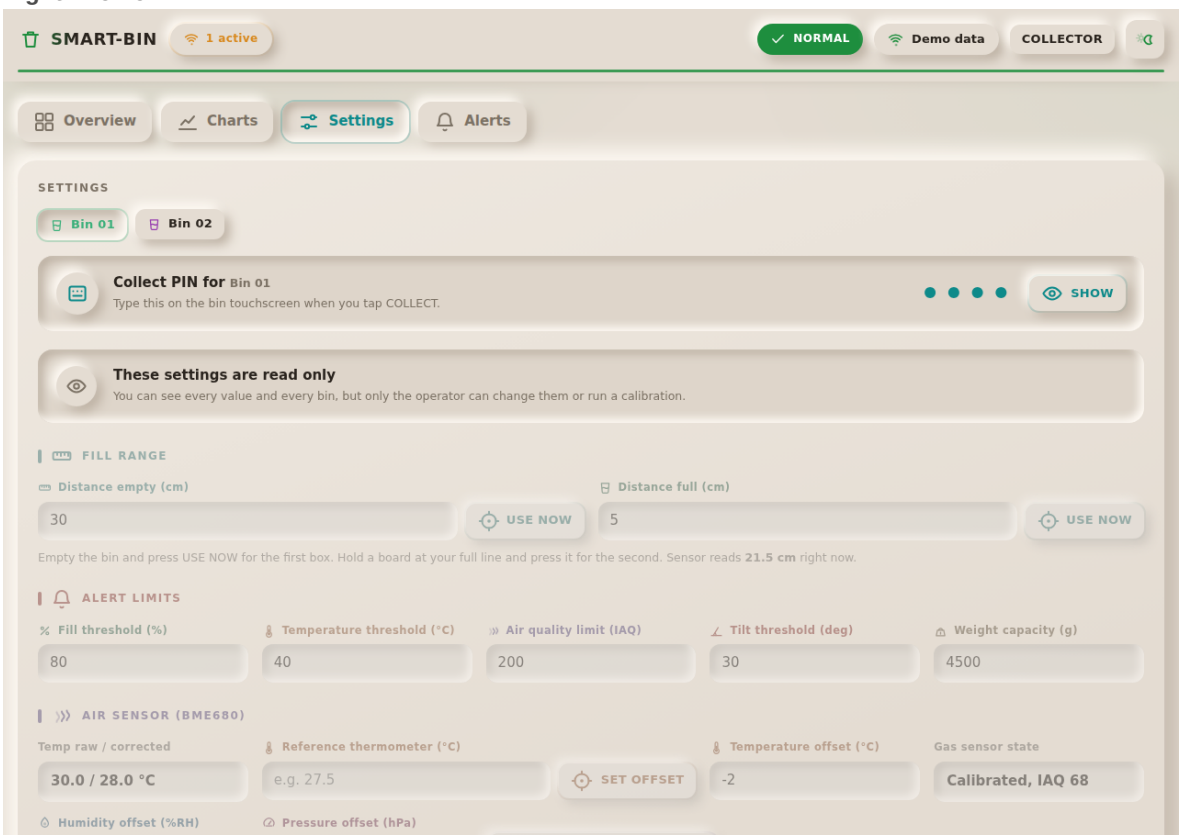
Dark Theme



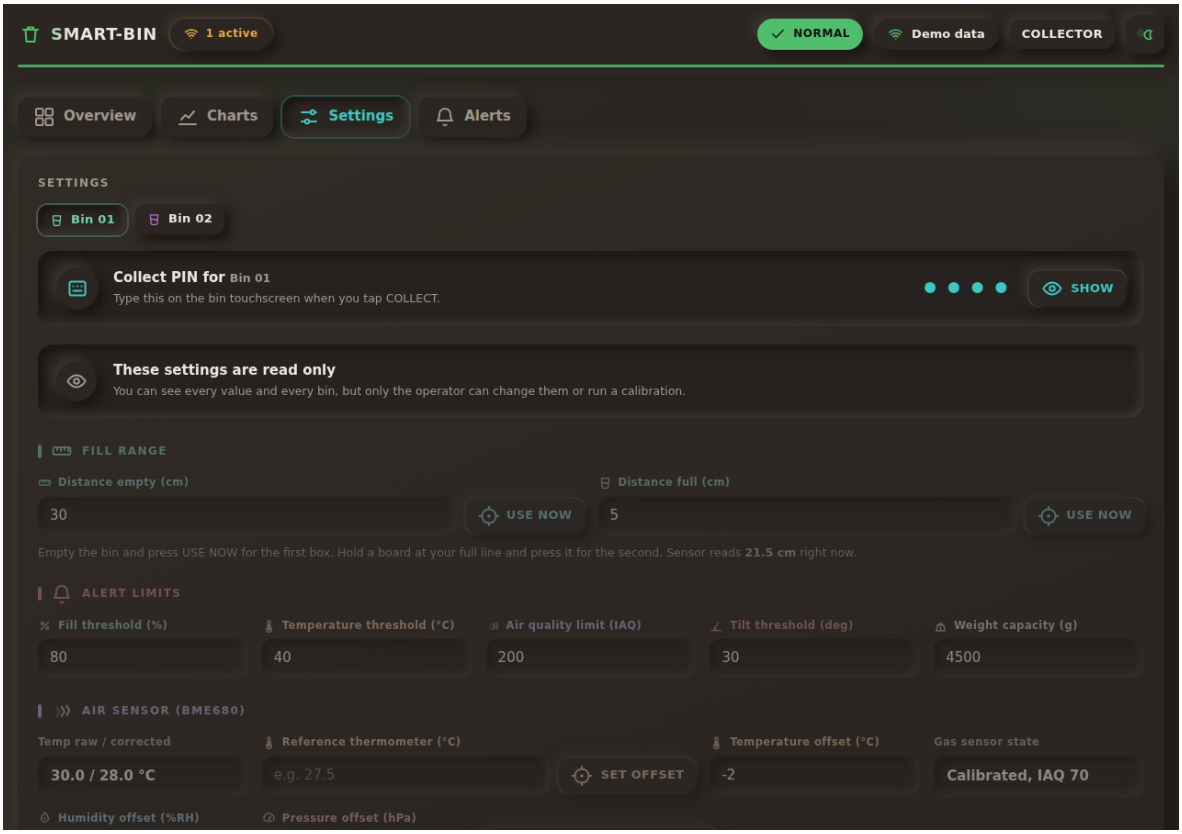
Collector Role - Settings

Read-only settings with collect PIN card

Light Theme



Dark Theme



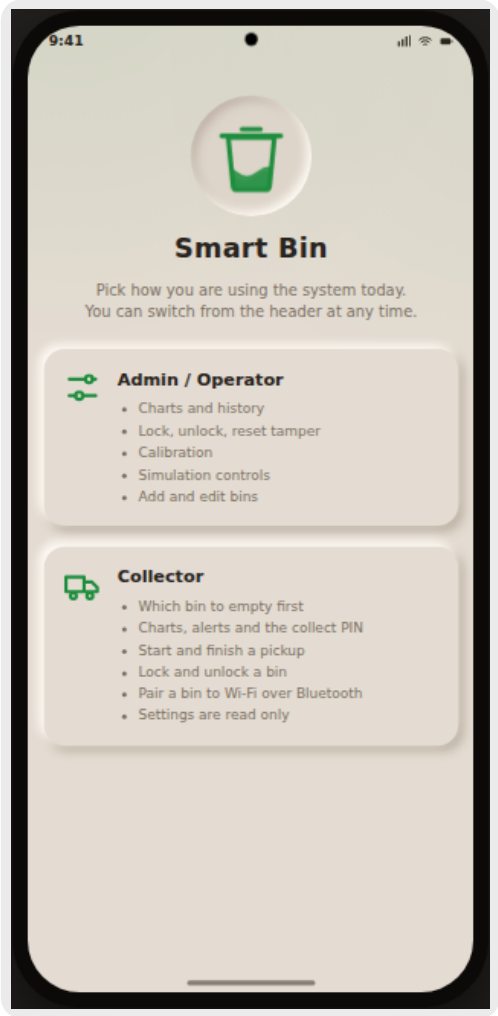
Smart Bin Mobile App

Android | 48 Screens | Light & Dark Themes

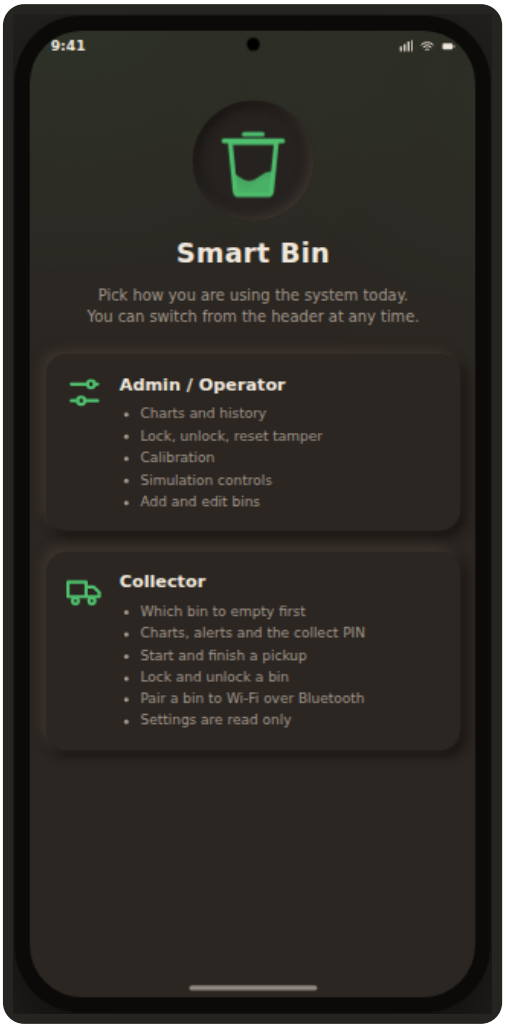
Start

First Launch - Role Picker

Same two roles as the dashboard gate



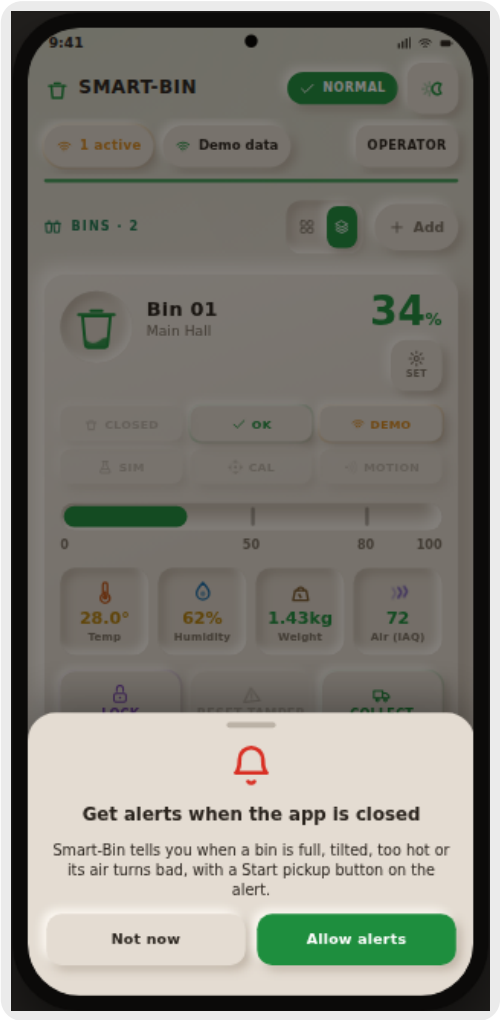
Light Theme



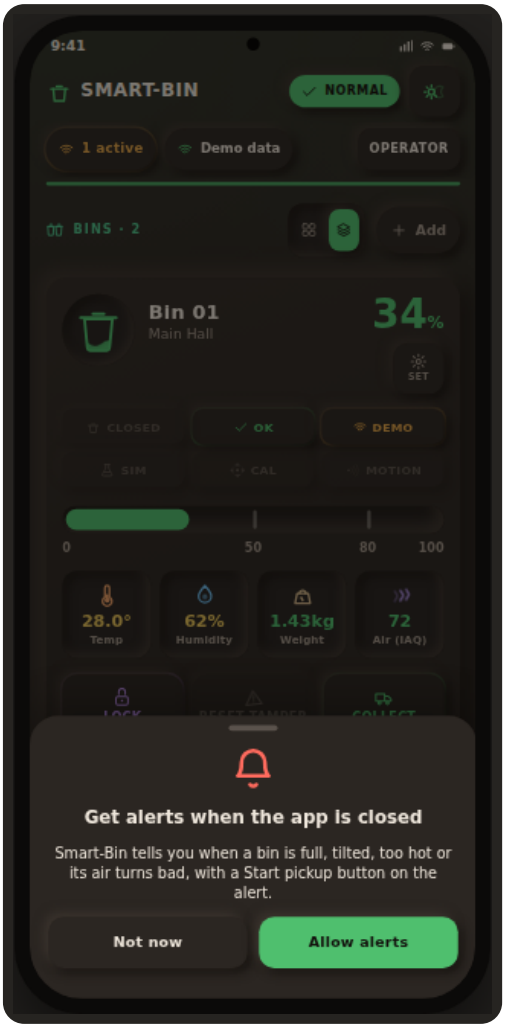
Dark Theme

Notification Permission Prompt

Custom explanation before the Android prompt



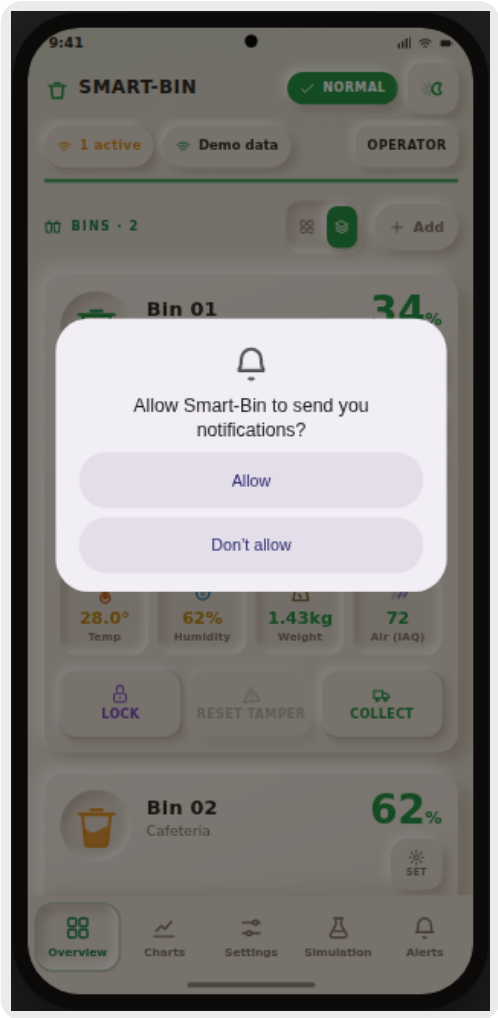
Light Theme



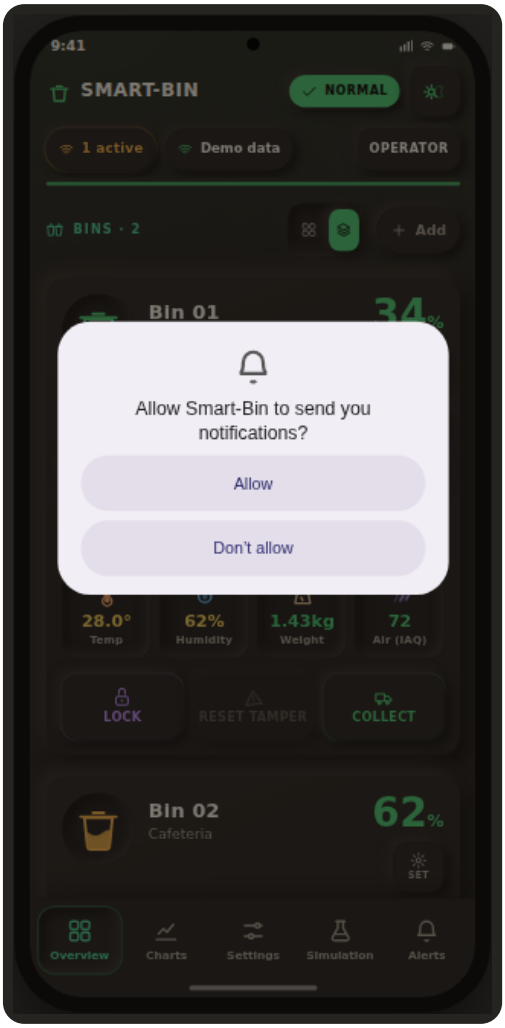
Dark Theme

Android Notification Dialog

System permission dialog on Android 13+



Light Theme

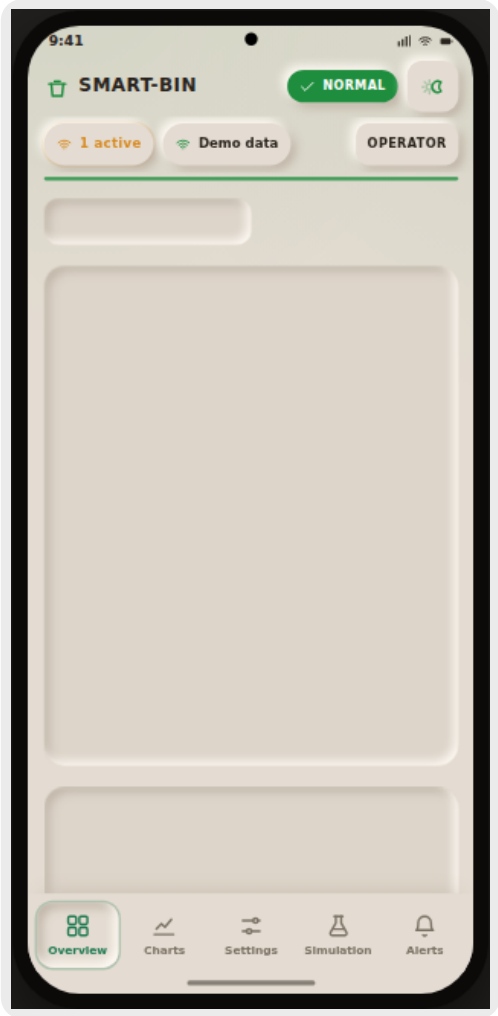


Dark Theme

Overview

Loading

Soft wells pulse while first readings arrive



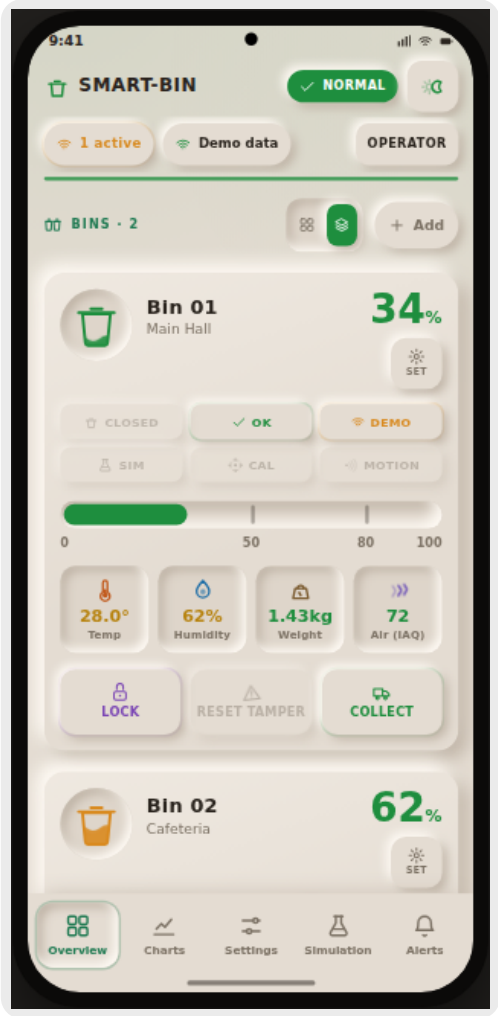
Light Theme



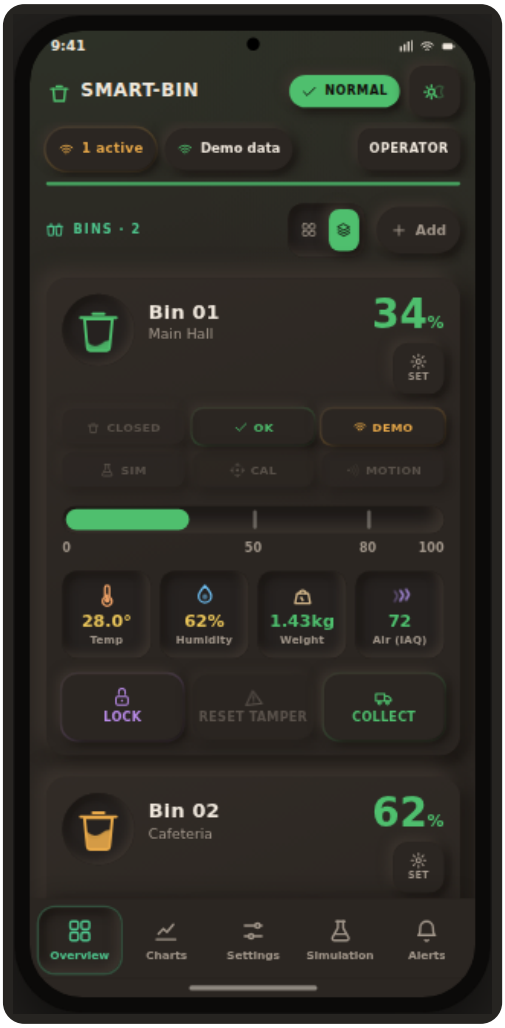
Dark Theme

Overview - Normal

Full cards with hero icon, chips, fill bar, and sensor tiles



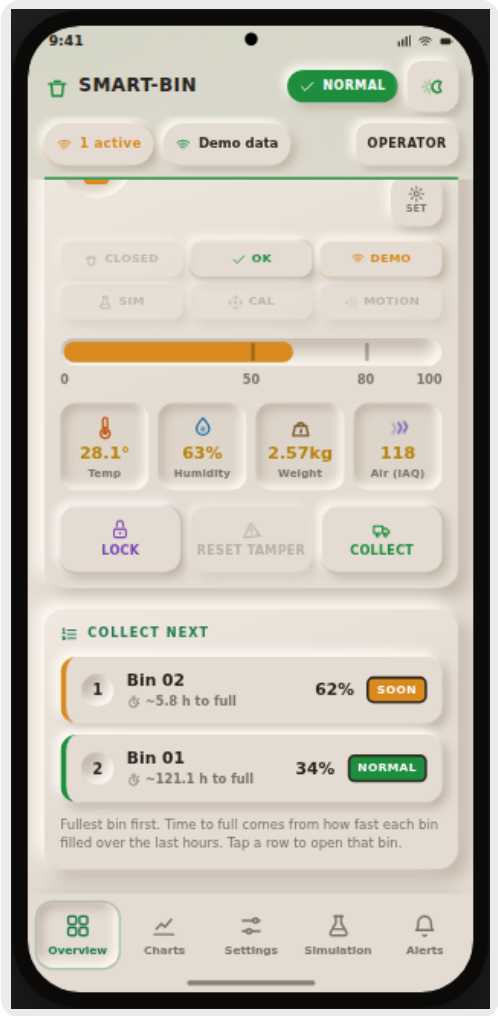
Light Theme



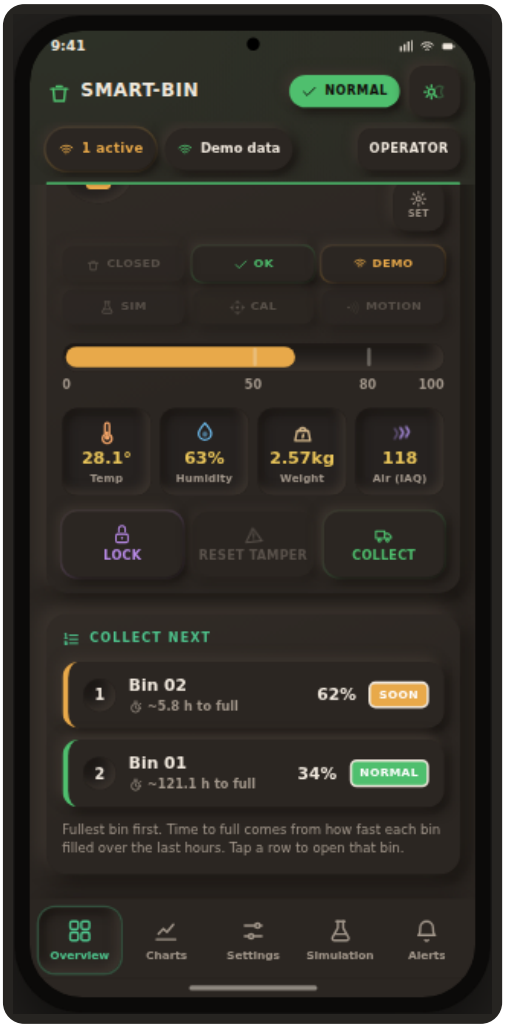
Dark Theme

Overview - Collect Next

Collection priority list, fullest first



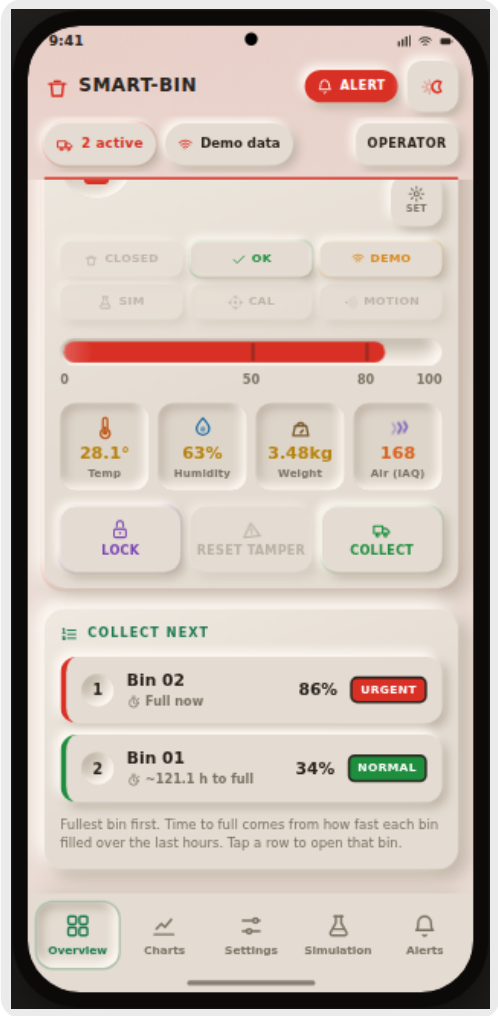
Light Theme



Dark Theme

Overview - Bin Full (Alert)

Red accent, glowing card, shining bar at 86% fill



Light Theme



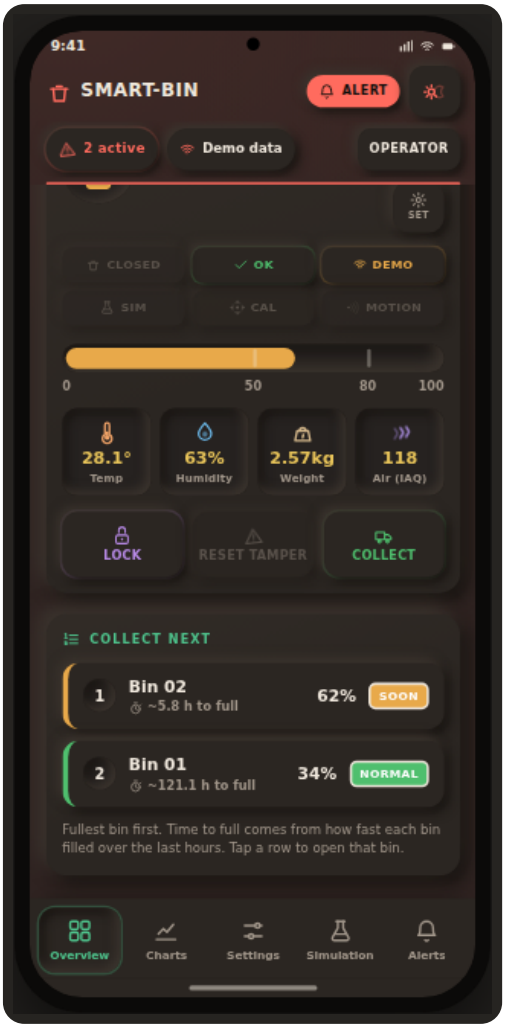
Dark Theme

Overview - Tamper

Tamper alert with RESET button in hard red



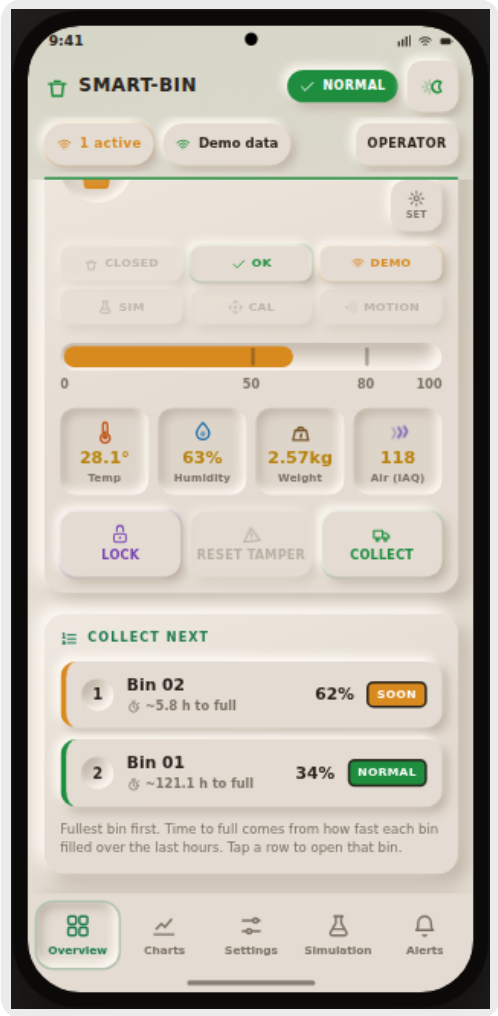
Light Theme



Dark Theme

Overview - Sensor Over Limit

Temperature 42.6 C and air 236 IAQ exceed limits



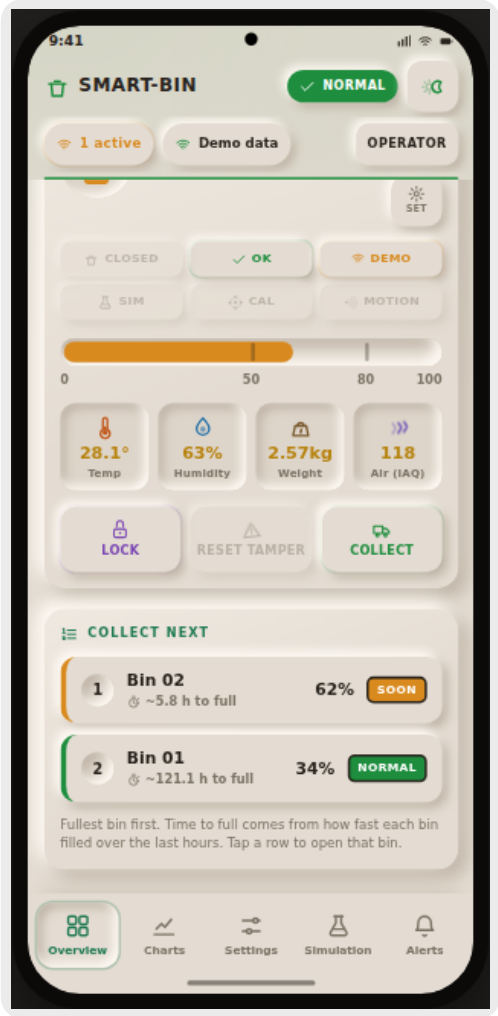
Light Theme



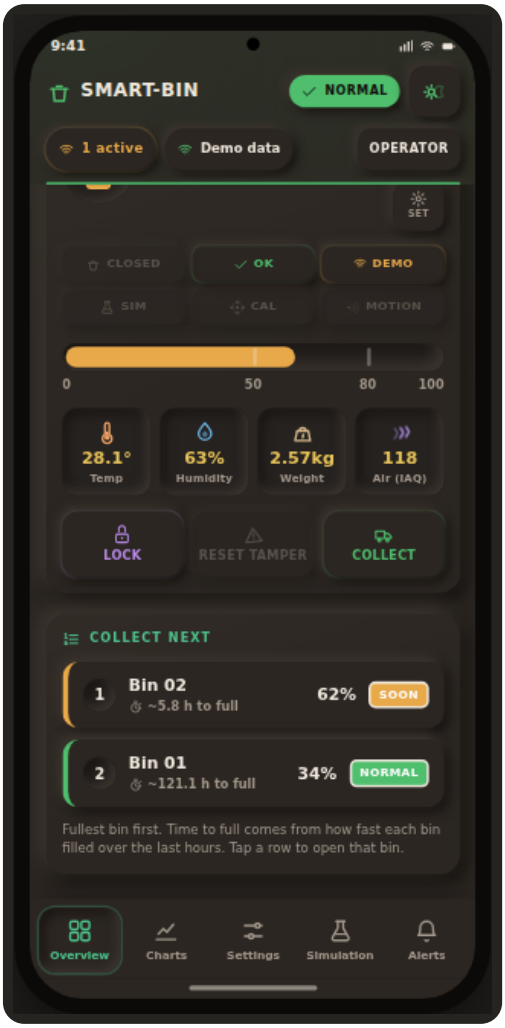
Dark Theme

Overview - Locked

Violet status chip with UNLOCK button



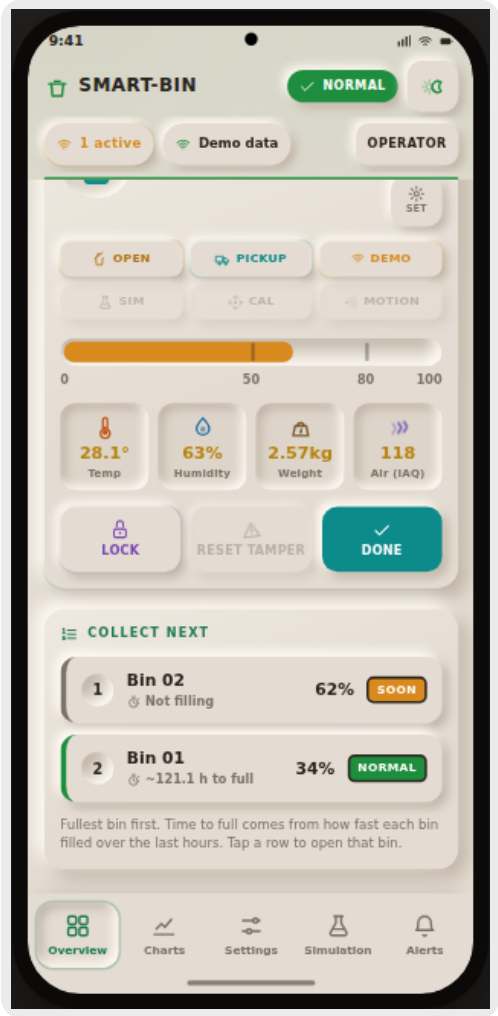
Light Theme



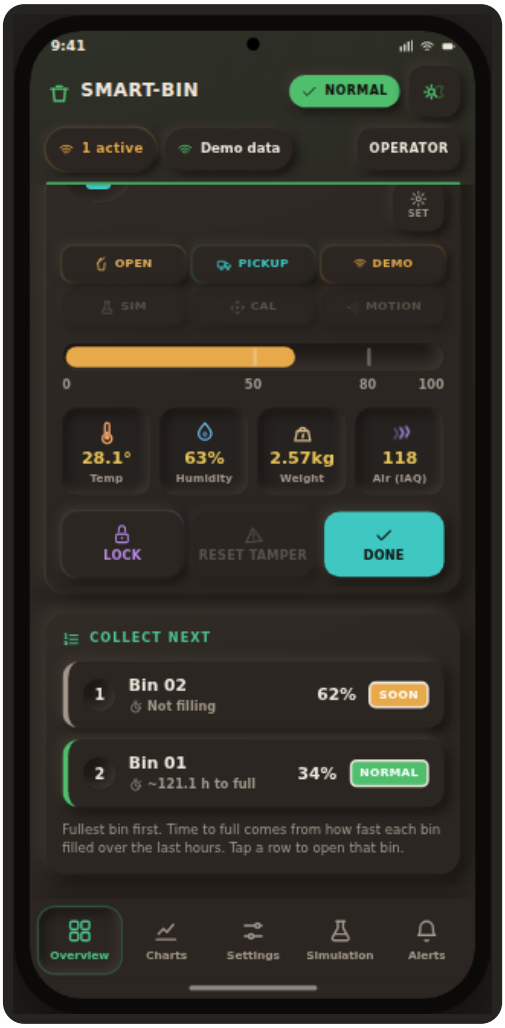
Dark Theme

Overview - Pickup in Progress

DONE button active, lid chip shows OPEN



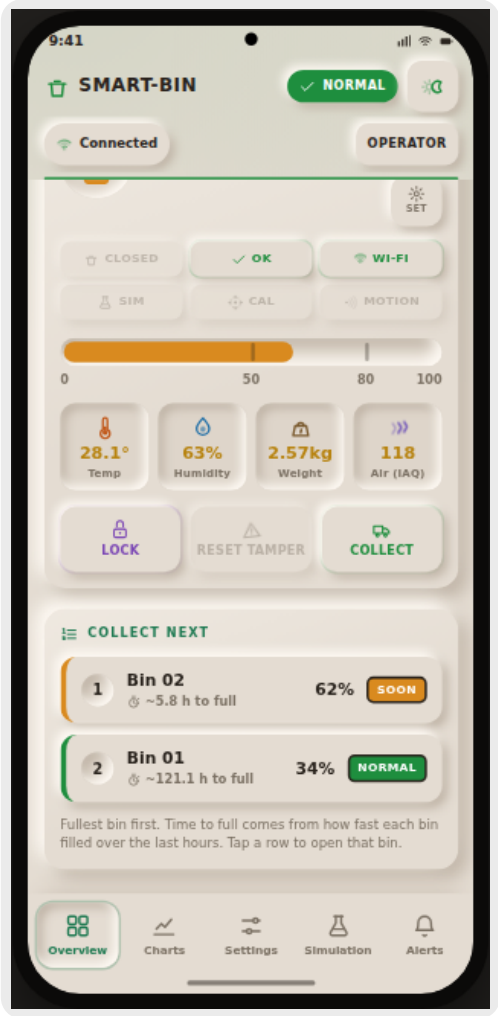
Light Theme



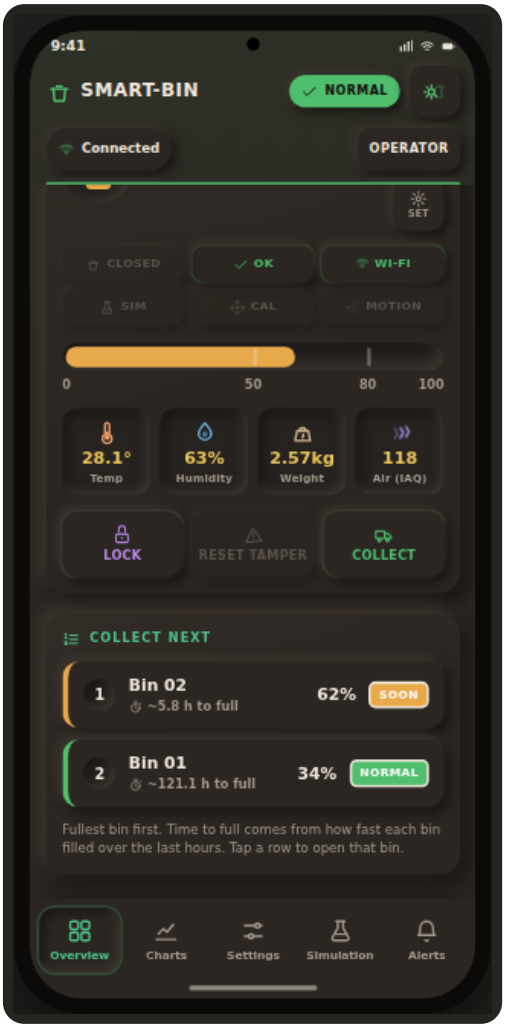
Dark Theme

Overview - Live over Wi-Fi

Connected to bins via IP address



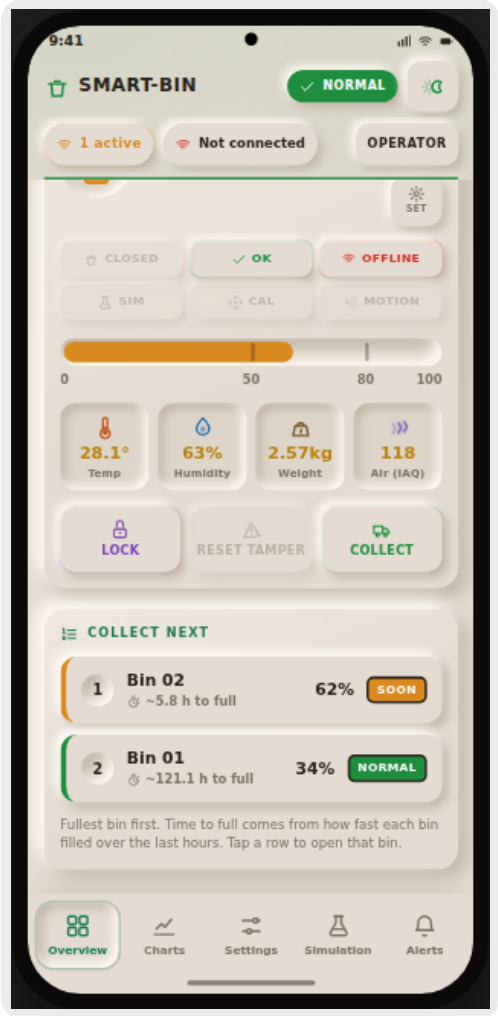
Light Theme



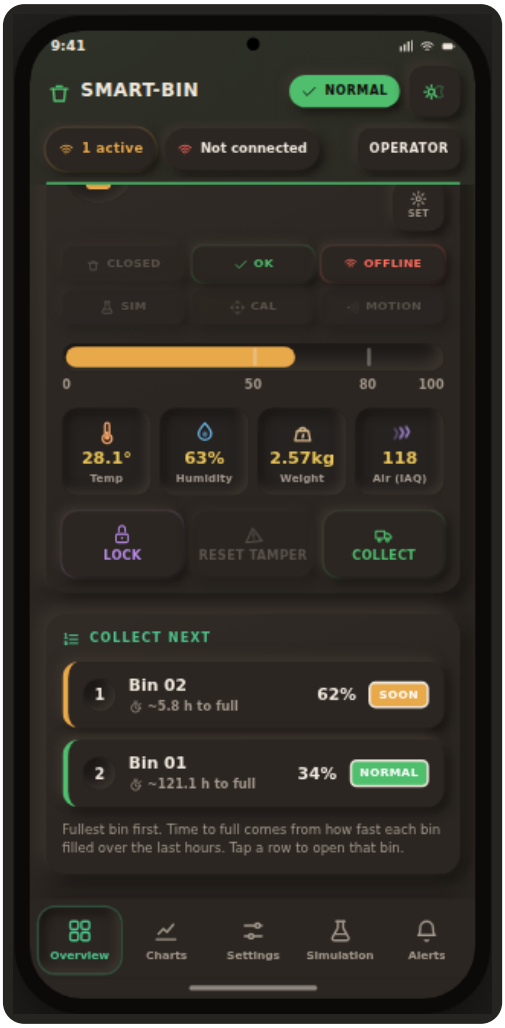
Dark Theme

Overview - Bins Offline

Red link chip, bins not responding



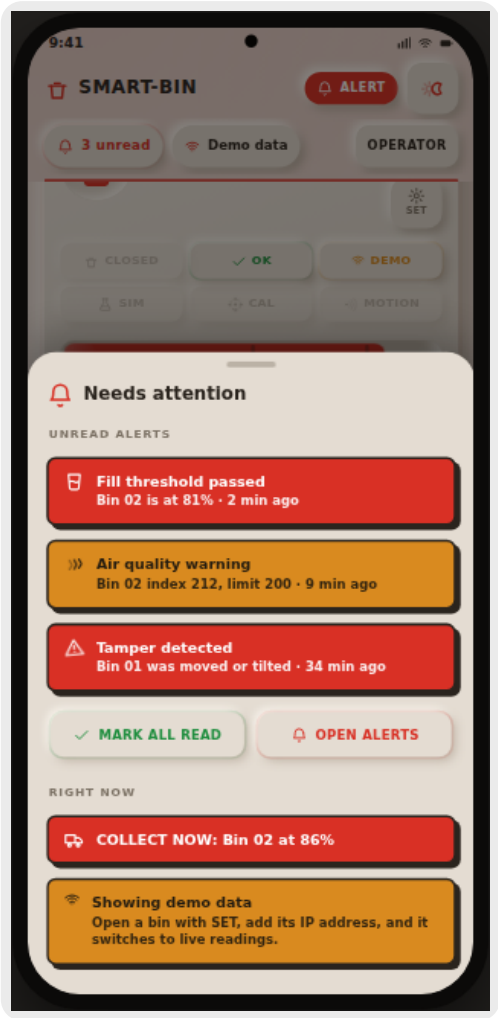
Light Theme



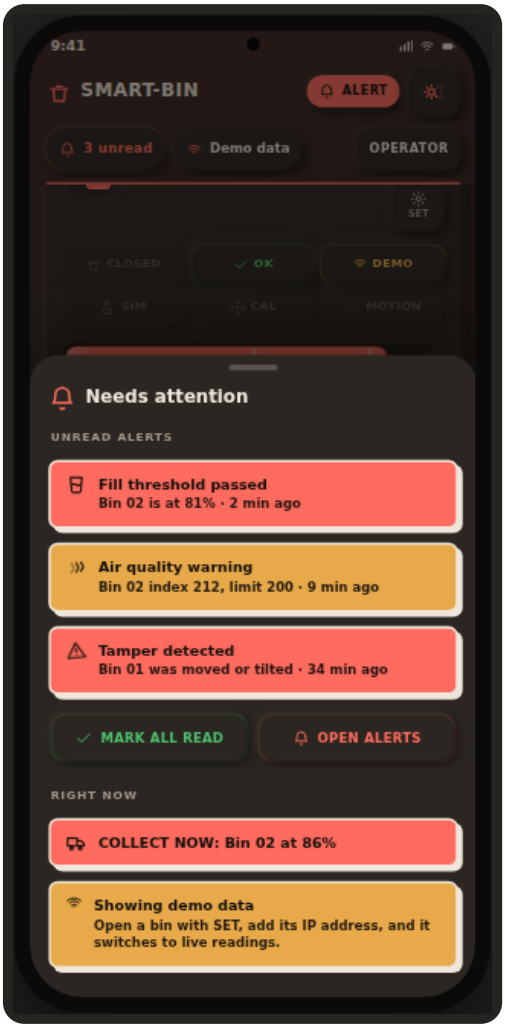
Dark Theme

Needs Attention Sheet

Dashboard alert dropdown as bottom sheet



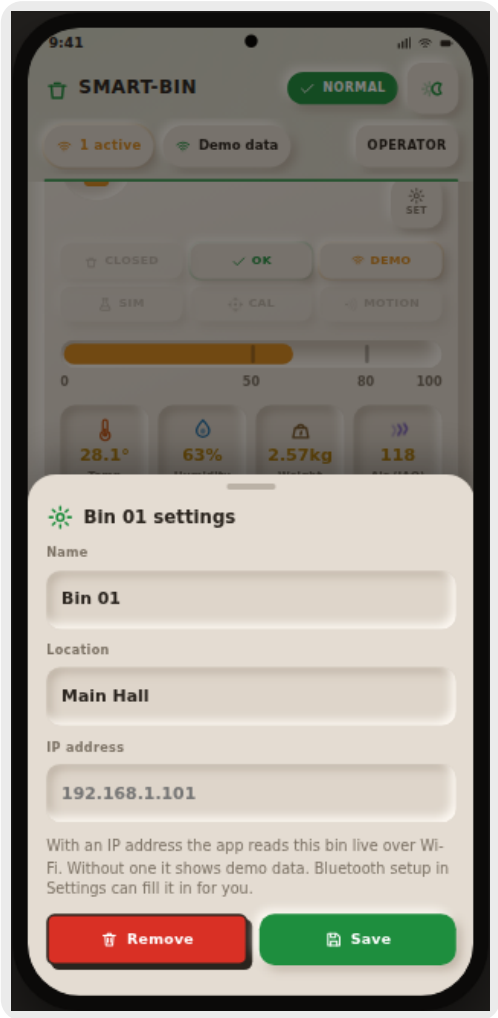
Light Theme



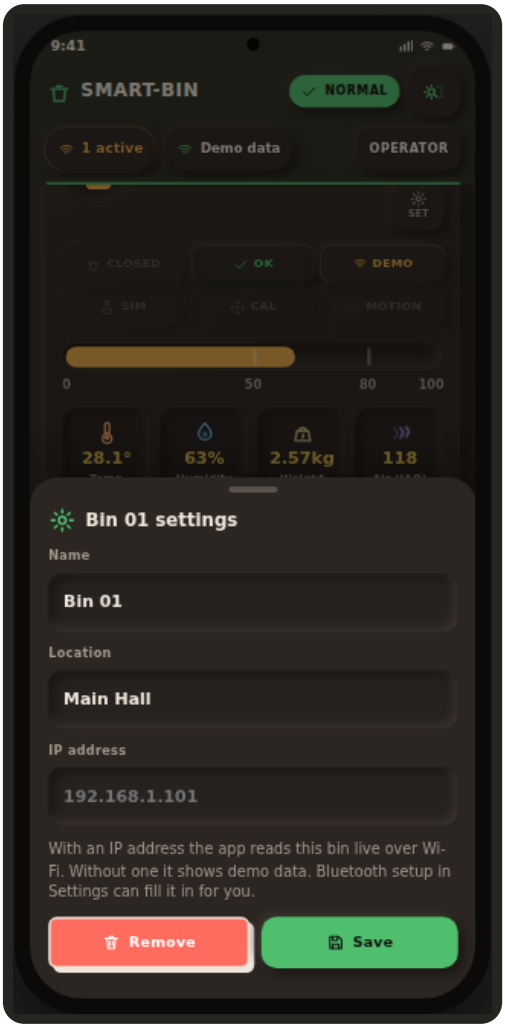
Dark Theme

Bin Settings Sheet

Name, location, and IP address editor



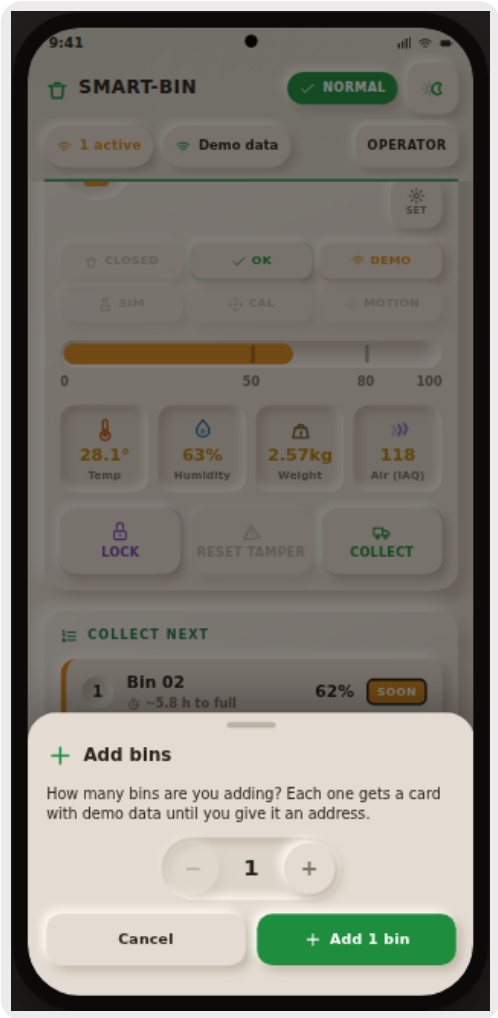
Light Theme



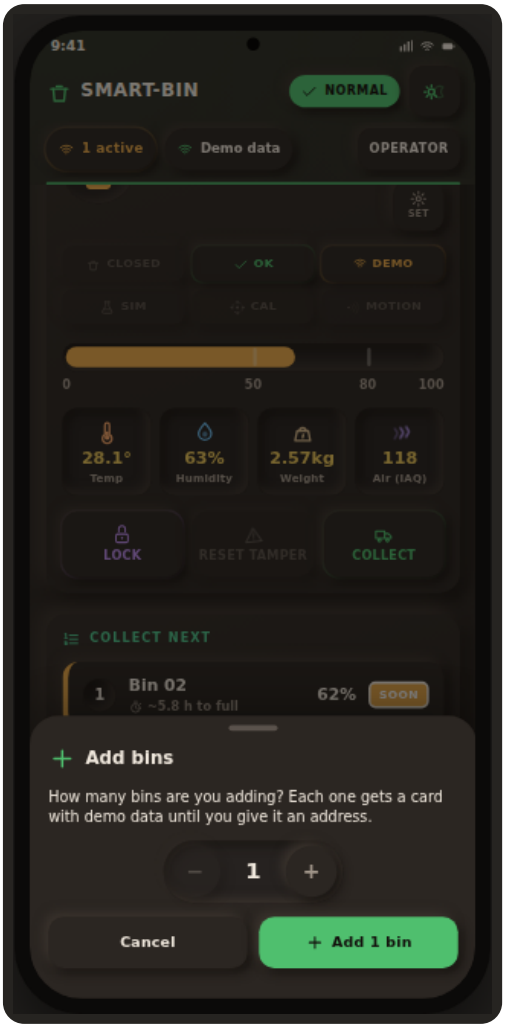
Dark Theme

Add Bins

Add bins with 44 dp buttons



Light Theme

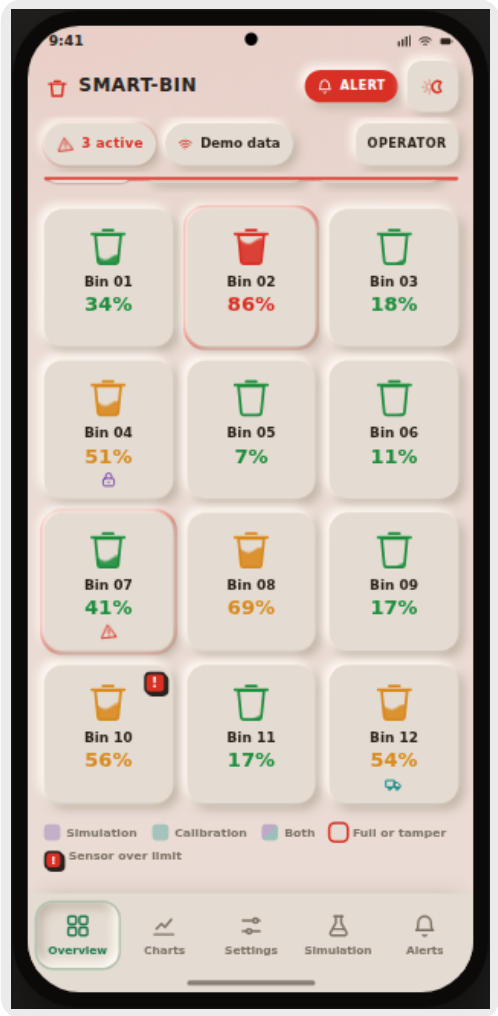


Dark Theme

Many Bins

Grid View - 12 Bins

Fill icons with color, status rings, and mini indicators



Light Theme



Dark Theme

Grid - Simulation & Calibration

Violet and teal tile colors for active modes



Light Theme



Dark Theme

Tap Tile - Full Card Sheet

Full card in sheet with arrow navigation



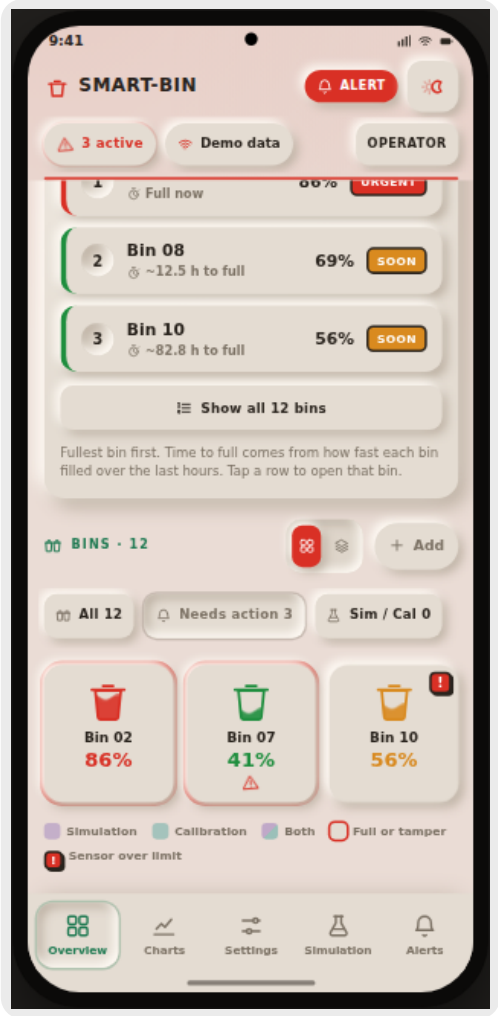
Light Theme



Dark Theme

Grid - Filter: Needs Action

Filter chips for full, tamper, or sensor alerts



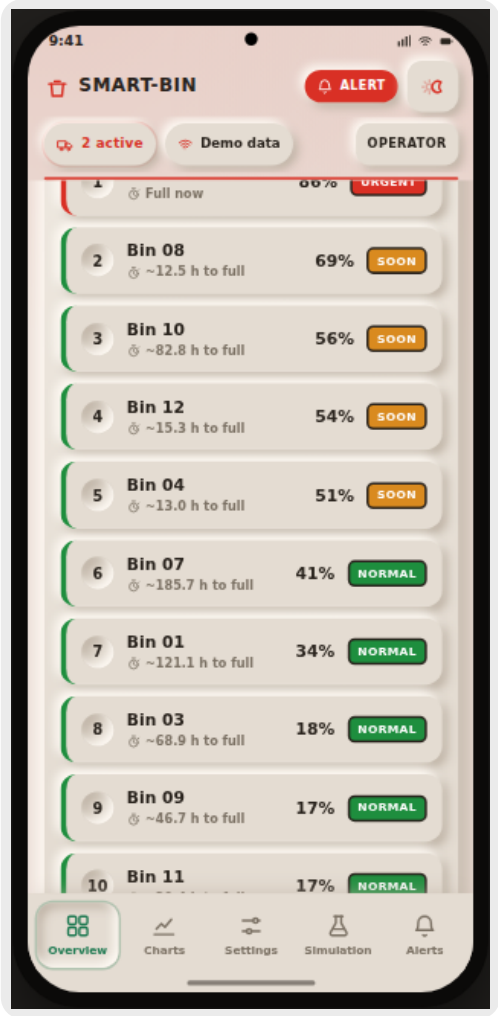
Light Theme



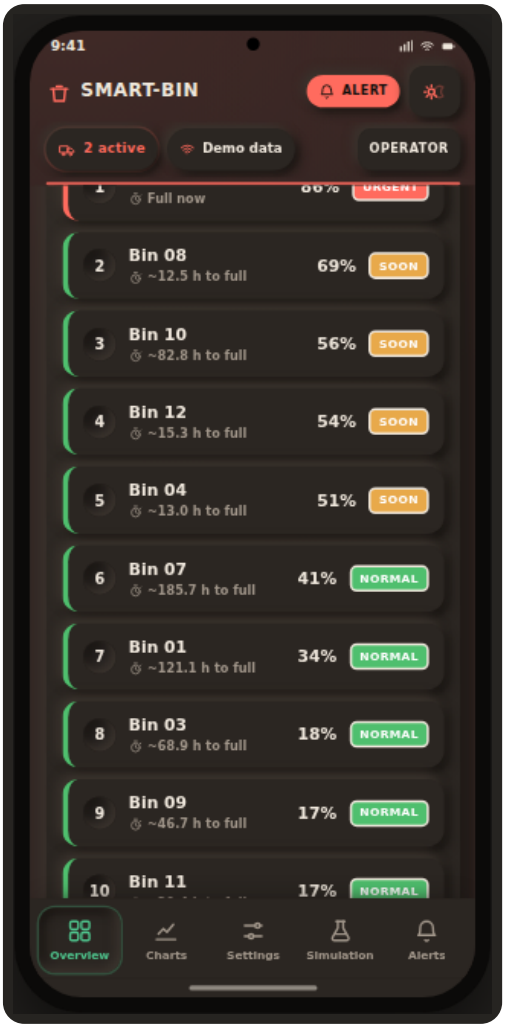
Dark Theme

Collect Next - All Bins

Priority list merged for all 12 bins



Light Theme



Dark Theme

Grid - Cards View

View switch to full cards for any number of bins



Light Theme



Dark Theme

Charts

Charts - Fill Level

Range and bin buttons with dashed limit line



Light Theme



Dark Theme

Charts - Environment (All)

0-200% scale, temperature and gas as share of limit



Light Theme



Dark Theme

Charts - Gas Only

BME680 range 0-500 IAQ



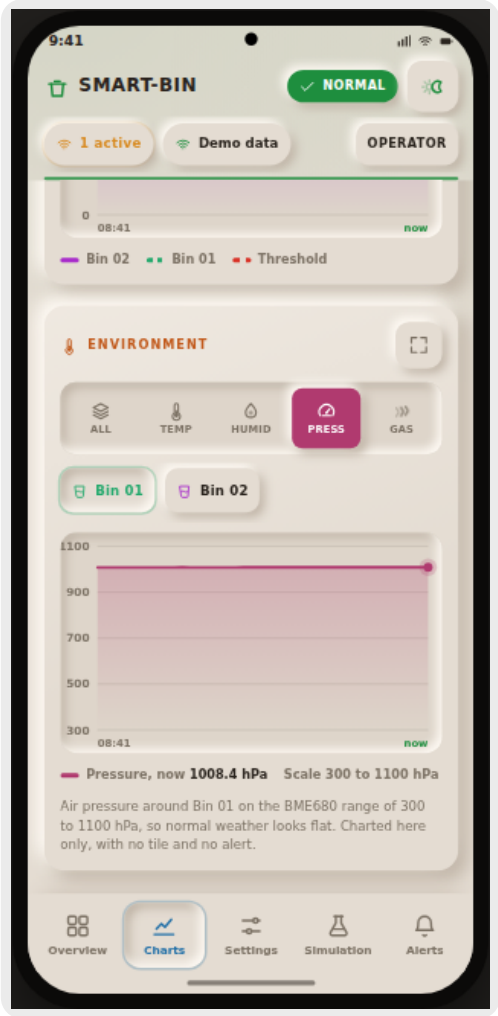
Light Theme



Dark Theme

Charts - Pressure

300-1100 hPa, no limit line



Light Theme



Dark Theme

Charts - 12 Bins

Bin picker for many bins, top 5 shown



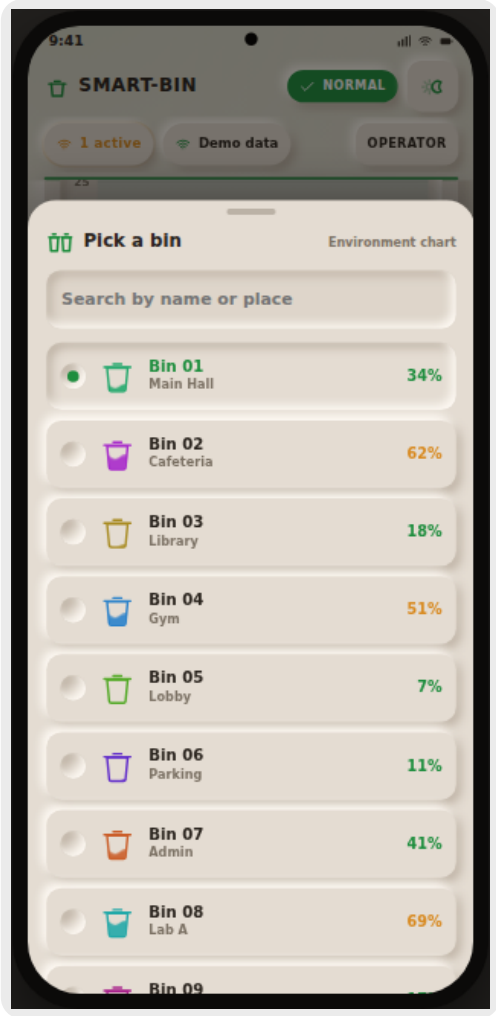
Light Theme



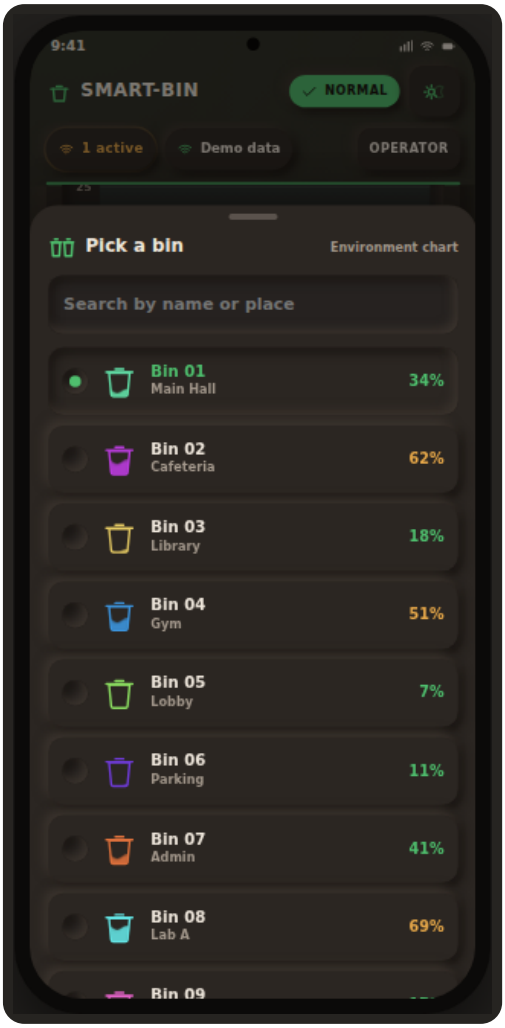
Dark Theme

Charts - Bin Picker Sheet

All bins with fill, place, and mode tags



Light Theme

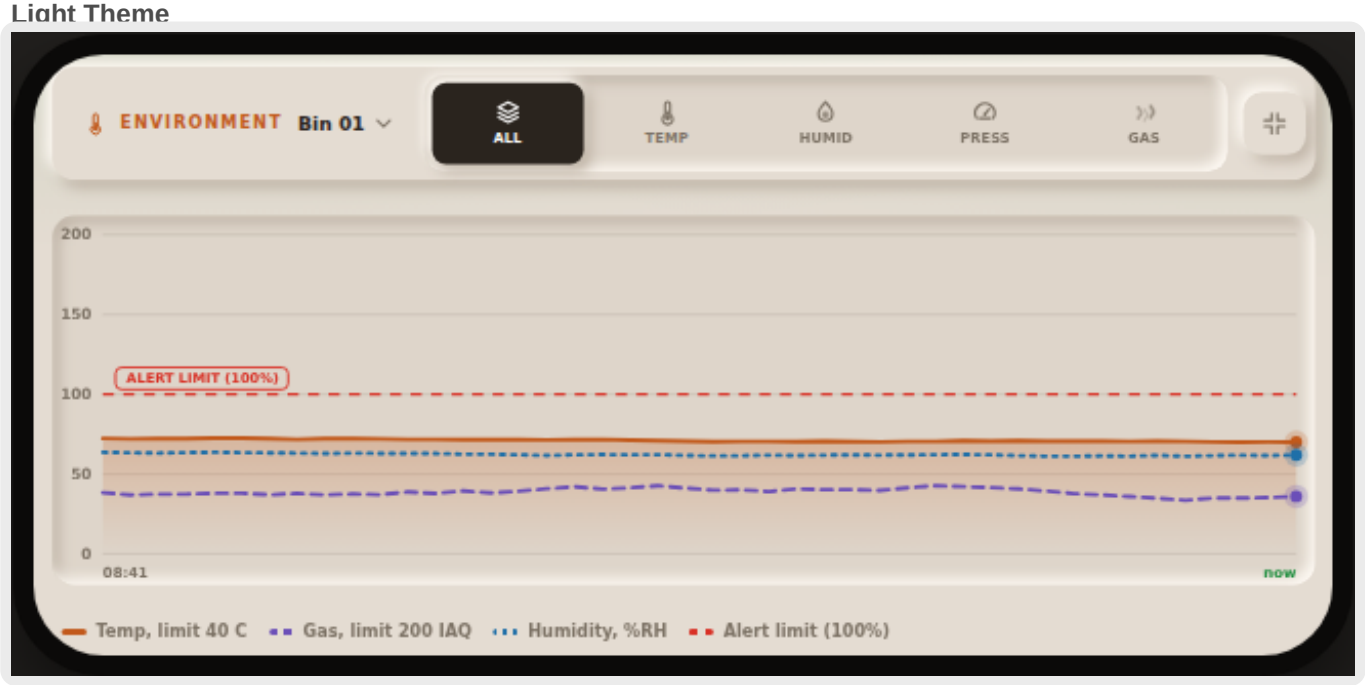


Dark Theme

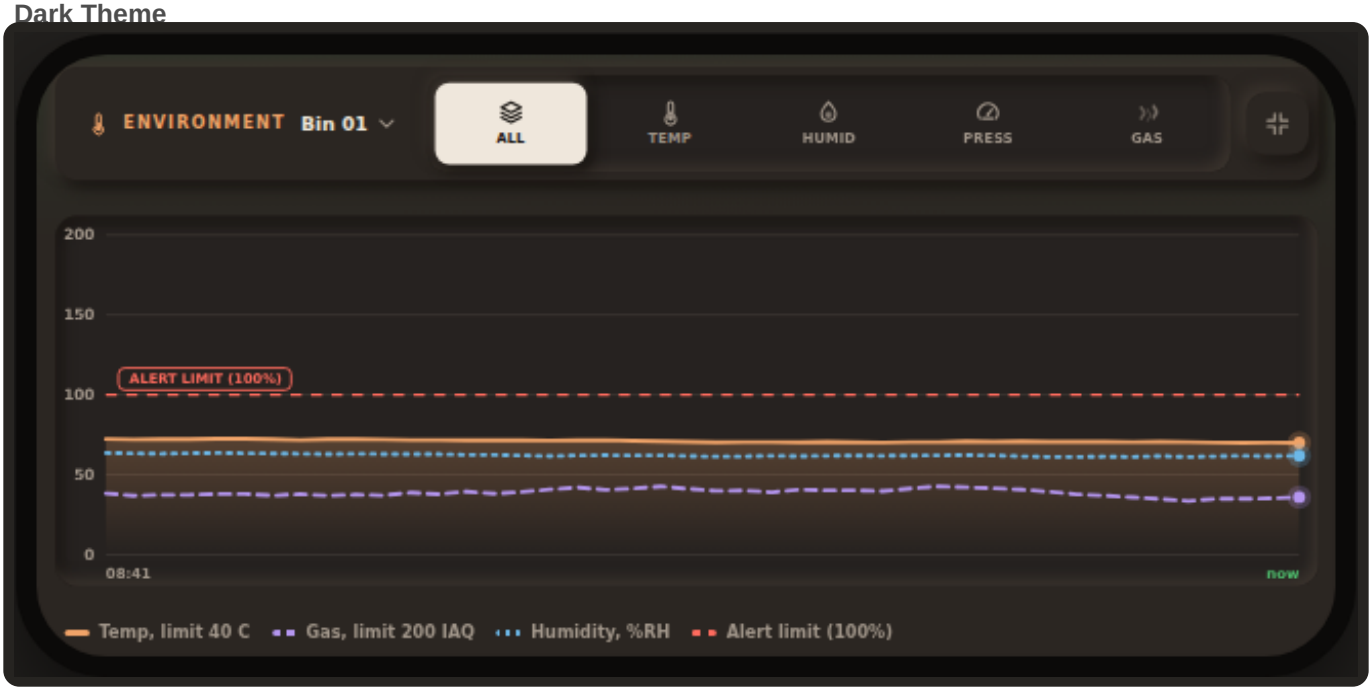
Charts - Landscape

Expanded chart in landscape orientation

Light Theme



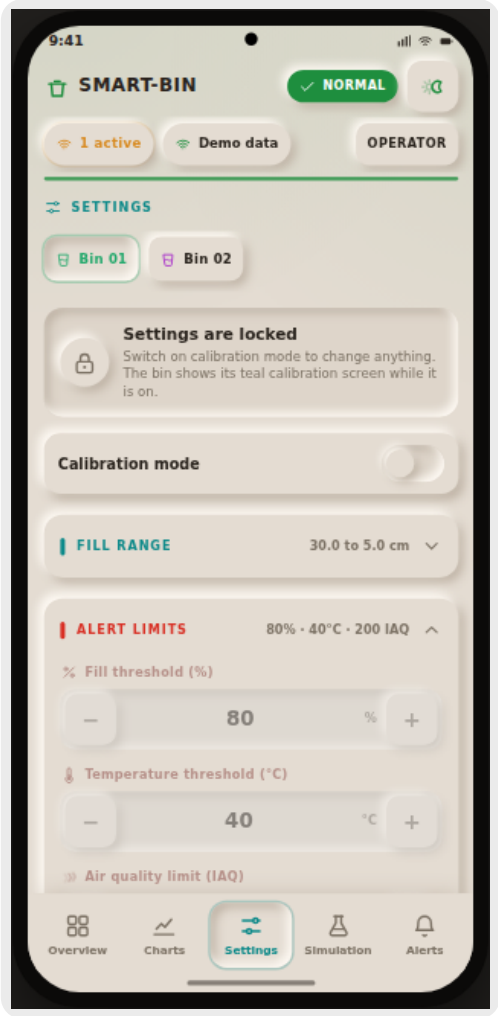
Dark Theme



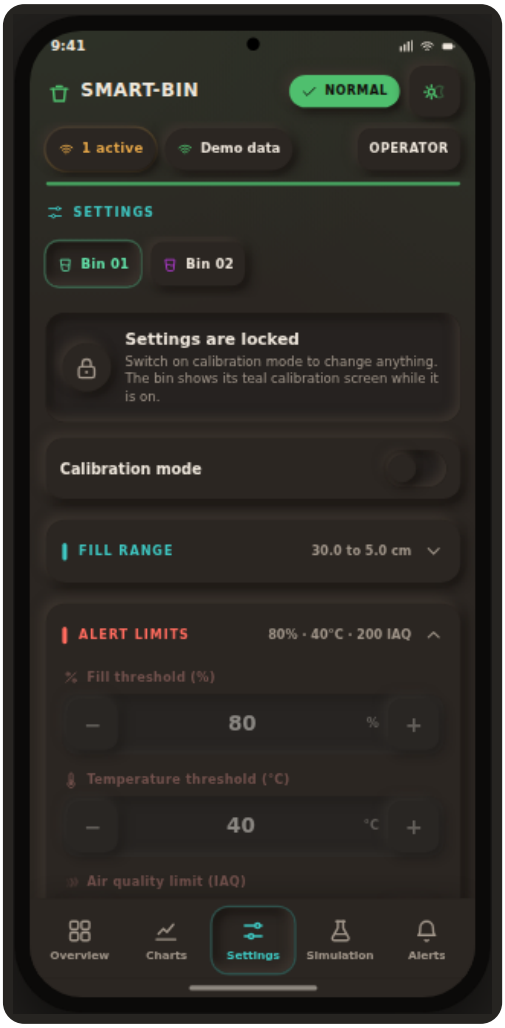
Settings

Settings - Locked

Folded sections with summaries



Light Theme



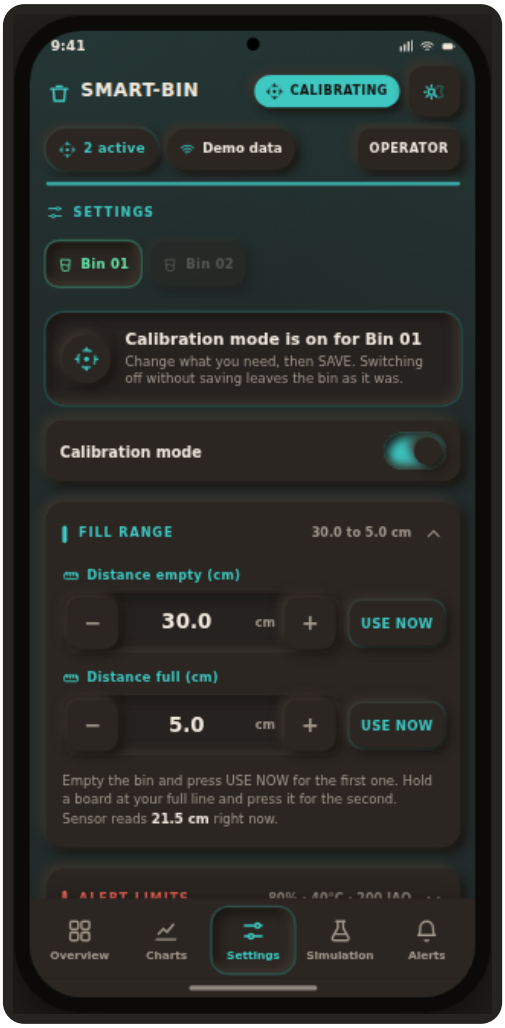
Dark Theme

Settings - Calibration Mode

Teal accent with stepper controls



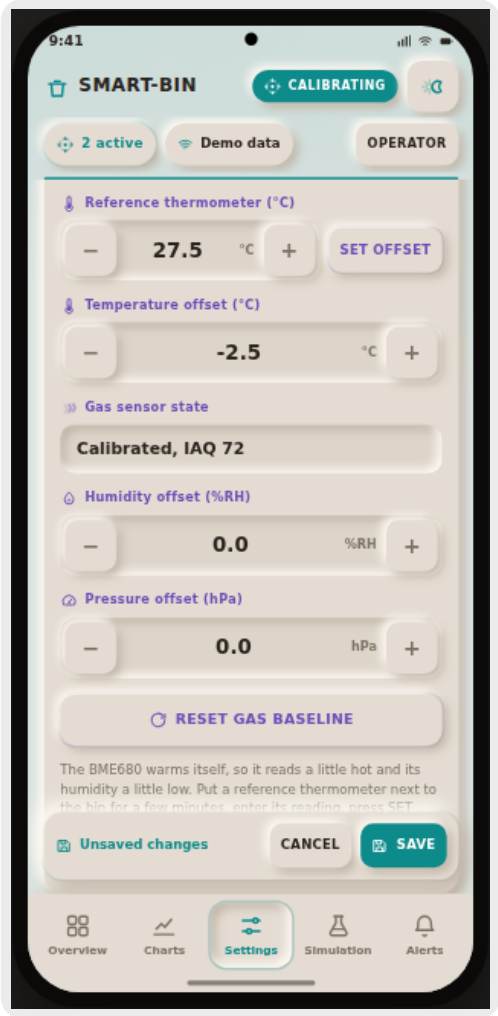
Light Theme



Dark Theme

Settings - Air Sensor (Unsaved)

BME680 offsets with sticky save bar



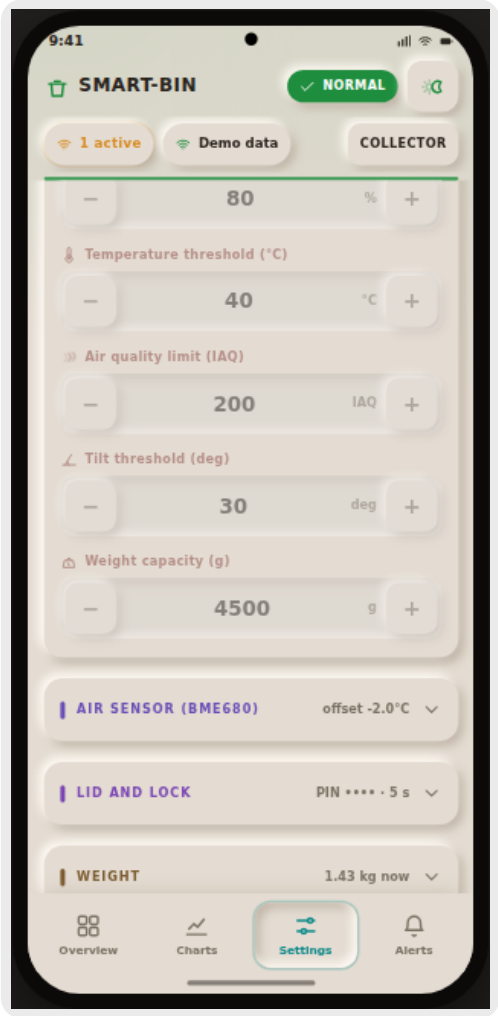
Light Theme



Dark Theme

Settings - Collector View

Read-only with collect PIN card



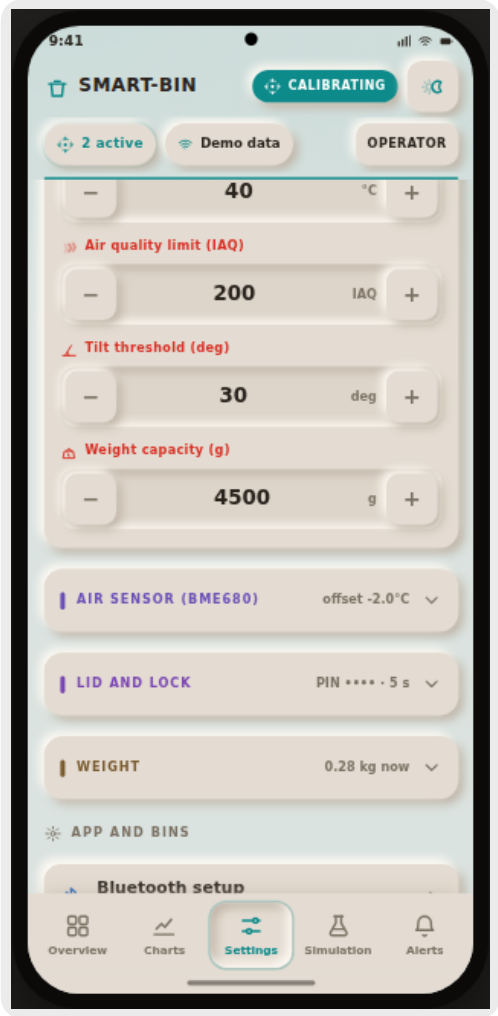
Light Theme



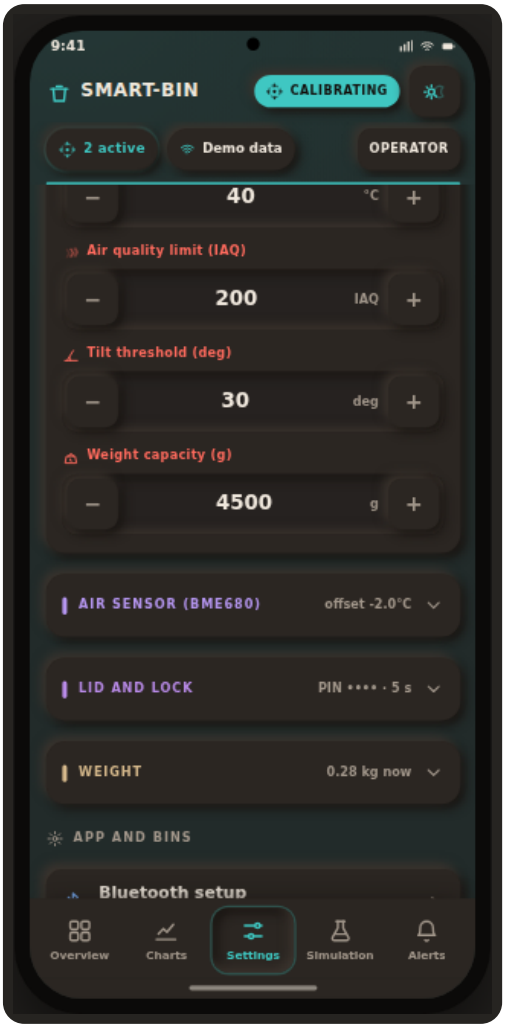
Dark Theme

Settings - 12 Bins

Bin picker during calibration



Light Theme



Dark Theme

Bluetooth Setup

Four-step Wi-Fi setup flow



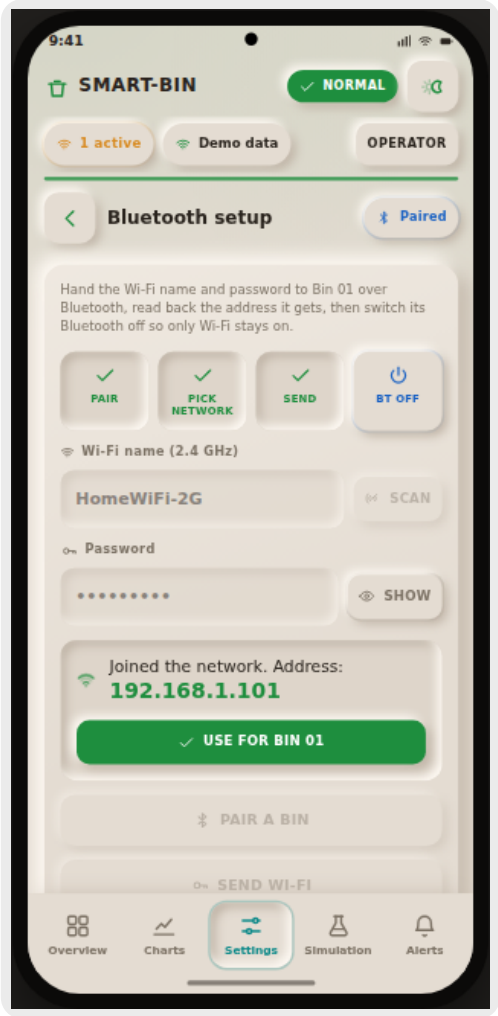
Light Theme



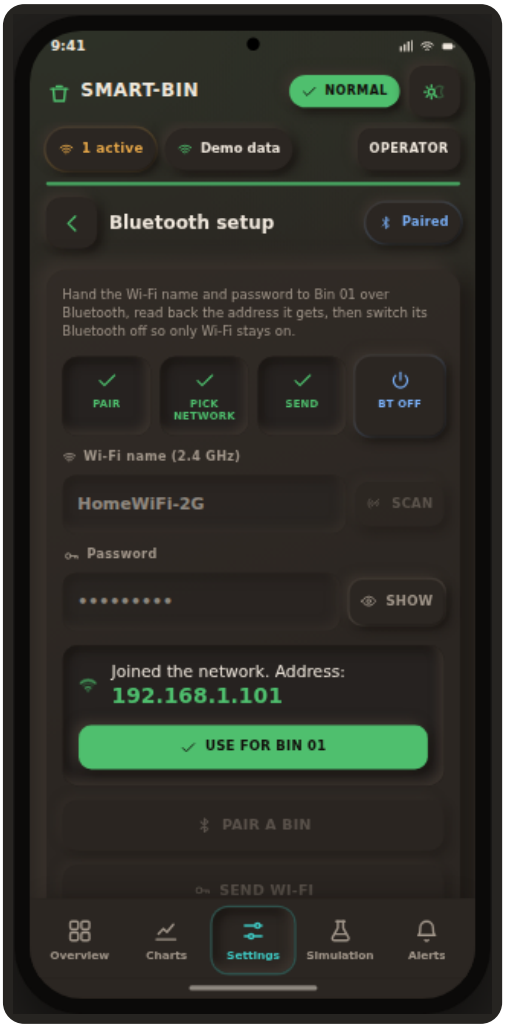
Dark Theme

Bluetooth - Joined

Bin address received, ready to use



Light Theme

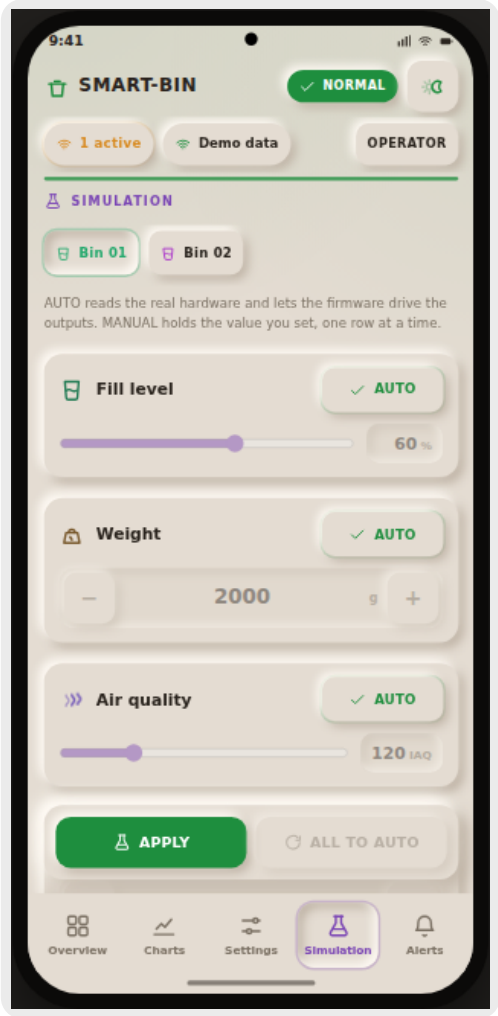


Dark Theme

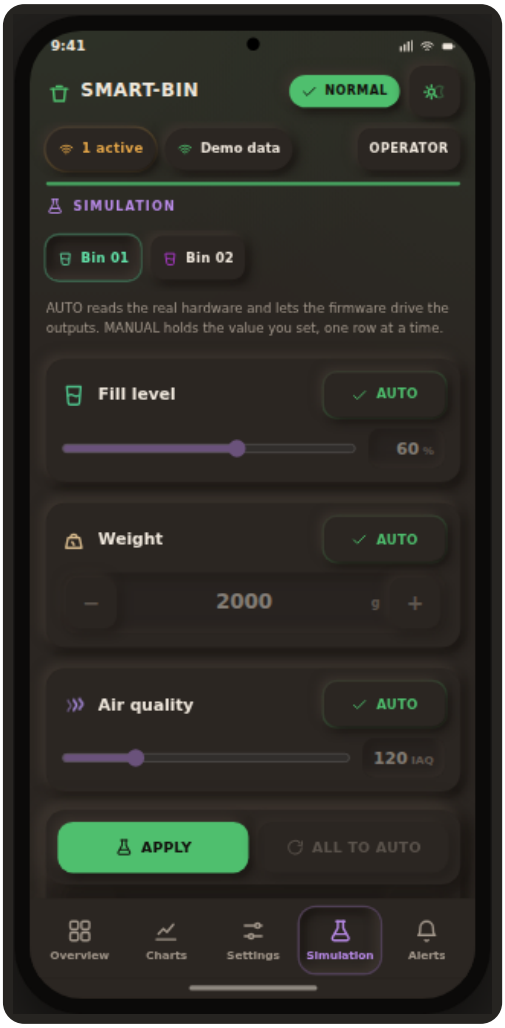
Simulation

Simulation - All AUTO

One card per sensor, values dimmed



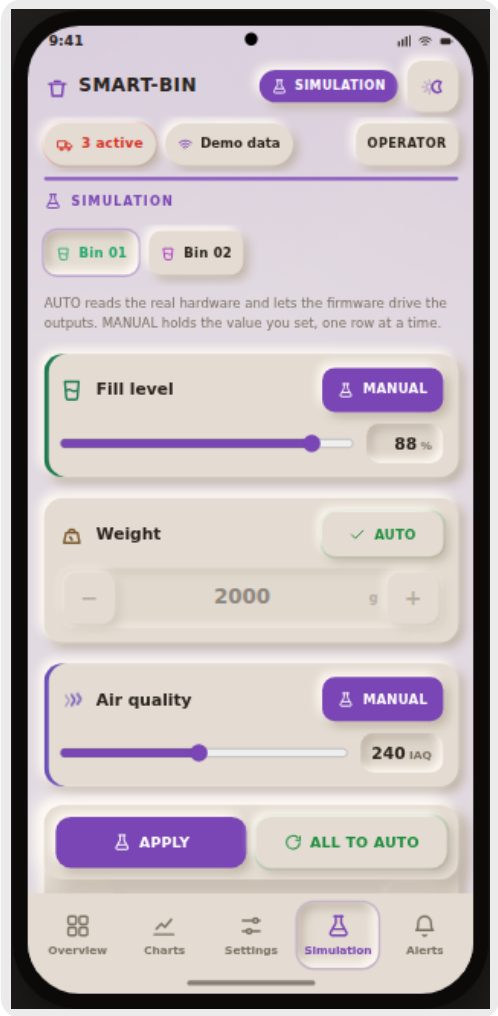
Light Theme



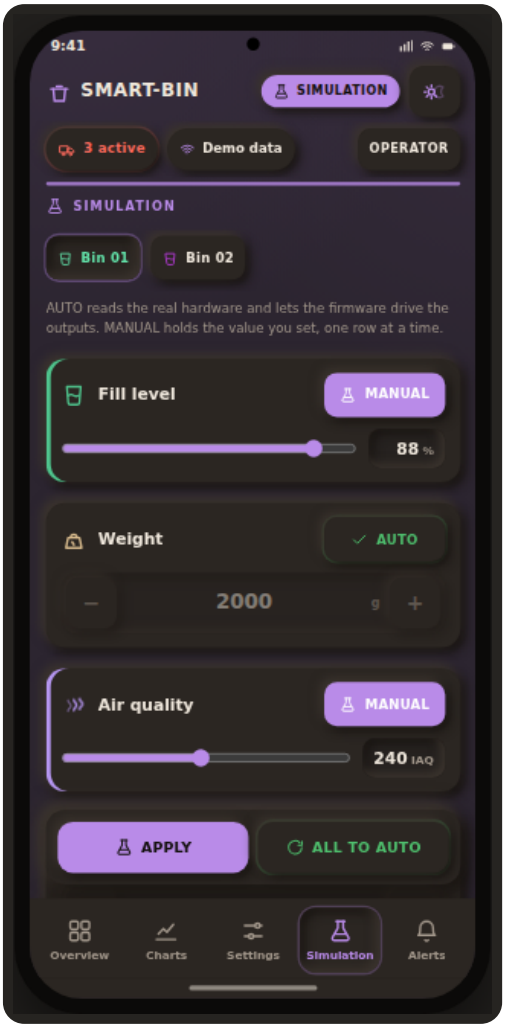
Dark Theme

Simulation - Manual Rows

Fill and air held by hand, violet accent



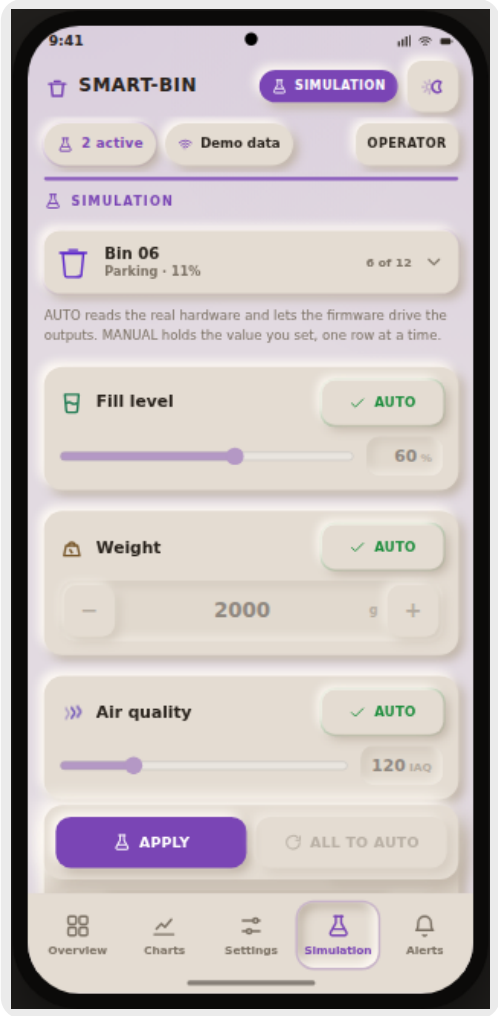
Light Theme



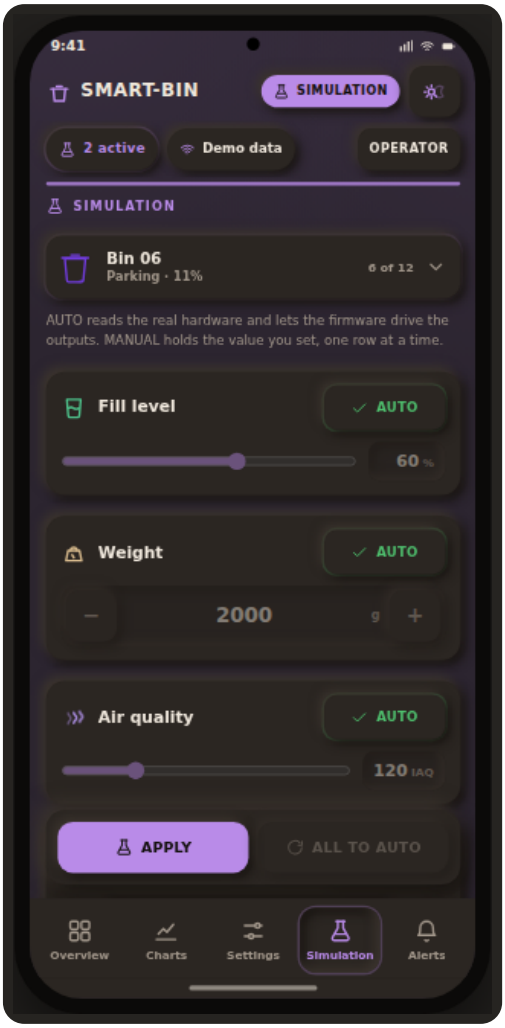
Dark Theme

Simulation - 12 Bins

Picker with SIM tags on active bins



Light Theme



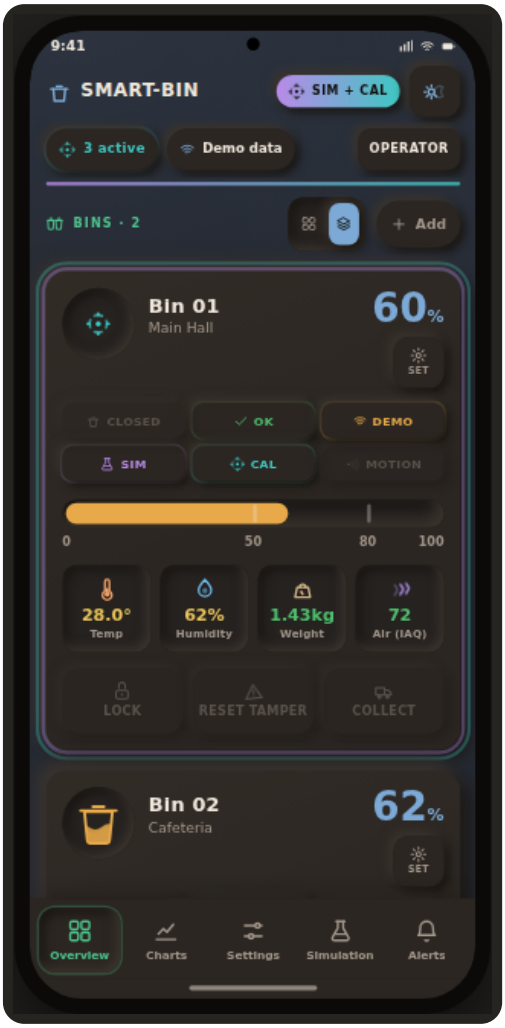
Dark Theme

Simulation + Calibration

Blended accent bar and double ring



Light Theme

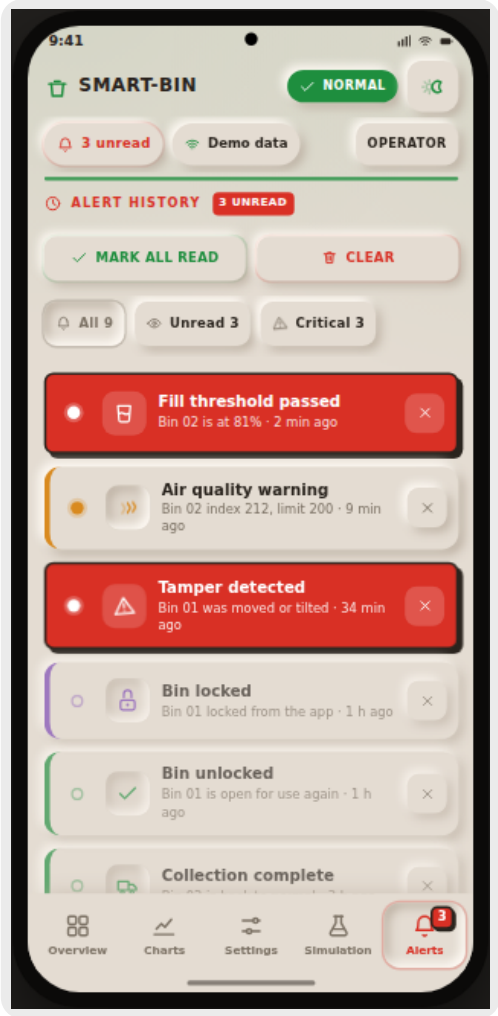


Dark Theme

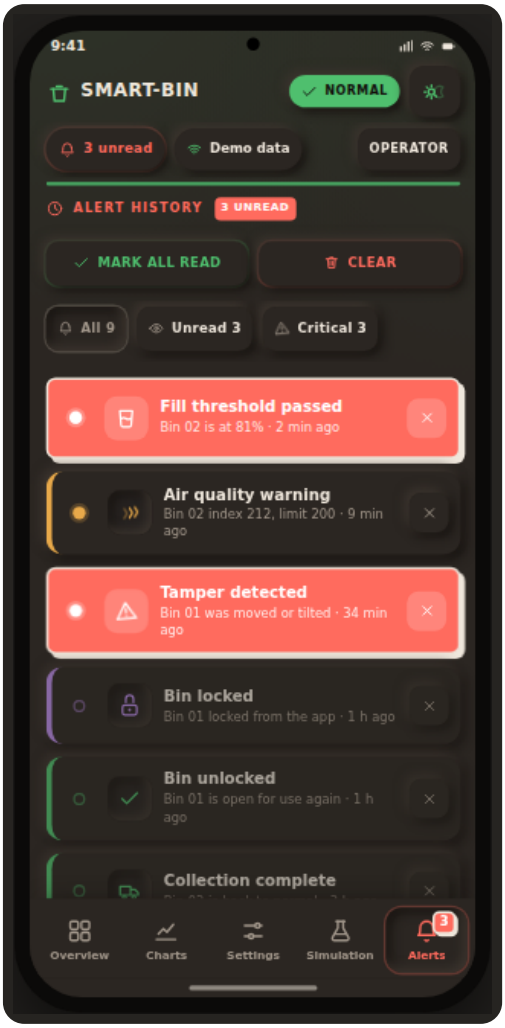
Alerts

Alert History

Critical alerts in hard red cards with unread dots



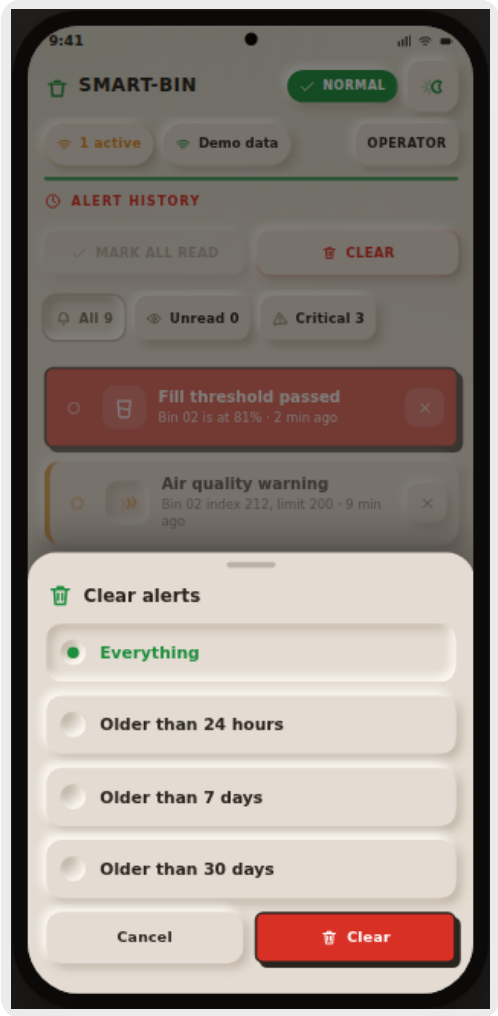
Light Theme



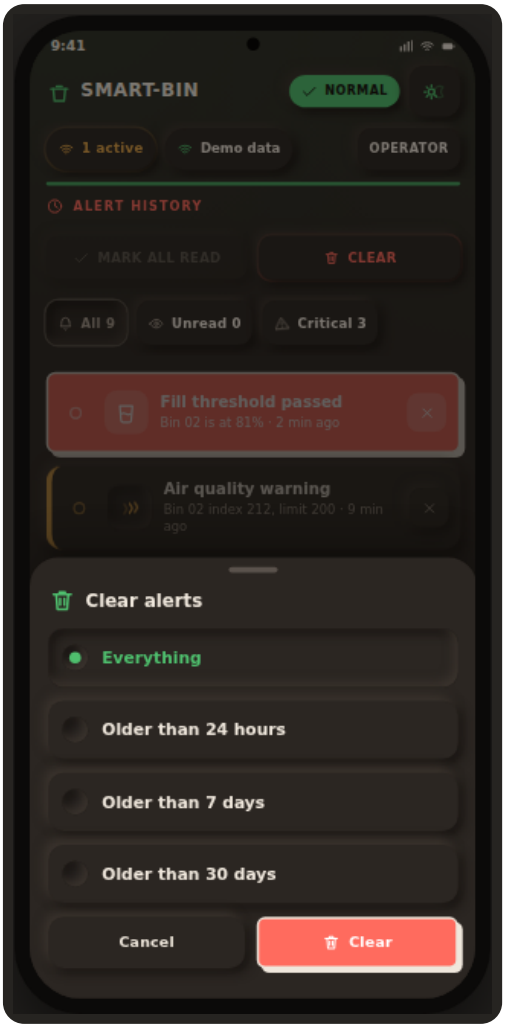
Dark Theme

Clear Alerts

Choice sheet for clearing alerts



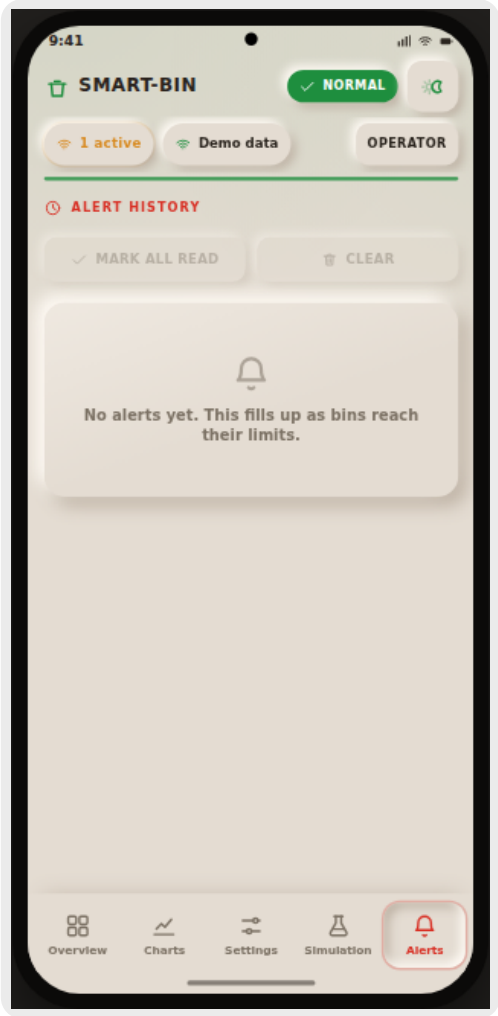
Light Theme



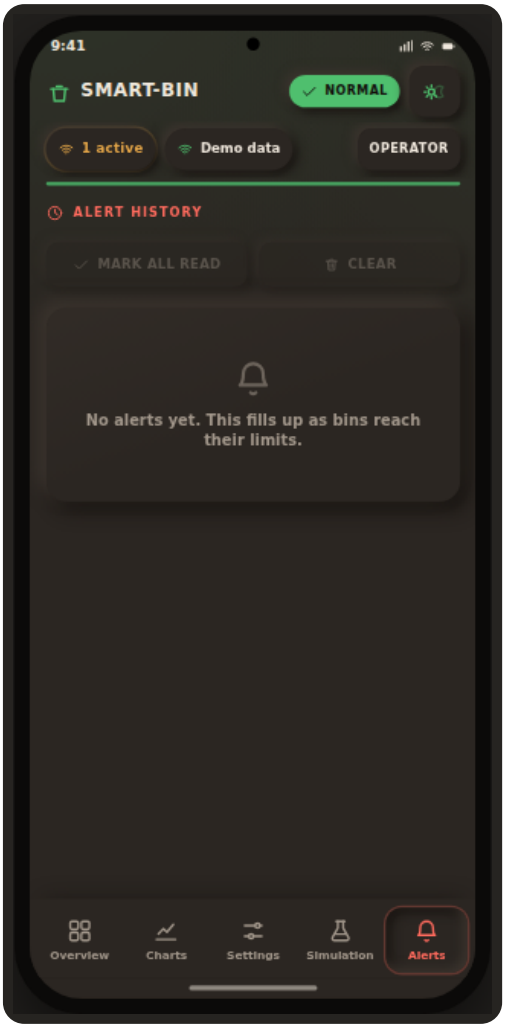
Dark Theme

No Alerts

Empty state



Light Theme

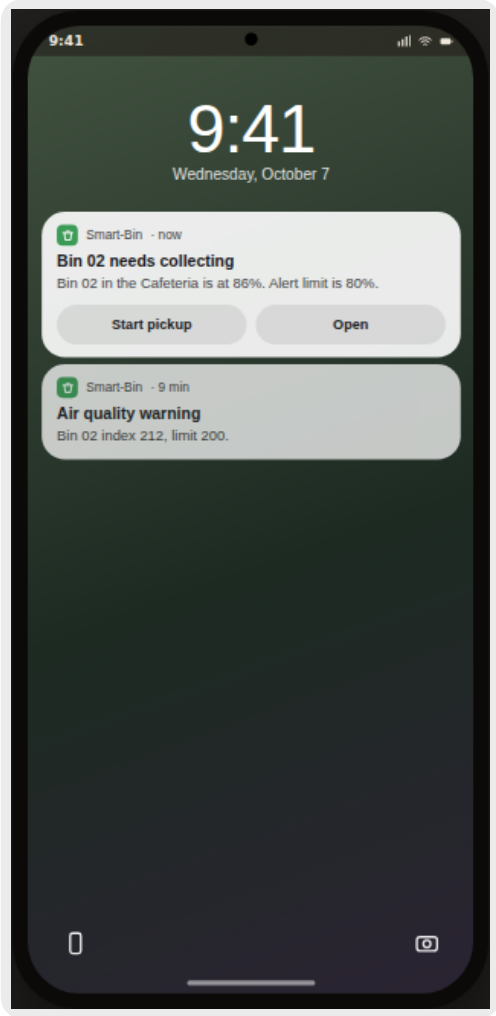


Dark Theme

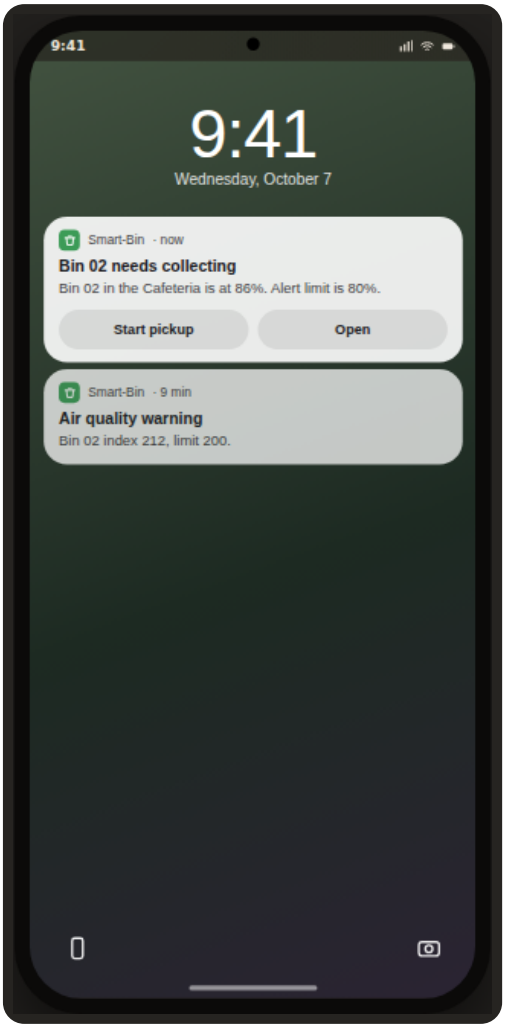
Push Notifications

Push - Lock Screen Alert

Notification with Start Pickup action



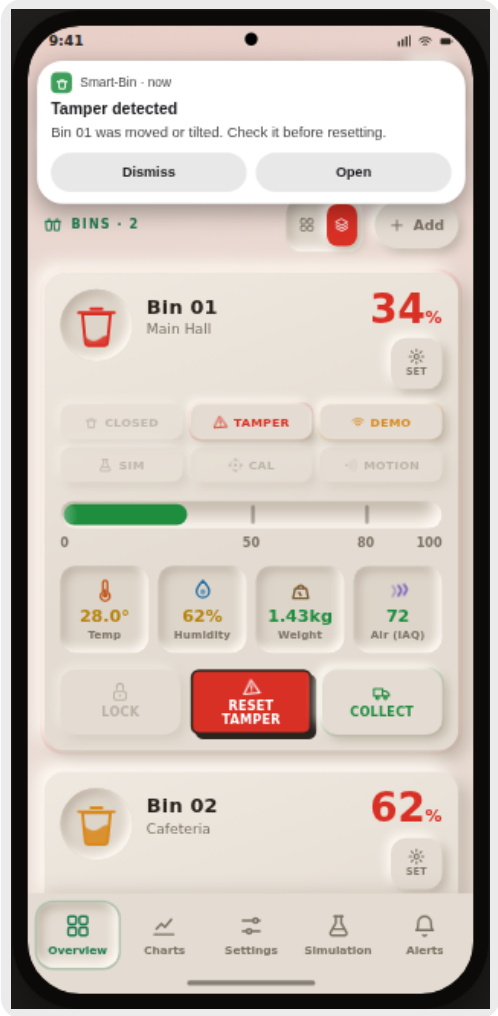
Light Theme



Dark Theme

Push - Heads-Up Banner

Alert banner while phone is in use



Light Theme

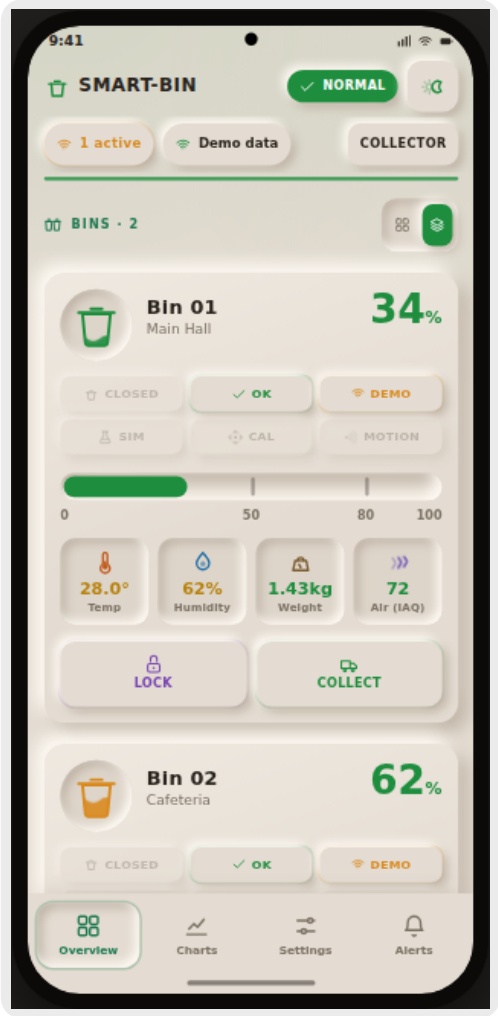


Dark Theme

Roles & Theme

Collector Overview

No Simulation tab, no SET, simplified controls



Light Theme



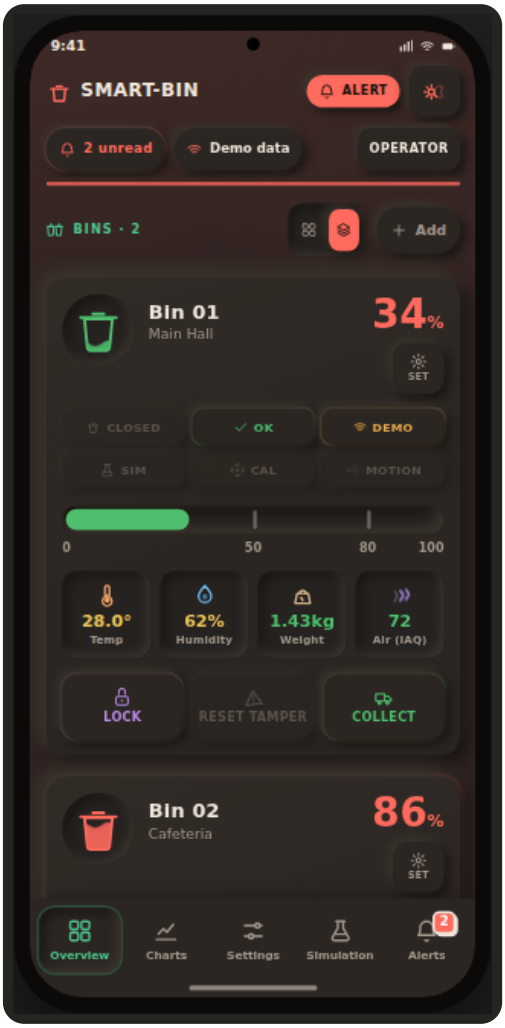
Dark Theme

Dark Theme - Alert

Dark tokens with alert state



Light Theme



Dark Theme

Dark Theme - Calibration

Teal mode in dark theme



Light Theme



Dark Theme